

Lab Manual

CSC418-DevOps for Cloud Computing



CUI

**Department of Computer Science
Islamabad Campue**

Lab Contents:

This lab emphasizes the core concepts of implementing DevOps using cloud computing. The topics include: Virtualization: Installing VMware Workstation with different flavours of linux OS on top of windows; Cloud: Service architecture and services using Azure; Containerization using Docker: Installation, configuration and Generating Docker Images; Cloud Deployment and Kubernetes; Installation and Uses of Git commands Pull and Push to GitHub Repository; Version Control Management using GIT; Git, and Docker for Cloud; DevOps Tools for CI / CD: Deploying Docker Images for multiple platform; DevOps Planning: Agile Project Management; Build using Maven; Testing in DevOps: Selenium; Creating Complete Jenkins Pipeline; Software Configuration Management using Puppet tool; and Software Configuration Management and provisioning using Ansible tool.

Student Outcomes (SO)

S.#	Description
2.	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines
3.	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations
4.	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to <i>complex</i> computing activities, with an understanding of the limitations
5.	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings.
9.	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional

Intended Learning Outcomes

Sr.#	Description	Blooms Taxonomy Learning Level	SO
CLO -1	Apply DevOps pipeline automation techniques for code deployment.	<i>Applying</i>	2,4
CLO -2	Develop cloud native applications using current DevOps tools.	<i>Creating</i>	3-5,9

Lab Assessment Policy

The lab work done by the student is evaluated using Psycho-motor rubrics defined by the course instructor, viva-voce, project work/performance. Marks distribution is as follows:

Assignments	Lab Mid Term Exam	Lab Terminal Exam	Total
25	25	50	100

Note: Midterm and Final term exams must be computer based.

List of Labs

Lab #	Main Topic	Page #
Lab 01	Virtualization: Installing VMware Workstation with different flavours of linux OS on top of windows.	
Lab 02	Cloud: Service architecture and services using Azure	
Lab 03	Containerization using Docker: Installation, configuration and Generating Docker Images	
Lab 04	Cloud Deployment and Kubernetes	
Lab 05	Installation and Uses of Git commands Pull and Push to GitHub Repository	
Lab 06	Version Control Management using GIT	
Lab 07	Git, and Docker for Cloud	
Lab 08	DevOps Tools for CI / CD: Deploying Docker Images for multiple platform	
	Mid-Term Exam	
Lab 09	DevOps Planning: Agile Project Management	
Lab 10	Build using Maven	
Lab 11	Testing in DevOps: Selenium	
Lab 12	Creating Complete Jenkins Pipeline	
Lab 13	Software Configuration Management using Puppet tool	
Lab 14	Software Configuration Management and provisioning using Ansible tool	
Lab 15	Final Term Exam	

Lab 01

I. Virtualization

Objective:

The objective of this part of lab-1 will learn the students about the Implementation of Virtualization in software VMware workstation 16 on windows 10 PC to Learn Virtualization Basics, Benefits of Virtualization in Cloud using Open Source Operating System.

Activity Outcomes:

The activities provide hands - on practice with the following topics

- About Virtualization.
- VMware workstation 16 on windows 10 PC

how to setup and install Kali Linux 2021.1 in VMware Workstation

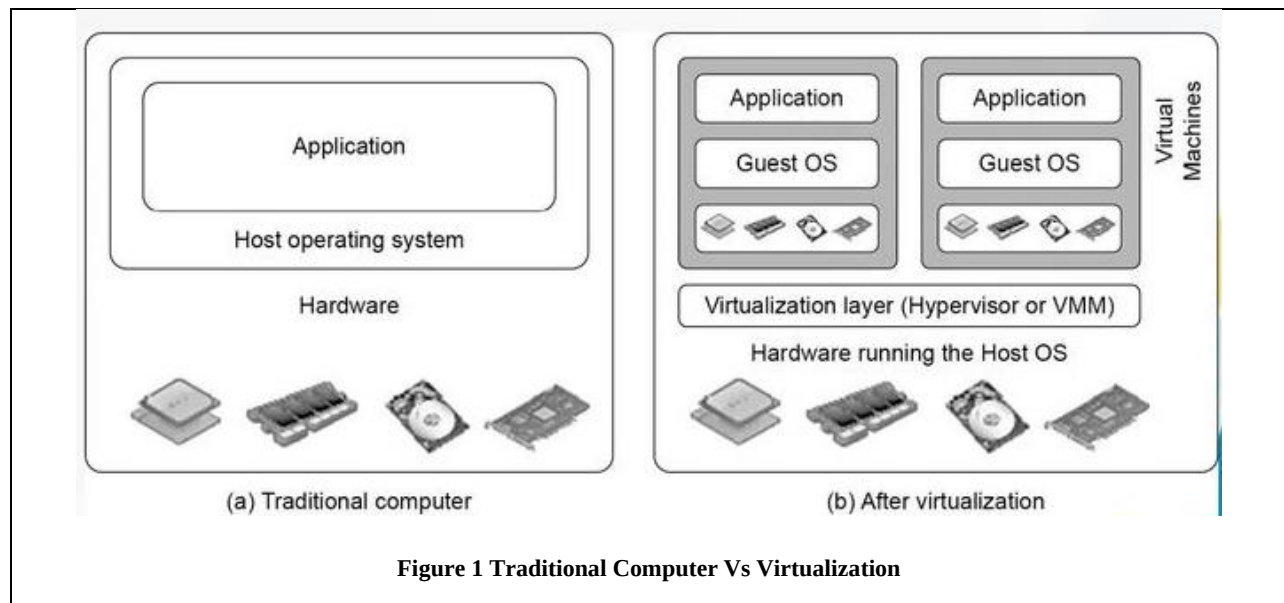
Instructor Note:

As pre-lab activity, read from the text book “Cloud Computing with Security: Concepts and Practices, Naresh Kumar Sehgal, Pramod Chandra P. Bhatt, John M. Acken, Springer, 2020.”

1. What is Virtualization?

Virtualization is the "creation of a virtual (rather than actual) version of something, such as a server, a desktop, a storage device, an operating system or network resources". In other words, Virtualization is a technique, which allows sharing a single physical instance of a resource or an application among multiple customers and organizations. It does by assigning a logical name to a physical storage and providing a pointer to that physical resource when demanded.

Virtual Machines, also called hypervisor or Virtual Machine Monitor have been around for quite sometime and allow us to run multiple operating system on the same physical machine. They are the backbone of the corporate IT infrastructure but can also be used on personal computers to experiment/work on various other operating systems without having to use other options such as dual boot or do a clean install on the existing PC.



In this Lab, will show to students how to install VMware Workstation Player in Windows 10.

About VMware Workstation Player

VMware Workstation Player is the limited capability **free (for non-commercial use) version of VMware Workstation Pro**. That is, it is just as VMware Workstation Pro with lesser functionality and free for personal use. But what is available is enough for most of the home users.

VMware Workstation Player formally known as VMware Player is a Virtualization software used to run multiple virtual machines on the same hardware. Its available for both Windows and Linux based operating systems. It runs on 64 bit operating system, which means that if you have 32 bit operating system, you will not be able to use it.

2. Solved Lab Activites

1.1 Activity 1 Installing VMWare

Lets get started. Installing the software is fairly straight forward. All you have to do is to download the installer from the website <https://www.vmware.com/products/workstation-pro.html>, run it and follow the instructions. Below are the steps:

Step 1- Download Link

Link for downloading the software is <https://getintopc.com>. Download Free Your Desired App search VMworkspace software for windows. Good thing is that there is no signup process. Click and download begins. Software is around 622 MB.

Step 2- Download the installer file

It should probably be in the download folder by default, if you have not changed the settings in your browser. File name should be something like VMware-workstation-full-16.1.0-17198959.exe. This file name can change depending on the version of the software currently available for download. But for now, till the next version is available, they will all be VMware Workstation 16 Pro.

Step 3- Run the Installer

Find the installer on your system and double click to launch the application. By default, the downloaded files are placed in the downloads folder.

Now you should see User Access Control (UAC) dialog box. Click yes to continue.

Initial Splash screen will appear. Wait for the process to complete.

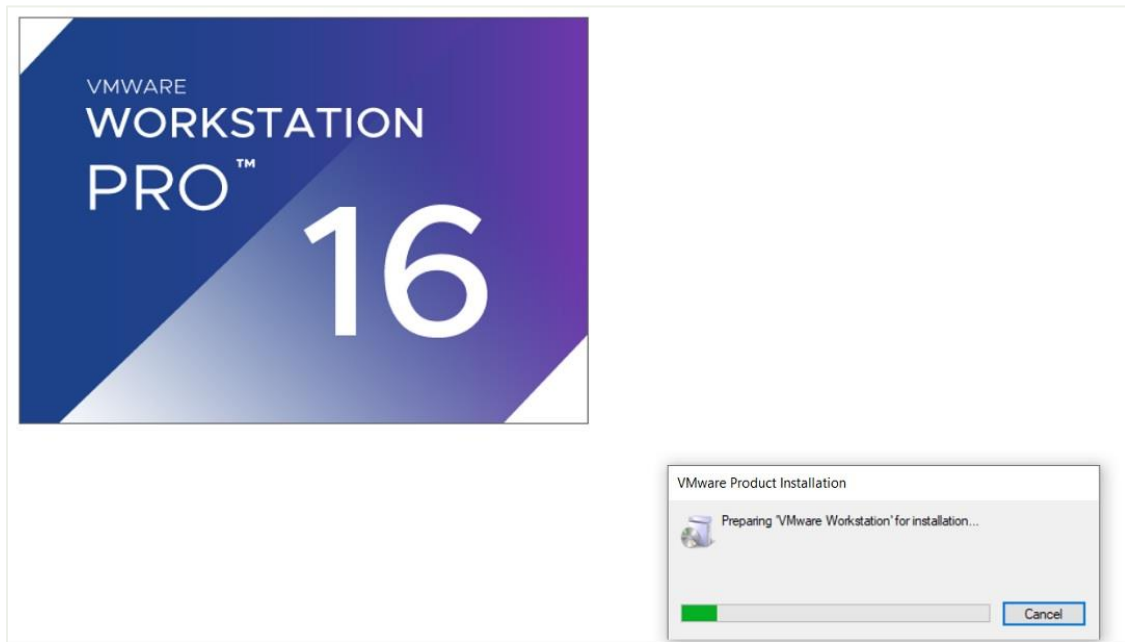


Figure 3 VMware Workstation 16 Pro Installation Splash Screen

Step 4- VMware Workstation Setup wizard

Now you will see VMware Workstation setup wizard dialog box. Click next to continue.

Step 5- End User Licence Agreement

This time you should see End User Licence Agreement dialog box. Check “I accept the terms in the Licence Agreement” box and press next to continue.

Step 6- Custom Setup options

Select the folder in which you would like to install the application. There is no harm in leaving the defaults as it is. Also select Enhanced Keyboard Driver check box.

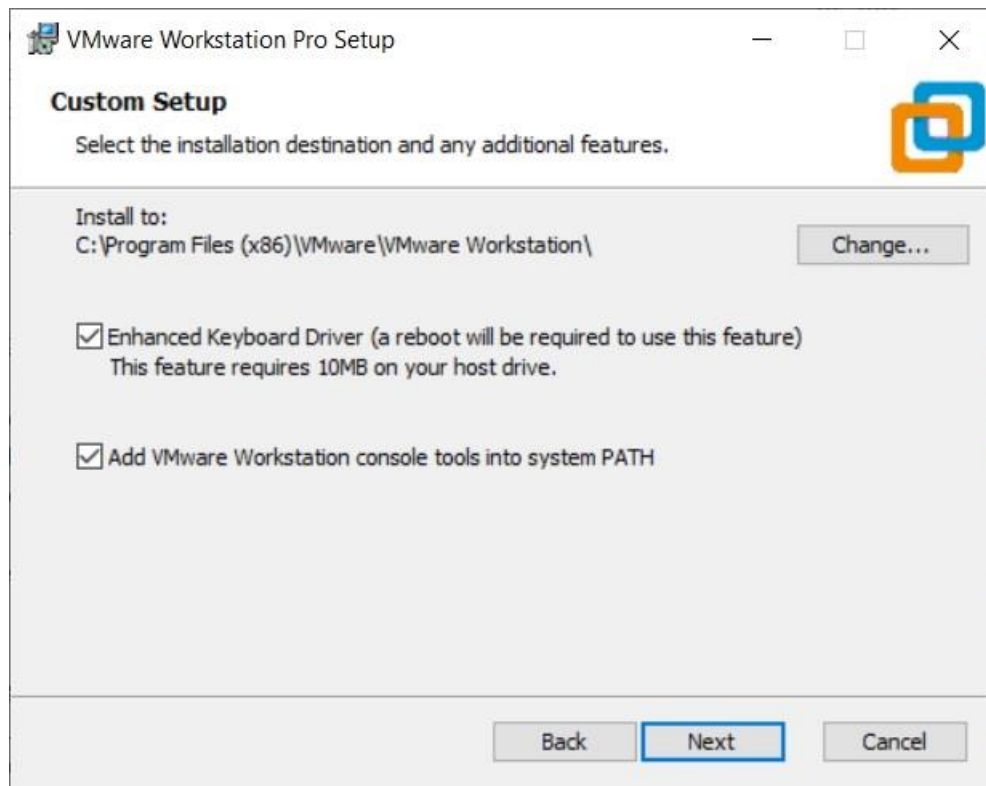


Figure 4 VMware Workstation 16 Pro installation – Select Installation Folder

Step 7- Application Shortcuts preference

Next step is to select the place you want the shortcut icons to be placed on your system to launch the application. Please select both the options, desktop and start menu and click next.

Step 8- Installation begins

Now you see the begin installation dialog box. Click install to start the installation process.

At the end you will see installation complete dialog box. Click finish and you are done with the installation process. You may be asked to restart your computer. Click on Yes to restart.

Step 9- Launch VMware Workstation

After the installation completes, you should see VMware Workstation icon on the desktop. Double click on it to launch the application.

Step 10- Licence Key

If you see the dialog box asking for licence key, click on trial or enter the licence key. If don't have the licence key, you will have 30 days trial. Click on trial to start your trial.

That's it, you can start using VMware Workstation.

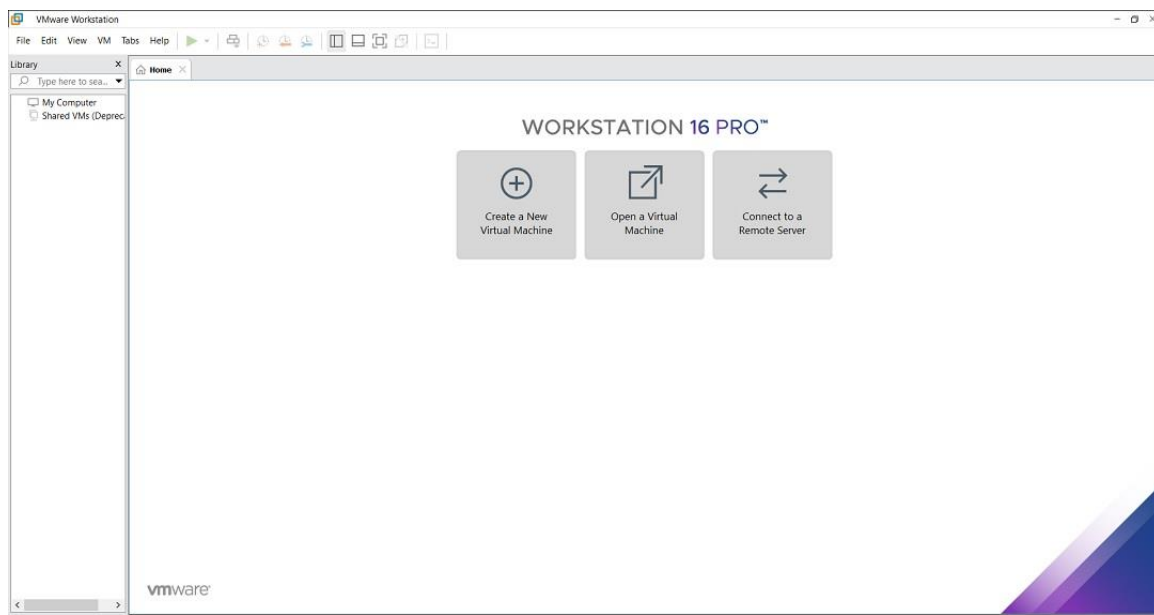


Figure 5 VMware Workstation 16 Pro home screen

2.2 Activity 2

Installing Kali Linux 2021.1 in VMware Workstation Player 16

Now will show you how to setup and install Kali Linux 2021.1 in VMware Workstation Player 16 which is the latest version of VMware Player.

Lets gets started with Installing Kali Linux 2021 in VMware Player 16

Step 1 – Download Kali Linux ISO image

To install the Kali Linux, we will have to first get the installer ISO image file. You can get it by visiting the official [download](#) page. Please download the 64 bit or 32 bit image depending on the system you have.



Image Name	Torrent	Size
 Kali Linux 64-Bit (Installer)	Torrent	4.0G
 Kali Linux 64-Bit (Live)	Torrent	3.4G
 Kali Linux 64-Bit (NetInstaller)	Torrent	379M

Figure 6 Kali Linux official download page

Step 2 – Locate the downloaded file

You can find the downloaded image file in the downloads folder, if you have not changed the default settings. The filename would be something like kali-linux-2021.1-installer-amd64.iso and would be around 4.0 GB. If you have downloaded through torrent, the ISO file will be downloaded in a folder, folder name would be something like kali-linux-2021.1-installer-amd64.iso.

Step 3- Open VMWare Player

Open VMware Player from Windows Start menu or from you desktop if you have VMware Player icon there.

Step 4 – Launch VMware Player – New Virtual Machine installation wizard.

To launch the wizard to create a new virtual machine, Click on Create a New Virtual Machine or File - > New Virtual Machine. Welcome to the new Virtual Machine Wizard dialog box will open. Select typical and click on next.

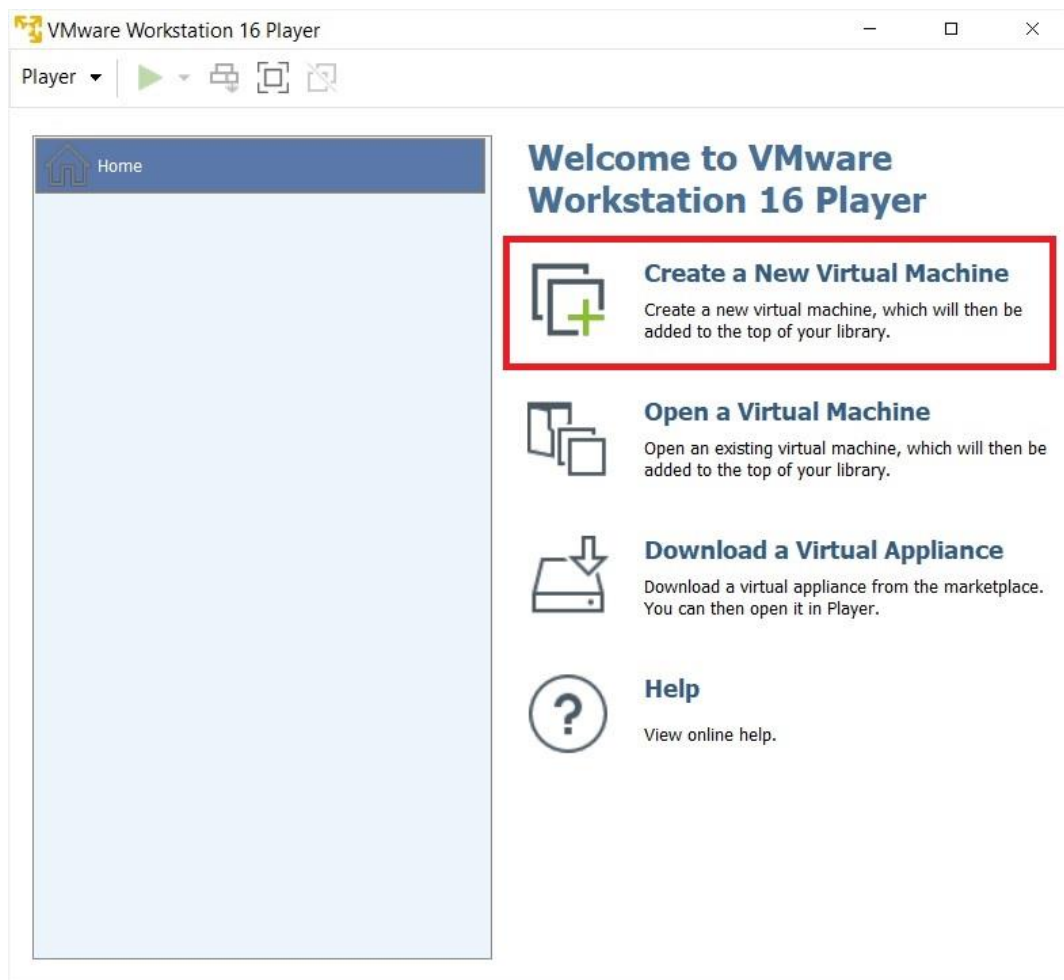


Figure 7 VMware Player 16 – Create new virtual machine

Step 5- Welcome to the new Virtual Machine Wizard dialog box appears

You will see new Virtual machine Wizard dialog box. Click Next.

Step 6- Select installation media or source

In this dialog box you will have to browse to the downloaded ISO file and click next. Generally, VMware Workstation detects the OS automatically and initiates what they call as the Easy Install. But in the case of Kali Linux this is not the case and you will see a warning(yellow triangle). Please ignore that and click next to continue.

Step 7- Select Guest Operating System

In this dialog box, you will be asked to select the Guest Operating System. Select Guest operating system as Linux and Version as Debian 10.x 64-bit or 32 bit depending on your system.

Step 8- Provide Virtual machine name

In this dialog box, you will be asked to provide the name of the virtual machine. You can provide any name you like. You can also change the location of the virtual machine. By default it is place in the Documents/Virtual Machine folder. Leaving it as the default is also fine.

Step 9- Specify disk capacity

This dialog box asks you to specify the disk capacity. This is the maximum amount of disk space it will utilize once the Virtual Machine is created. You can leave it to the default but if you are running low on disk space, you can reduce it to 20 GB. This is generally sufficient if you are not planning to install heavy disk using software's such as Photoshop. Such software's reduce the performance of your Virtual Machine if your Computer is not powerful enough.

Check Split Virtual Disk into multiple files. This is the default option. Say if you specify 60 GB, all of 60 GB will not be utilized or say 60 GB will not be blocked at once. These Virtual Disks expand according to the usage with a Maximum size you specifed as the disk capacity. On a fresh install normally it takes 10 GB of space which will grow according to the software's you install in the VM.

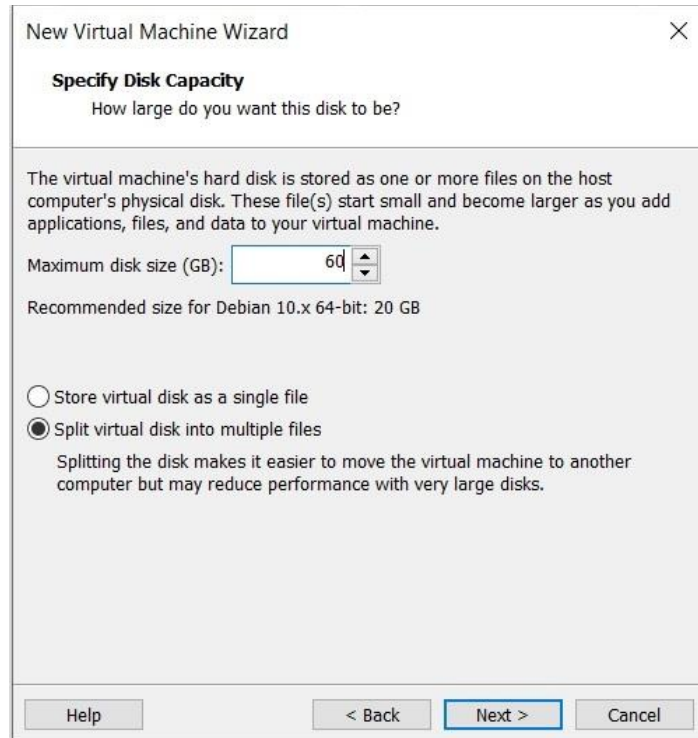


Figure 8 Kali Linux Installation – Specify Disk Space

Step 10- Ready to create Virtual Machine Dialog Box

This is the final dialog box and what you see is all the options you have selected in previous dialog boxes.

Customise Hardware – Normally we increase the RAM and memory before clicking finish. This helps to finish the installation process faster. If you have sufficient RAM and CPU on your host Windows machine, we suggest you should increase RAM and CPU. To increase the RAM, before clicking Finish, click on customise hardware. Increase the memory using the slider.

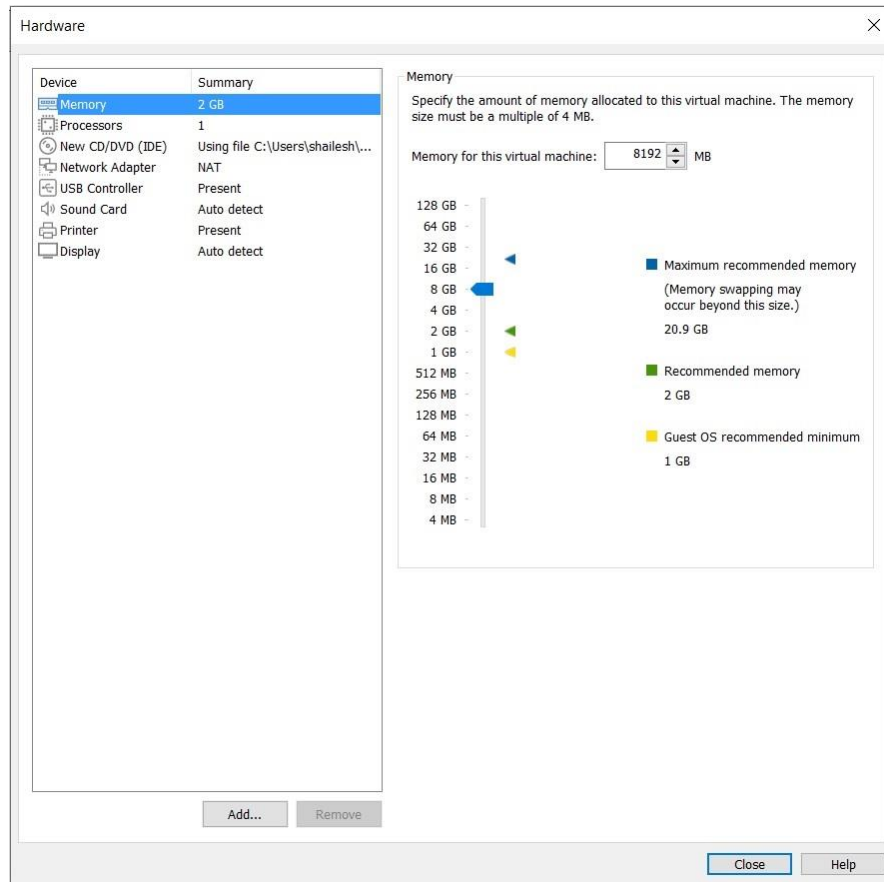


Figure 9 Kali Linux Installation – Customise Hardware – Memory

To increase the CPU, Enter the number of cores. I normally enter 4. Based on your CPU you can enter the number of cores. Click on Finish to start the installation process.

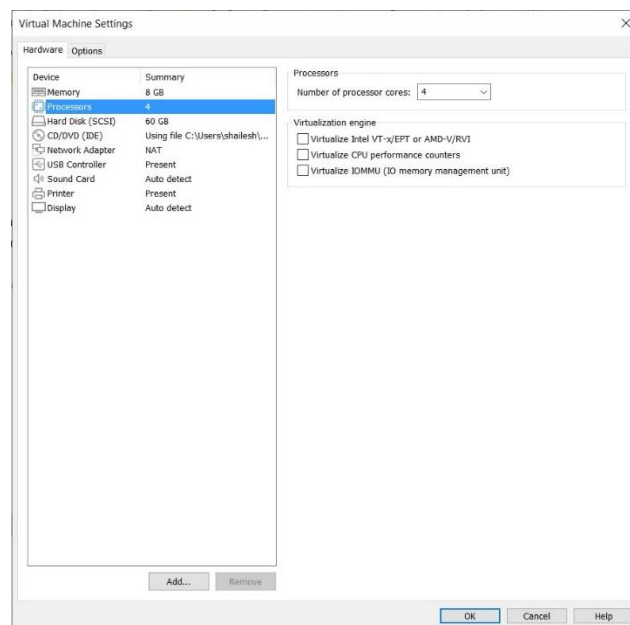


Figure 10 Kali Linux Installation – Customise Hardware – Processor

Click on Display to change the display settings. Check Accelerated 3D Graphics, Use host settings for monitor.

Click on OK and Finish to start the installation process

Step 11 – Play Virtual Machine

Now you will have to click on play virtual machine to start the process.

Step 12 – Select Graphical Install from Boot Menu

Here you will see many options. Select Graphical Install using the down arrow key and click continue.

Step 13 – Select a Language

In this dialog box you will be asked to select a language. Please select a language and continue. This option sets your language in the Kali Linux Operating system. Default is English.

Step 14 – Select Location

In this dialog box you will be asked to select a Location. Please select a location and continue. This option sets your Location in the Kali Linux Operating system. later on, you will be able to set the time zone based on the location you choose here.

Step 15 – Configure the Keyboard

In this dialog box you will be asked to select the keyboard layout. Please select a Keyboard layout using the arrow keys and click continue. This option sets your Keyboard in the Kali Linux Operating system. By default it is set to American English.

After you click continue, you will see the installation progresses for some then you see the Network Configuration dialog box, which is the next step.

Step 16 – Configure the Network – Enter Hostname

In this dialog box you will be asked to enter the hostname for your system. This being a home network, we can set anything. Enter any anything and click continue.

Step 17 – Configure the Network – Enter domain name

In this dialog box you will be asked to enter the domain name for your system. This being a home network, we can set anything like example.com

Step 18 – Set User account and password

In this dialog box you will be asked create an account other than the root user. Please note that this is the user other than the root user. Please type your name and click continue

Now you will be asked to provide the user name again. I use the same name as the account name in the previous screen.

You will be asked to setup a password of the user you created. Enter the password. We will login with this username and password once the installation completes.

Step 19 – Partition Disk

In this dialog box you are asked how you would like to partition your disk. Select Guided – Use entire disk and click continue. This is the default option.

In this dialog box you are asked to select a disk to partition. Select sda, VMware Virtual disk. There should be only one option. Click Continue.

Step 20 – Select partition scheme

In this dialog box you are asked to select a scheme for disk partition. Select the default, All files in one partition and click Continue.

Step 21 – Disk partition Overview

In this dialog box you are see the summary of your disk partition. Select the Finish Partitioning and Write changes to disk which should be selected by default. All you have to do is to click continue.

Step 22 – Disk partition Confirmation

In this dialog you are asked to confirm Write changes to disk. Select yes and click continue.

Step 23 – Installation starts

Now the actual installation starts. Wait for Configure the Package Manager Dialog box to appear.

Installation process will continue. Wait for sometime and let the process continue.

Step 24 – Software Selection

In this screen, you will have to specify what software you want to install. I accept the large, that is install all that is a part of default and additional tools. Click continue to proceed.

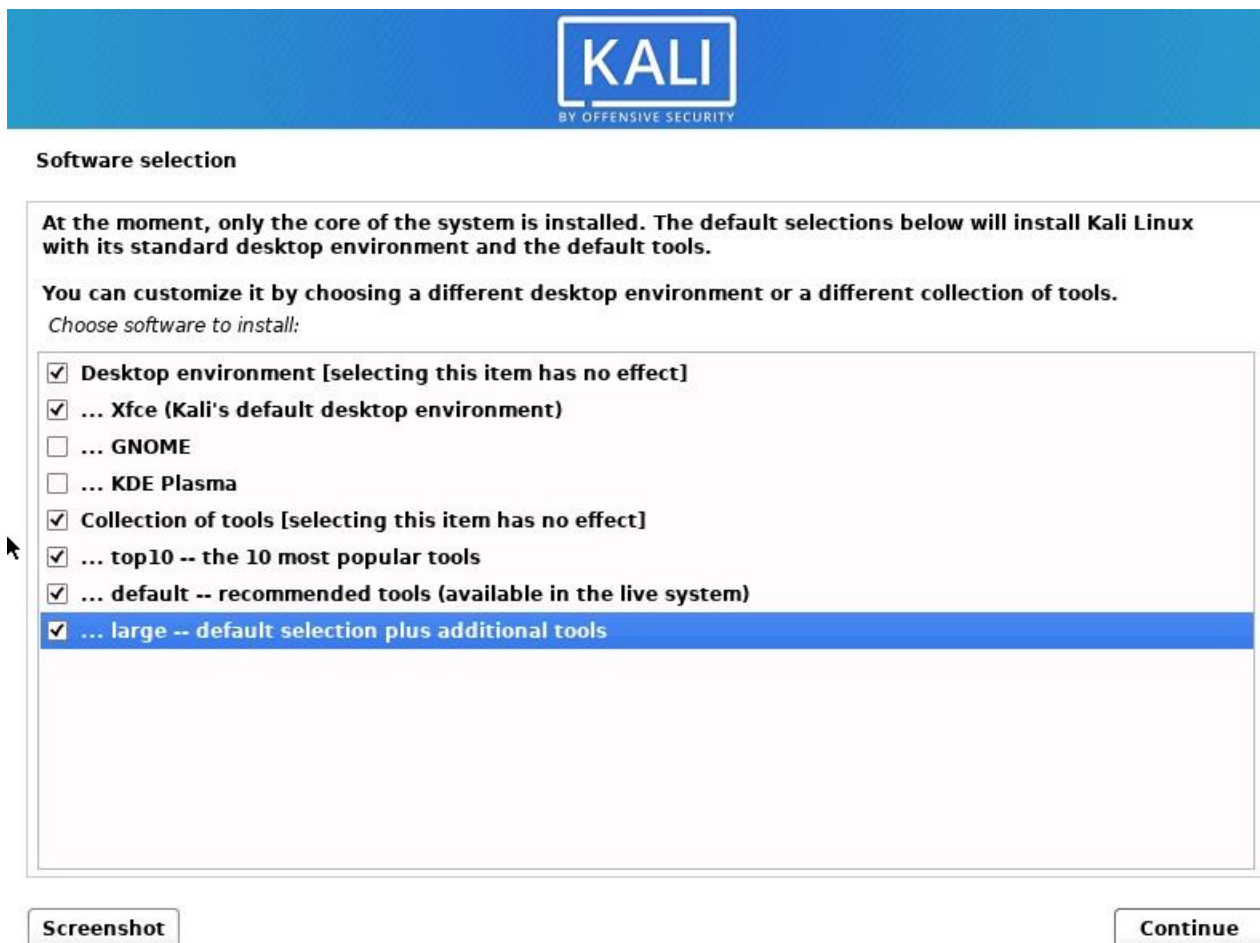


Figure 11 Kali Linux 2021 installation – Software selection

Step 25 – Install GRUB boot loader

In this dialog box you will be asked if you would like to install the GRUB boot loader. Select Yes and click Continue.

Step 26 – Select device for GRUB boot loader Installation

In this dialog box you will be asked to select boot loader device for GRUB installation. Select /dev/sda and click Continue.

Step 27 – Installation Complete

Installation will continue and now you will see Installation complete dialog box. Click continue to finalize the installation and wait for the VM to reboot. After reboot you will see the boot loader screen.

Process Complete

After the VM starts, you will see the Grub boot loader. Select Kali GNU/Linux.

Login with your credential you created. Enter your username and Password

You will be logged into Kali Desktop 2021.

Step 28 – Create root user

By default, only user account is setup during the installation stage. After installation you have to setup the root account by setting up a password for the root account.

1. Open Terminal
2. Type command `sudo su` and enter password to gain root access
3. Type command `passwd root` and enter the password for the root user. You can set the password of your choice

That's it, you should be able to login as root user.

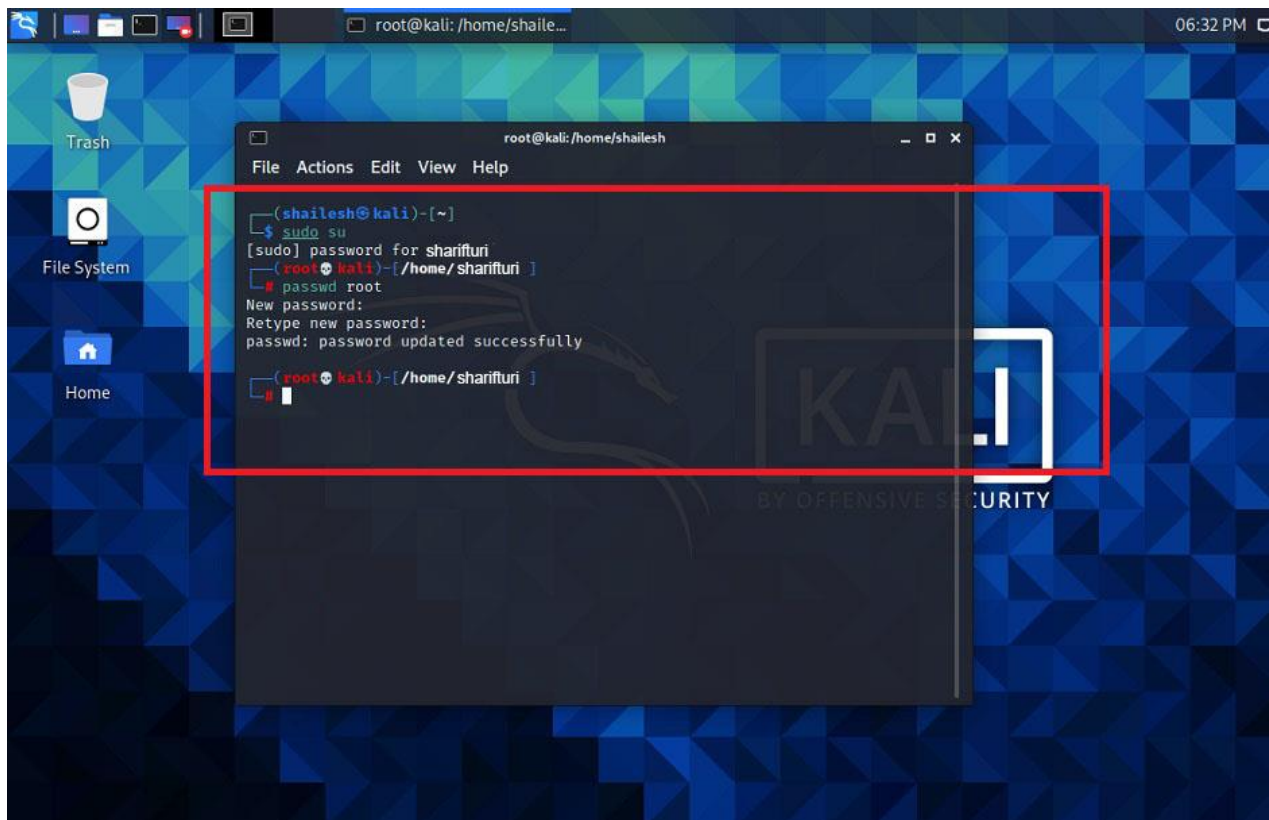


Figure 12 Kali Linux – change root password

Install VMware Tools (Optional for VMware 16 – Skip if you are able to switch to full screen automatically)

In VMware Player 16, VMware tools gets installed automatically. Manual steps for installing VMware tools is not required. You should **automatically be able to switch to the full screen mode**. Try to switch to full screen first. It should work, If not, follow the below steps, else ignore.

1. Open Terminal
2. Execute the command

```
sudo apt-get install open-vm-tools-desktop fuse
```

3. Enter your password
4. Enter your password, if asked
5. Enter Y to accept whenever asked
6. Wait for the process to complete and restart. You are done. Login and now you will see the desktop in full screen mode

You can check the version of installed VMware Tools. In terminal, execute the command

```
vmware-toolbox-cmd -v
```

This will show you the version number.

If you don't have internet connection, you can install VMware tools directly from the VM. If you have internet connection, follow the next instruction which is easier.

1. Goto VM-> Install VMware Tools
2. You will see VMware tools Icon on the desktop. If you don't, then click on Places in the menu, Places->VMware Tools. Double click to open.
3. Copy the file with the file name something like this VMware-tools....tar.gz to the desktop.
4. Right click on this file on the desktop and click Extract here. You will see that the file is extracted to the folder vmware-tools-distrib
5. Open terminal by going to System Tools -> terminal
6. Goto the folder vmware-tools-distrib on desktop by entering the command `cd /Desktop/vmware-tools-distrib`
7. Now run this command:

```
sudo ./vmware-install.pl -d
```

-
8. Enter your password and press Enter. You will be asked if you want to proceed with the legacy installer. Enter Y and press Enter.
 9. Now you will be asked series of questions, just accept the default by pressing Enter for all and wait for the installation to complete.
 10. Wait for the installation to complete. You will see more prompts, except the defaults by clicking enter. Once completed, restart your VM by going to VM->Power->Restart Guest
 11. Once the system restarts, you will be prompted to login. Once you login, you can see the desktop in full screen mode which stretches up to the complete screen.

Process Complete

You are done, You can start working on Kali Linux 2021. Thanks, and enjoy now.

3) Graded Lab Tasks

Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.

Lab Task 1:

Run command to check the IP address of virtual machine.

Lab Task 2:

Walk through the environment and set IP address so that this virtual machine can be accessed from actual system.

Lab 02

Cloud: Service architecture and services using Azure

Objective:

In this lab, I will show you the Cloud Service providers (CSP) names and their offers various services.

1. How to Create Azure VM (Virtual Machine).

Activity Outcomes:

The activities provide hands - on practice with the Cloud Service Providers Companies.

Instructor Note:

<https://docs.microsoft.com/en-us/documentation/>

1. Cloud Service Provider Companies

Cloud Service providers (CSP) offer various services such as Software as a Service, Platform as a service, Infrastructure as a service, network services, business applications, mobile applications, and infrastructure in the cloud. The cloud service providers host these services in a data center, and users can access these services through cloud provider companies using an Internet connection.

There are the following some popular Cloud Service Providers Companies.

Microsoft Azure

Azure is also known as Windows Azure. It supports various operating systems, databases, programming languages, frameworks that allow IT professionals to easily build, deploy, and manage applications through a worldwide network. It also allows users to create different groups for related utilities.

Features of Microsoft Azure

- Microsoft Azure provides scalable, flexible, and cost-effective
- It allows developers to quickly manage applications and websites.
- It managed each resource individually.
- Its IaaS infrastructure allows us to launch a general-purpose virtual machine in different platforms such as Windows and Linux.
- It offers a Content Delivery System (CDS) for delivering the Images, videos, audios, and applications.

Amazon Web Services (AWS)

Amazon Web Services (AWS) is a secure cloud service platform provided by Amazon. It offers various services such as database storage, computing power, content delivery, Relational Database, Simple Email, Simple Queue, and other functionality to increase the organization's growth.

Features of AWS

AWS provides various powerful features for building scalable, cost-effective, enterprise applications. Some important features of AWS

is given below-

- AWS is scalable because it has an ability to scale the computing resources up or down according to the organization's demand.
- AWS is cost-effective as it works on a pay-as-you-go pricing model.
- It provides various flexible storage options.
- It offers various security services such as infrastructure security, data encryption, monitoring & logging, identity & access control, penetration testing, and DDoS attacks.
- It can efficiently manage and secure Windows workloads.

2. Solved Lab Activities

1. How to Create Azure VM (Virtual Machine)

We can create a windows virtual machine (VM) in Microsoft Azure for free or How to create a free Azure VM.

Note: To create an Azure VM for free, you can sign up for a 30-days Azure free trial.

Follow the below steps to create a virtual machine in Azure.

Here, we will discuss How to Create Azure VM (Virtual Machine) with the new version of Windows Server 2021, Datacenter or how to create a VM in Azure portal step by step.

Make sure you have a subscription to azure before starting. If you don't have a subscription then create a free Azure account.

Assuming you have an Azure subscription now. Follow the below steps to create a windows virtual machine.

Step-1:

Login to <https://portal.azure.com>.

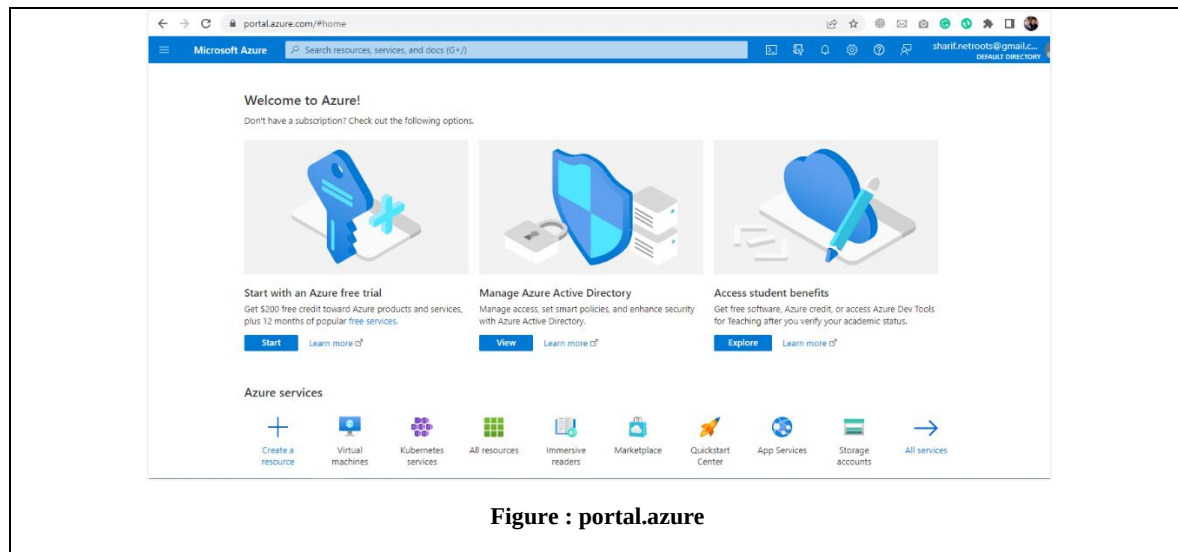


Figure : portal.azure

Step-2:

Now click on “Virtual machines” option from the left side Menu and then click on “Create” link.

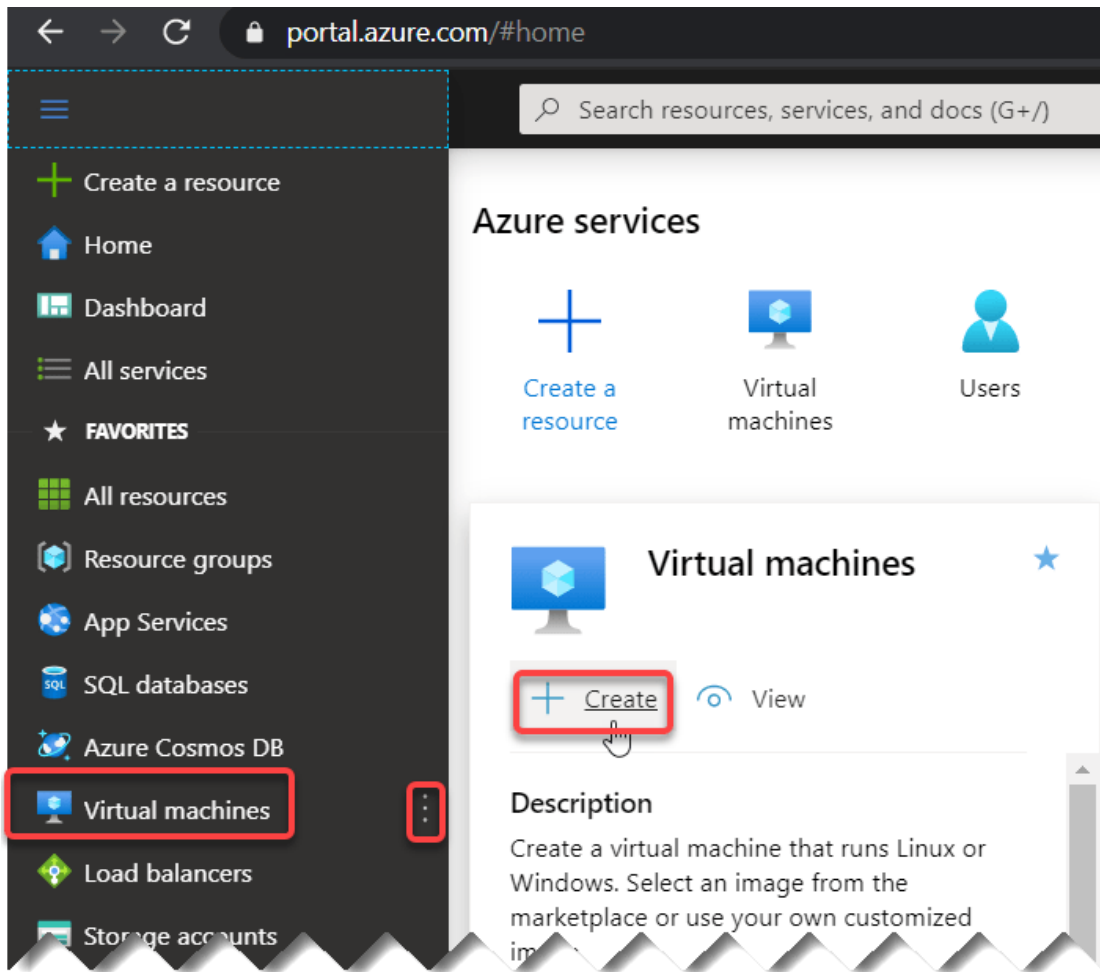
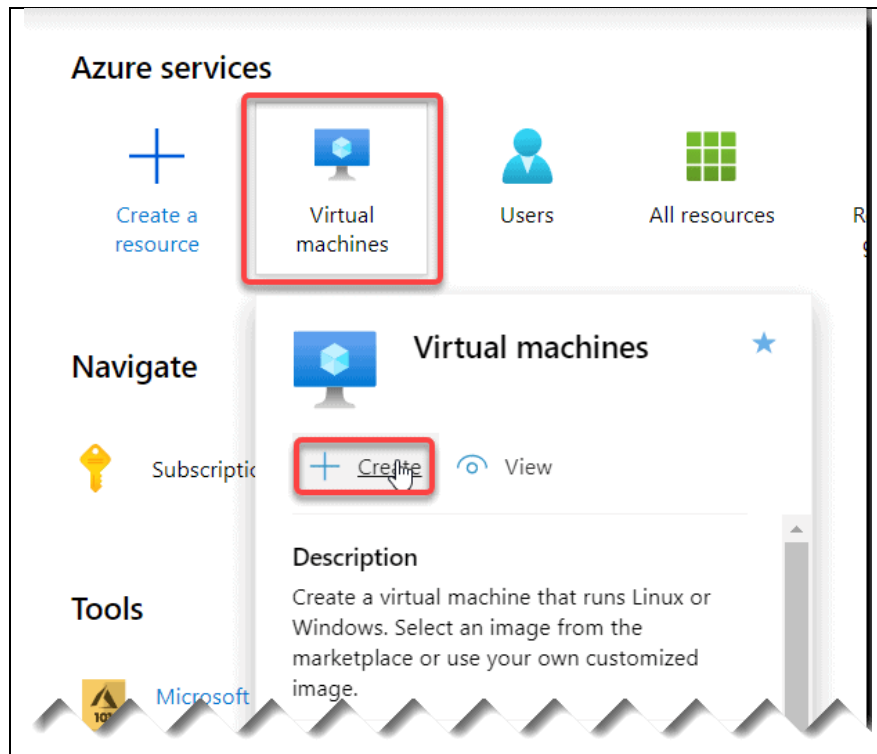


Figure 2 Virtual machines Creation

Or Go to “Virtual machines” option from the “Azure services” section.



One more way is to go to the same option from the **“Create a resource”** link from the left side menu and then select the **“Compute”** and **“Virtual machine”**.

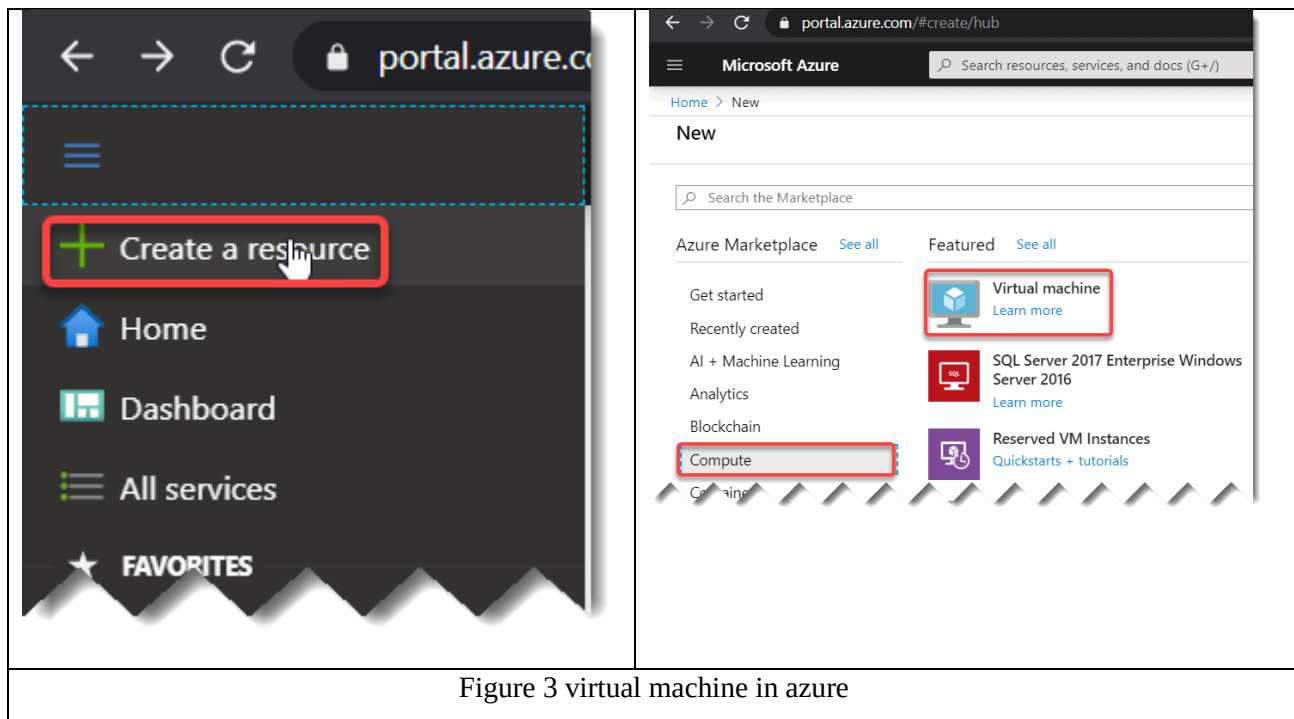


Figure 3 virtual machine in azure

Step- 3:

Fill the details required. A few of the options you can keep as it is but based on your business need you can change those.

- **Subscription:** Choose your correct subscription.
- **Resource group:** You can choose the existing resource group or you can create a new one by clicking the “**Create New**” option. It acts like a container that stores the resources related to an Azure solution.
- **Virtual machine name:** Provide a name for your virtual machine.
- **Region:** This the location where you are going to create all the resources related to the virtual machine.
- **Image:** Choose the Windows Server 2019 Datacenter. You can change it based on your business need.
- **Size:** The size you want to assign based on your requirement. I have chosen Standard D2s v3 as per my business requirement.

In the Administrator account section, Choose a **Username**, **Password** that you will use to login to the VM once created.

Select Inbound port: Choose **HTTP(80),RDP(3389)**. This is a very important option.

All other options you can keep as it is. Now click on “**Next:Disks >**” button.

The screenshot shows the 'Create a virtual machine' page in the Microsoft Azure portal. The 'Basics' tab is active. The following fields are highlighted with red boxes:

- Subscription:** Visual Studio Enterprise
- Resource group:** newresgroup
- Virtual machine name:** MyNewVM
- Region:** (US) West US
- Availability options:** No infrastructure redundancy required
- Image:** Windows Server 2019 Datacenter

Azure Spot instance  ☐ Yes ☒ No

Size *  **Standard D2s v3**
2 vcpus, 8 GiB memory(Rs.5,645.28/month)
[Change size](#)

Administrator account

Username *  

Password *  

Confirm password *  

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports *  ☐ None ☒ Allow selected ports

Select inbound ports * 

 This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

If we do not have a [windows server](#) license, you can select **No** license.

Save money

Save up to 49% with a license you already own using Azure Hybrid Benefit. [Learn more](#) 

Already have a Windows Server license? * ☐ Yes ☒ No 

In the next screen (**Disks**), you can add the disk architecture that you want. I kept the default option as it is. You can change it based on your business need. Click on **Next: Networking** button.

Now you can fill the below details in the Networking screen.

- **Virtual Network:** Choose a **virtual network** or else you can create a new one by clicking the “**Create New**” link.
- **Subnet:** You can keep the default option as it is.
- **Public inbound ports:** choose **Allow selected ports**.
- **Select inbound ports:** This is a very important option. Choose the option as **HTTP(80), RDP(3389)** here. Once you choose this option in the Basics tab it will automatically show the same option here. Keep the option as it is.

portal.azure.com/#create/Microsoft.VirtualMachine

Microsoft Azure Search resources, services, and docs (G+)

Home > Virtual machines > Create a virtual machine

Create a virtual machine

Basics Disks **Networking** Management Advanced Tags Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution. [Learn more](#)

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network * ⓘ (new) newresgroup-vnet [Create new](#)

Subnet * ⓘ (new) default (10.0.0.0/24)

Public IP ⓘ (new) MyNewVM-ip [Create new](#)

NIC network security group ⓘ ☐ None ☒ Basic ☐ Advanced

Public inbound ports * ⓘ ☐ None ☒ Allow selected ports

Select inbound ports * HTTP (80), RDP (3389)

⚠ This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab

Accelerated networking ⓘ ☐ On ☒ Off The selected VM size does not support accelerated networking.

Load balancing

You can place this virtual machine in the backend pool of an existing Azure load balancing solution. [Learn more](#)

Place this virtual machine behind an existing load balancing solution? ☐ Yes ☒ No

[Review + create](#) [< Previous](#) [Next : Management >](#)

Now for other tabs all the options you can keep as it is. Once you fill all the above details click on the “**Review + Create**” button. Azure will validate the above details internally and will show a message “**Validation passed**”.

portal.azure.com/#create/Microsoft.VirtualMachine

Microsoft Azure Search resources, services, and docs (G+)

Home > Virtual machines > Create a virtual machine

Create a virtual machine

✓ Validation passed

Resource group	newresgroup
Virtual machine name	MyNewVM
Region	West US
Availability options	No infrastructure redundancy required
Username	sharifhuri
Public inbound ports	RDP, HTTP
Already have a Windows license?	No
Azure Spot	No

Disks

OS disk type	Premium SSD
Use managed disks	Yes
Use ephemeral OS disk	No

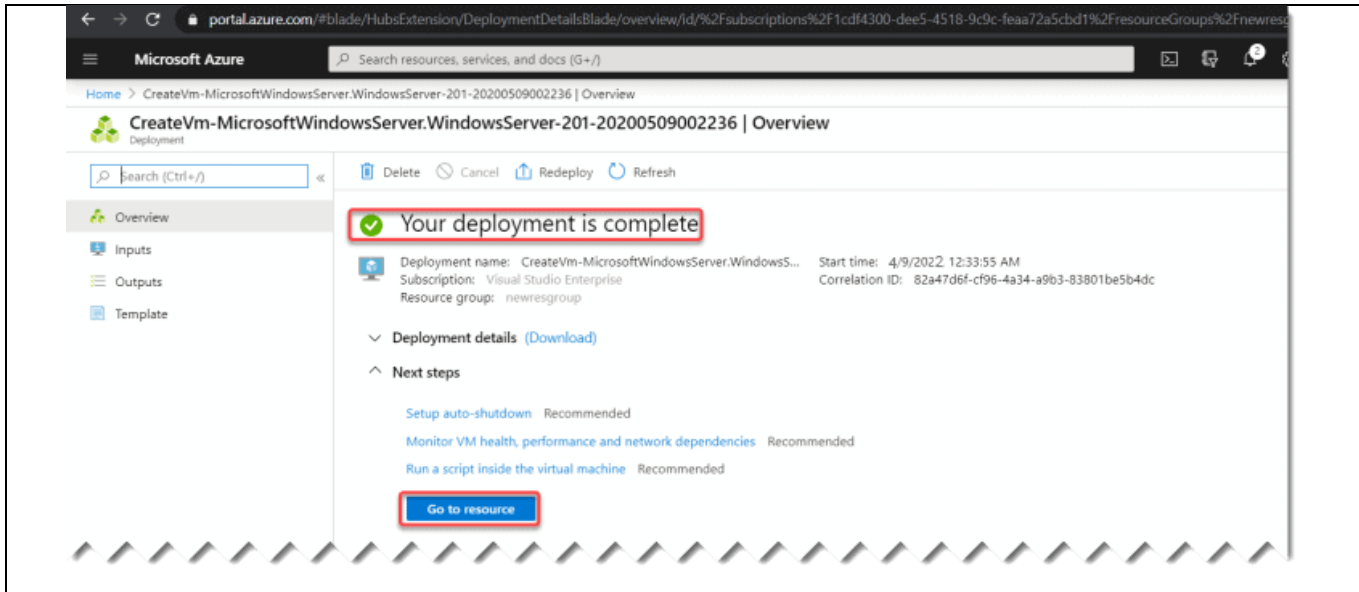
Networking

Virtual network	(new) newresgroup-vnet
Subnet	(new) default (10.0.0.0/24)
Public IP	(new) MyNewVM-ip
Accelerated networking	Off
Place this virtual machine behind an	No

[Create](#) [< Previous](#) [Next >](#) [Download a template for automation](#)

Step- 4:

Now click on the “Create” button. It will show you “Your deployment is complete“.Now click on “Go to resource”



Step- 5:

Congratulations, Now you have created the VM successfully. Click on the **Connect** button to connect to the VM.

3) Graded Lab Tasks

Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.

Lab Task 1:

Repeat all above steps for:

1. To Create Azure VM (Virtual Machine)

Lab 03

Containerization using Docker: Installation, configuration and Generating Docker Images

Objective:

In this lab, I will show you how to install Docker in Windows 10. We will be using community edition (CE) as this is a free version. You can use this edition to install it on your personal computer to learn and build application around it. Knowledge of Docker is a must if you want to build Cloud Native Microservices based applications.

Activity Outcomes:

The activities provide hands - on practice with the following topics

- Describe about the docker
- Docker installation and configuration
- Docker lifecycle

Instructor Note:

<https://docs.docker.com/get-started/overview/>

1. Docker overview:

Docker is an open platform for developing, shipping, and running applications in an isolated environment called a "container". Docker enables you to separate your applications from your infrastructure so you can deliver software quickly. With Docker, you can manage your infrastructure in the same ways you manage your applications.

Docker objects

When you use Docker, you are creating and using images, containers, networks, volumes, plugins, and other objects. This section is a brief overview of some of those objects.

Docker Images

An image is a read-only template with instructions for creating a Docker container. Often, an image is based on another image, with some additional customization. For example, you may build an image that is based on the ubuntu image, but installs the Apache web server and your application, as well as the configuration details needed to make your application run.

You might create your own images or you might only use those created by others and published in a registry. To build your own image, you create a Dockerfile with a simple syntax for defining the steps needed to create the image and run it. Each instruction in a Dockerfile creates a layer in the image. When you change the Dockerfile and rebuild the image, only those layers which have changed are rebuilt. This is part of what makes images so lightweight, small, and fast, when compared to other virtualization technologies.

What are containers?

Containers are packages of software that contain all of the necessary elements to run in any environment. In this way, containers virtualize the operating system and run anywhere, from a private data center to the public cloud or even on a developer's personal laptop. From Gmail to YouTube to Search, everything at Google runs in containers. Containerization allows our development teams to move fast, deploy software efficiently, and operate at an unprecedented scale.

A container is a runnable instance of an image. You can create, start, stop, move, or delete a container using the Docker API or CLI. You can connect a container to one or more networks, attach storage to it, or even create a new image based on its current state.

By default, a container is relatively well isolated from other containers and its host machine. You can control how isolated a container's network, storage, or other underlying subsystems are from other containers or from the host machine.

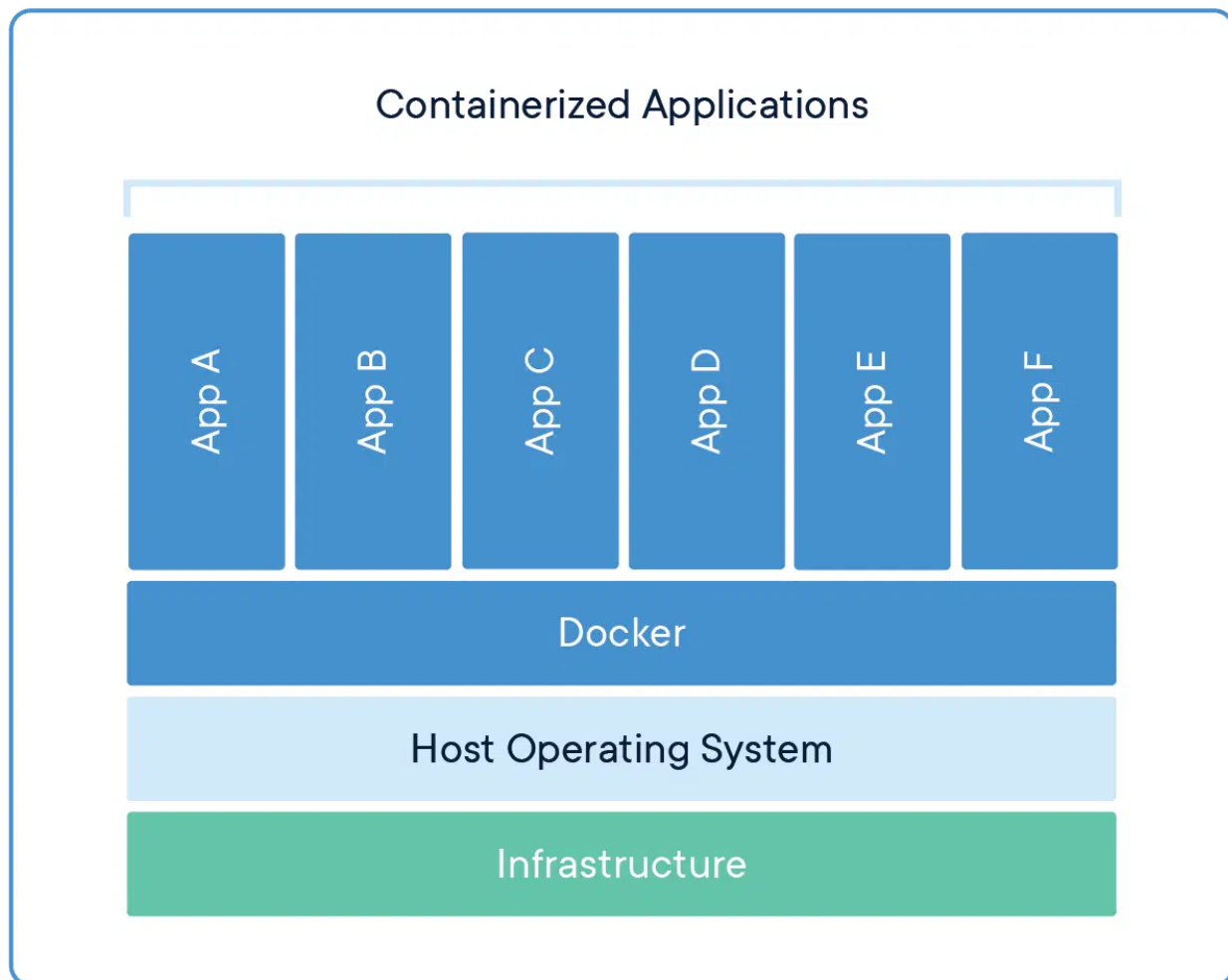
A container is defined by its image as well as any configuration options you provide to it when you create or start it. When a container is removed, any changes to its state that are not stored in persistent storage disappear.

Comparing Containers and Virtual Machines

Containers and virtual machines have similar resource isolation and allocation benefits but function differently because containers virtualize the operating system instead of hardware. Containers are more portable and efficient.

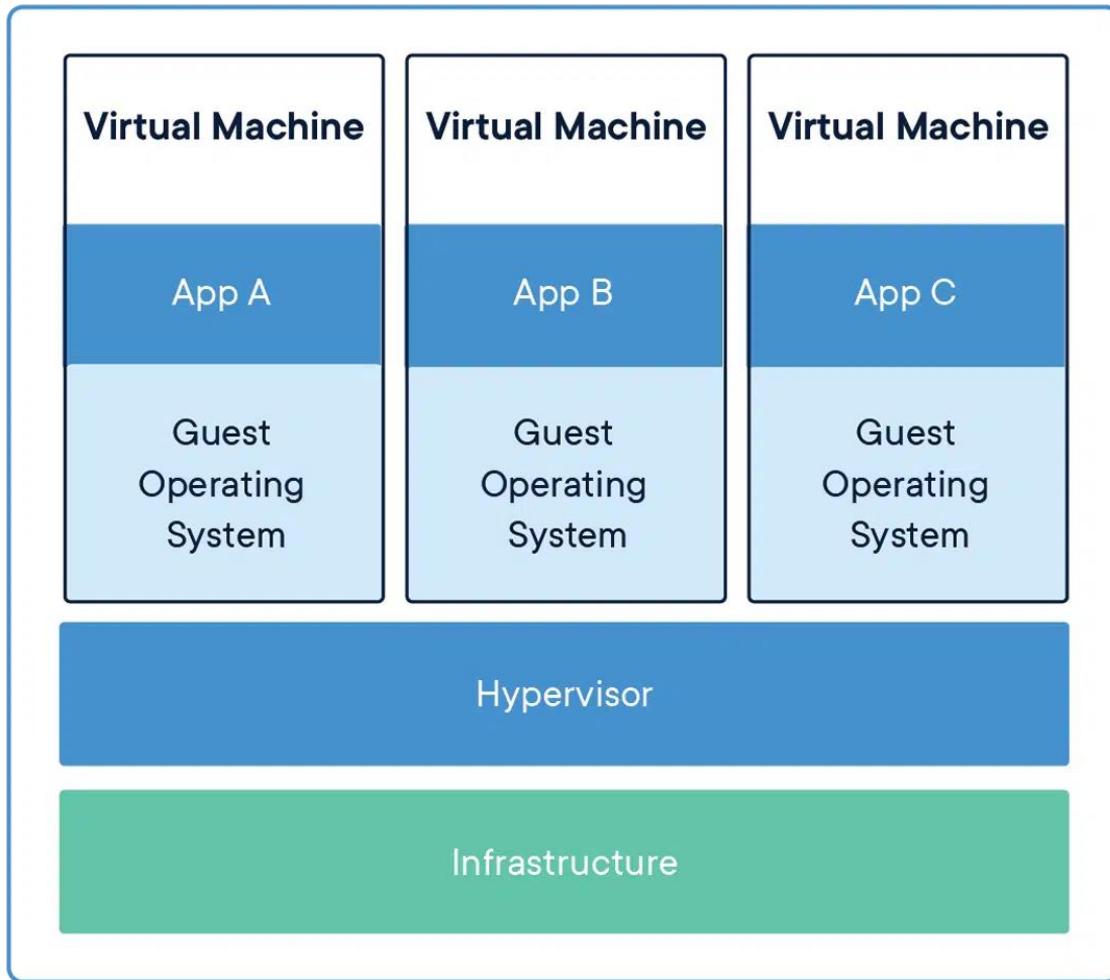
Containers

Containers are an abstraction at the app layer that packages code and dependencies together. Multiple containers can run on the same machine and share the OS kernel with other containers, each running as isolated processes in user space. Containers take up less space than VMs (container images are typically tens of MBs in size), can handle more applications, and require fewer VMs and Operating systems.



Virtual Machines

Virtual machines (VMs) are an abstraction of physical hardware turning one server into many servers. The hypervisor allows multiple VMs to run on a single machine. Each VM includes a full copy of an operating system, the application, necessary binaries, and libraries – taking up tens of GBs. VMs can also be slow to boot.



2. Solved Lab Activities

Prerequisites: Docker requires Hyper-V enabled in Windows 10. Without this Docker will not run after you install Docker first of all to enable Hyper-V in Windows.

Please note that Docker needs Hyper-V only in Windows 10. It will not work with VMware Workstation or Virtual Box. You can refer to the docker documentation.

Step 1 – Download Docker

Officially Docker installer, community edition can be downloaded from [Docker Store](#). You will have to create an account to be able to download it. Having an account for Docker is a very good thing. It allows you to download docker images in the future.

You can also reach the Docker Store download page from the Docker official page. Go to the Docker [Official home Page](#).

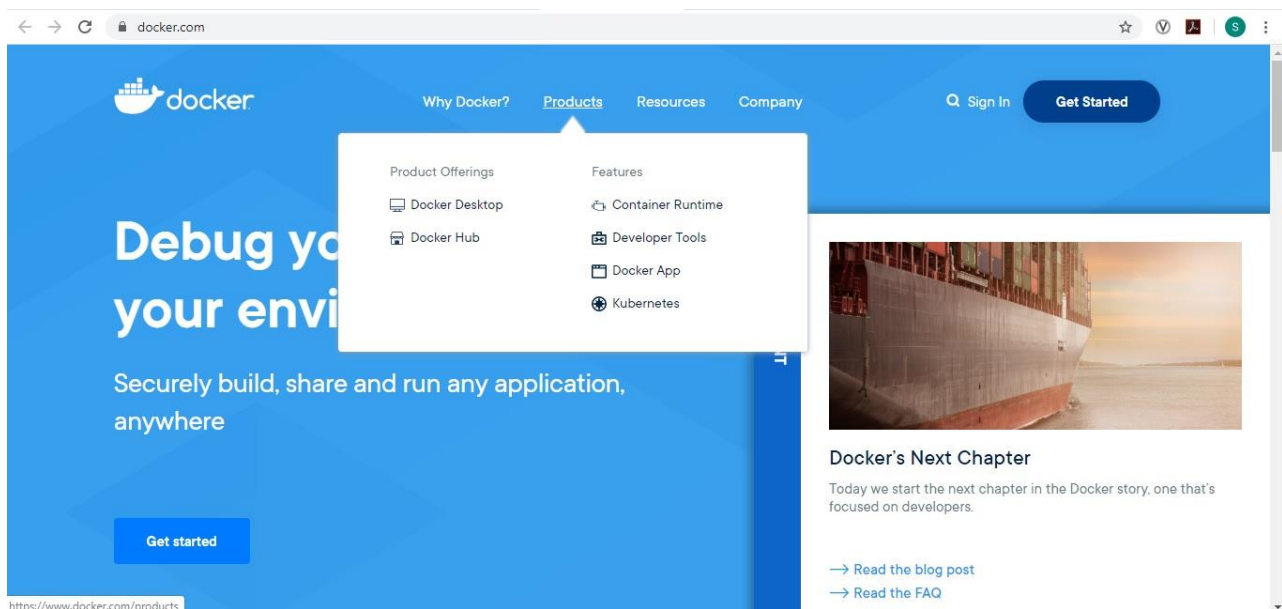


Figure 4 Docker Official home Page

Click on Products-> Docker Desktop on the menu bar. This will take you to the docker [desktop product page](#).

Click on Download Desktop for Mac and Windows.

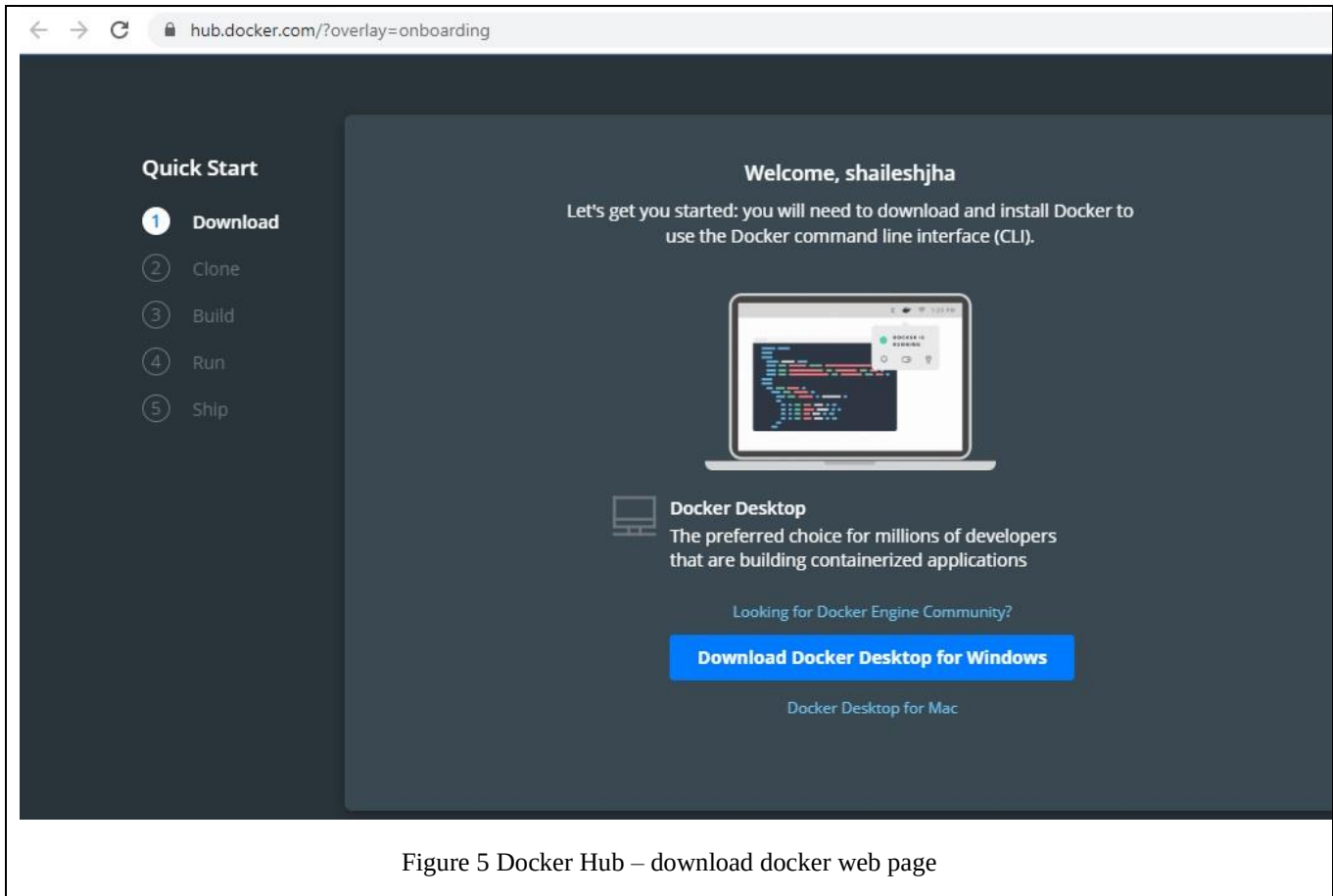


Figure 5 Docker Hub – download docker web page

The installer file for windows is around 499 MB.

Step 2 – Run the installer

Double click on the downloaded installer file to start the installation wizard. You will see Windows UAC – User Access Control asking for permission to allow the program to run. Click yes to continue.

Now you will see Docker installer downloading additional files required. If you dont have internet connection, installer will move on to the next step. Wait for the process to complete and you will see the configuration screen.

Step 3 – Configuration Settings

In this dialog box you will be asked if you want to create desktop icon for Docker. I leave this checked. Second option is if you want to use Linux or Windows Container. This option can be changed later on. I leave it as default, that is unchecked.

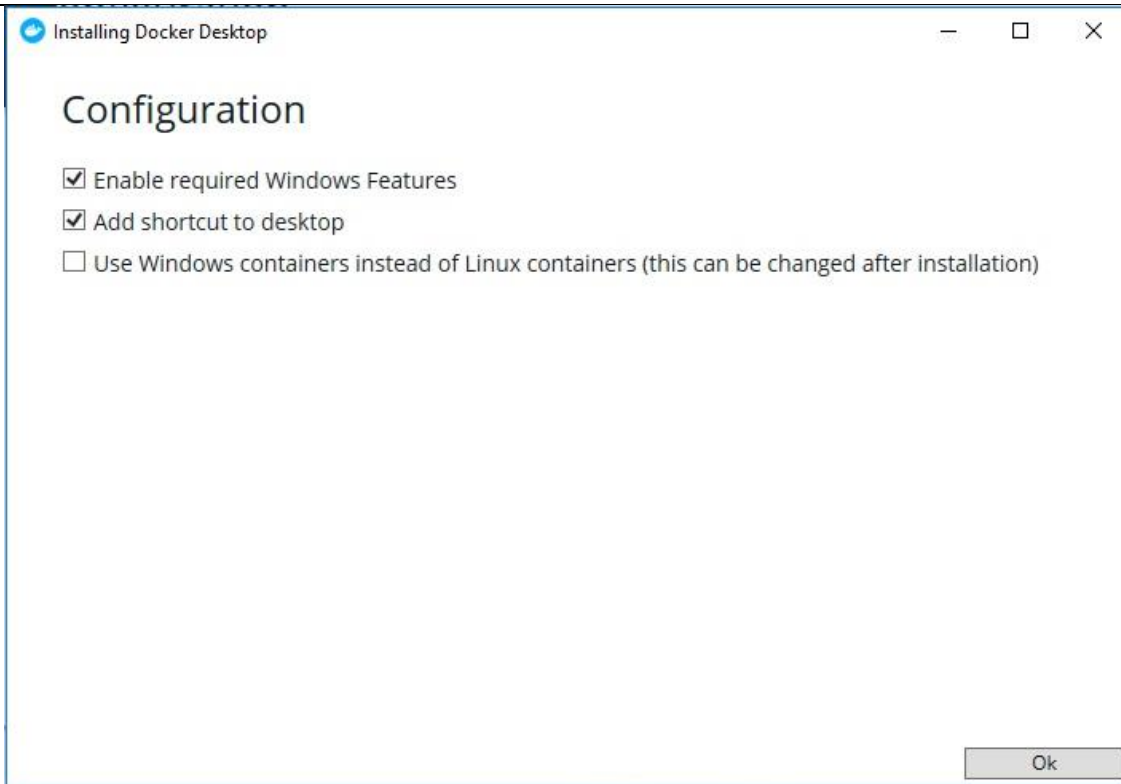


Figure 6 Docker Installation – Configuration settings

Click OK to start the installation. Wait for the installation to complete.

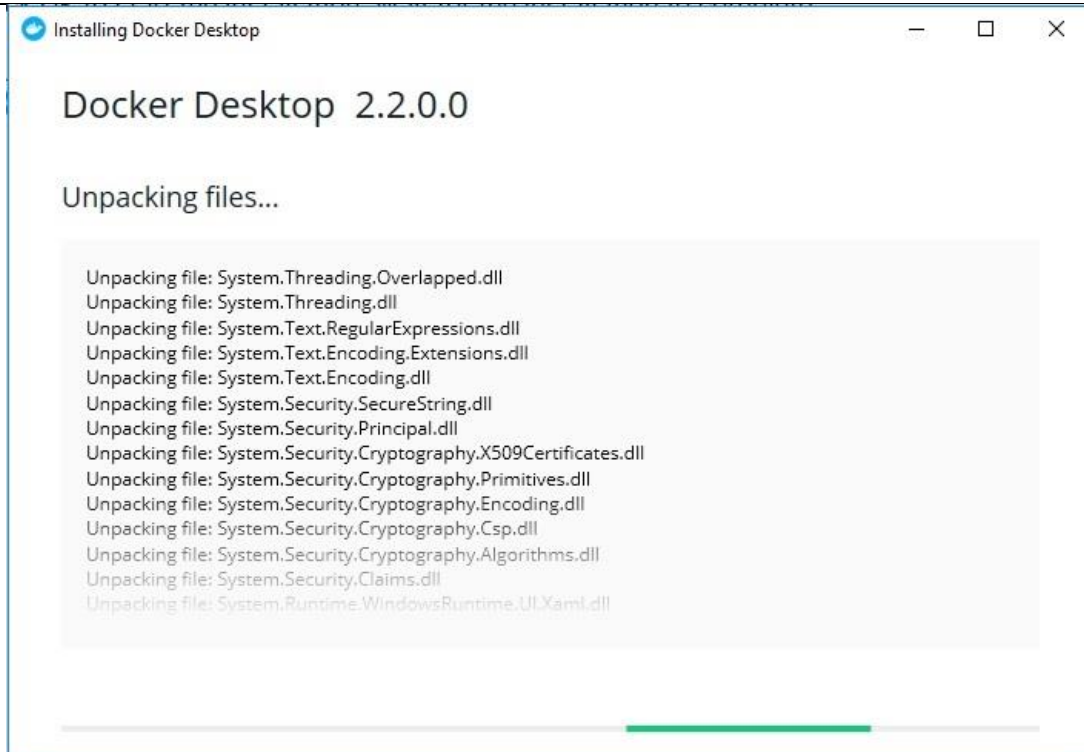


Figure 7 Docker Installation progress

Once the installation completes, click on Close and restart to complete the installation.

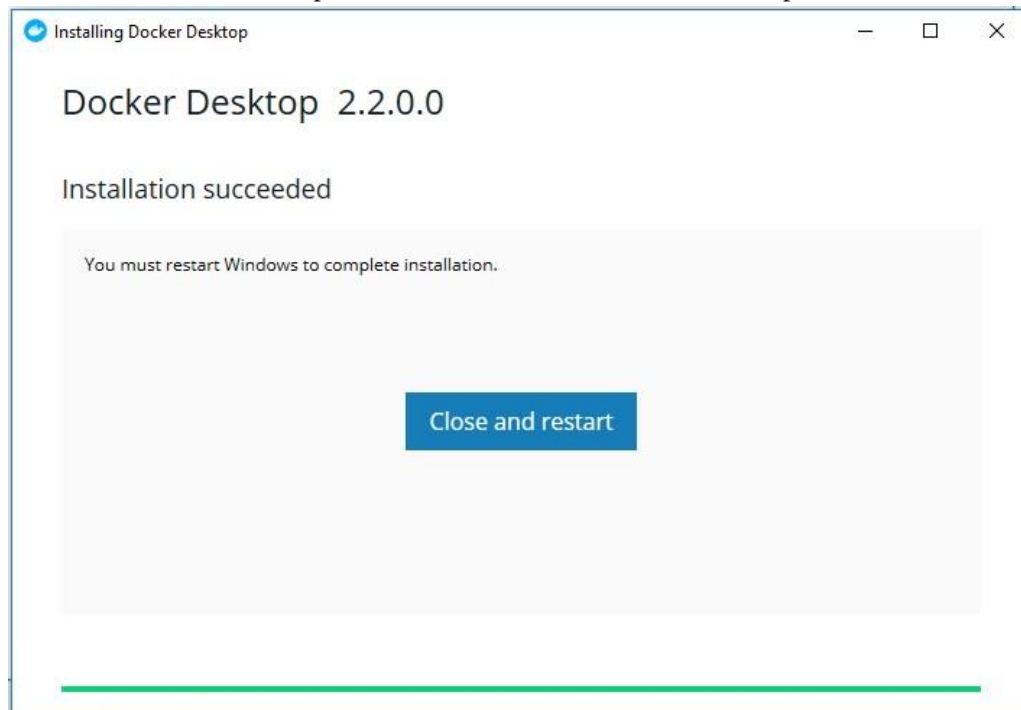
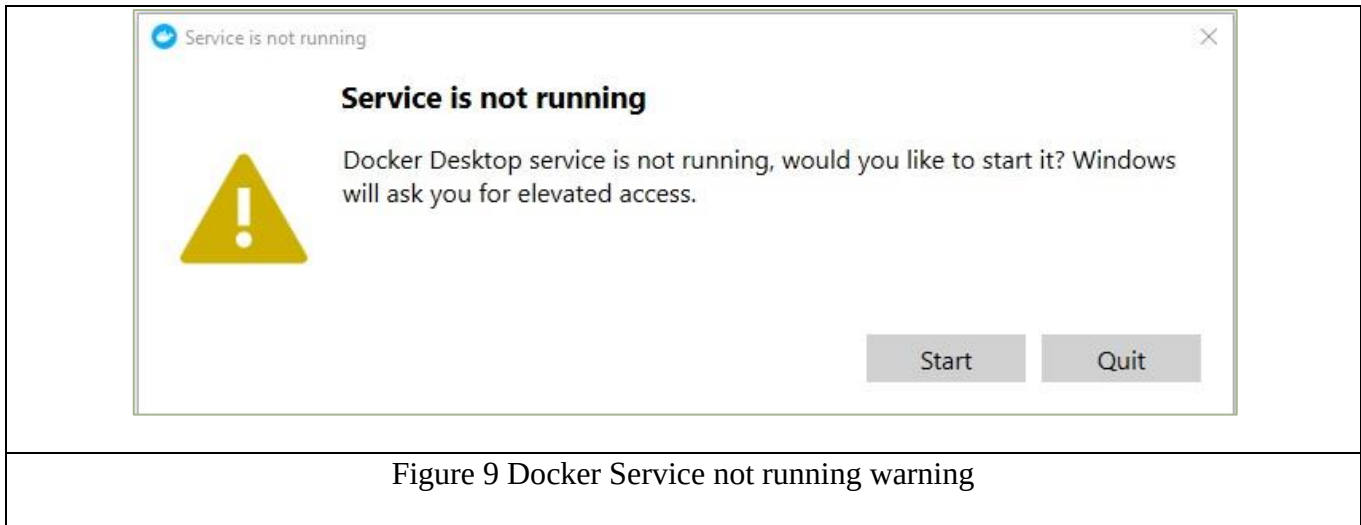


Figure 8 Docker Installation Complete

Step 4 – Run Docker

Once the system restarts, run Docker by double clicking the icon created on the desktop or from start menu. You will see a docker icon appear on your windows task bar. If you hover your mouse over it, it will say “Docker is Starting”. You will see a warning asking your permission to start the docker service. Click Start to continue and wait for docker service to start.



Once docker service starts for the first time, you will see a welcome screen asking you to login to Docker Hub. Enter the docker hub credentials you have created and click on sign in. This is optional but it is highly recommended that you do it.

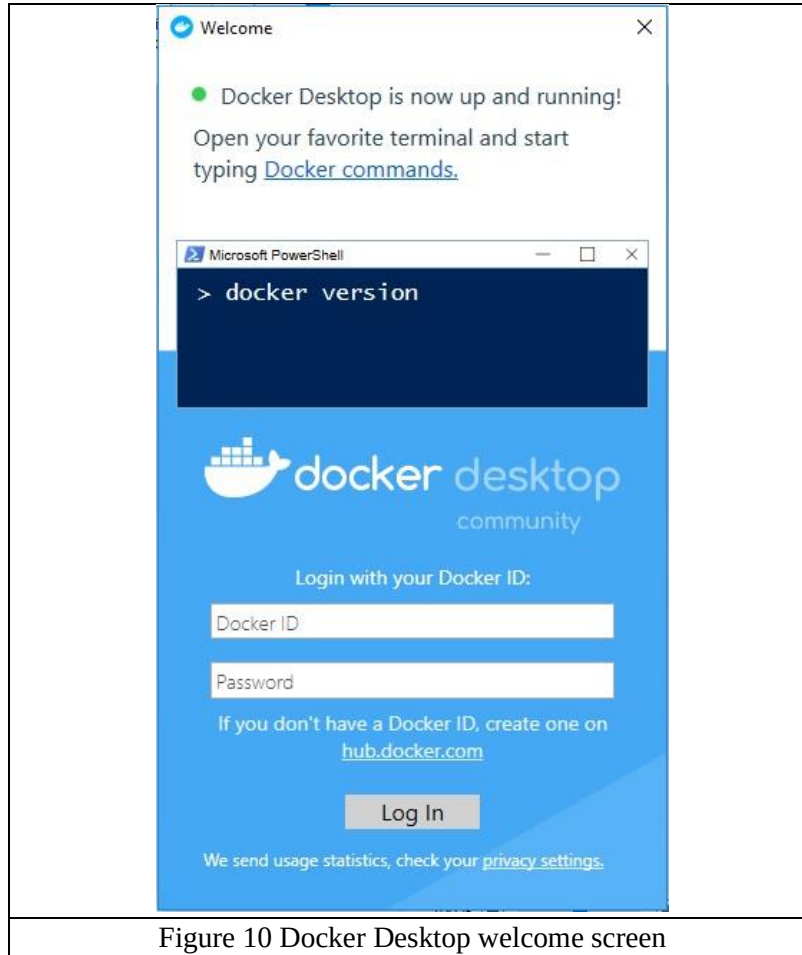


Figure 10 Docker Desktop welcome screen

Step 5 – Disable Start Docker at Startup

By default, docker will automatically start when you turn on/login to your computer. Since Docker requires a lot of RAM, I don't recommend this. We can start docker manually when we want to use it.

To disable starting docker at startup, right click on the docker Icon in the task bar. Click on Settings, under General Tab, uncheck, Start Docker when you login.

Click on apply and restart to make the changes. This will restart the docker service again (I will not restart Windows, only the docker service).

Activity 2:

Check the version of Docker Installed

Most of the time you will be working with command line to work with Docker. Let this be your first command to check Docker version and see what you get.

Open Powershell or command prompt and enter the command:

docker version

You should see something like this.

```
Command Prompt
Microsoft Windows [Version 10.0.19042.631]
(c) 2020 Microsoft Corporation. All rights reserved.

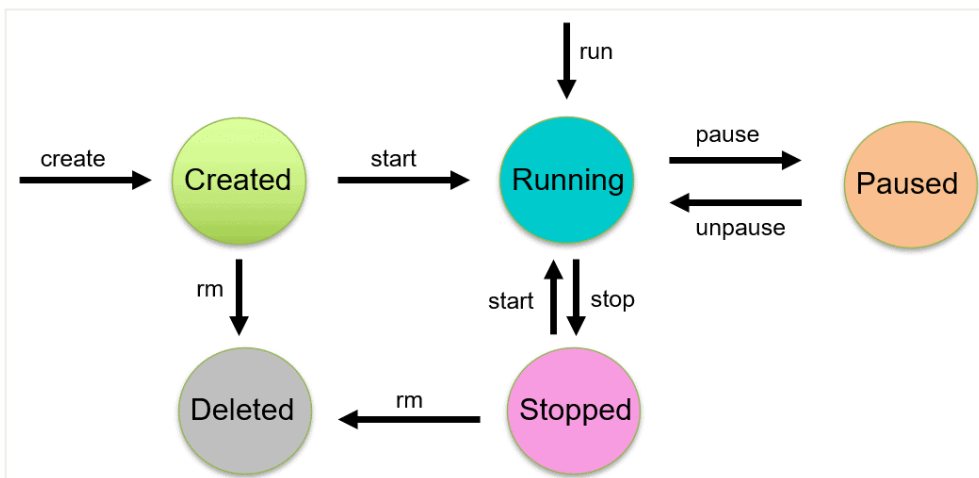
C:\Users\DELL>docker version
error during connect: This error may indicate that the docker daemon is not running.: Get "http://%2F%2F.%2Fpipe%2Fdocker_engine/v1.24/version": open //./pipe/docker_engine: The system cannot find the file specified.
Client:
 Cloud integration: v1.0.22
 Version:          20.10.12
 API version:      1.41
 Go version:       go1.16.12
 Git commit:       e91ed57
 Built:            Mon Dec 13 11:44:07 2021
 OS/Arch:          windows/amd64
 Context:          default
 Experimental:     true

C:\Users\DELL>
```

Figure 11 Docker Version Check – Windows Command Line

Activity 3:

Docker Container Lifecycle Management Architecture



Docker Management Commands

- **Docker Images**

A Docker Image is a read-only file with a bunch of instructions. When these instructions are executed, it creates a Docker container.

It use to describe about the container registry are public or private.

```
docker build -t ImageName:TagName dir
```

```
$ docker build -t comsatsimage
```

- **Create Containers**

Using this will create a new Docker container with the specified Docker image

```
$ docker create --name <container name> <image name>
```

```
$ docker create --name c1 ubuntu
```

- **Start Container**

To start a stopped container, we can use the docker start command.

```
docker start <container name>
```

- **Run Container**

The **docker run** command will do the work of both “**docker create**” and “**docker start**” command.

This command will create a new container and run the image in the newly created container.

```
$ docker run -it --name <container name> <image name>
```

- **Pause Container**

If we want to pause the processes running inside the container, we can use the “*docker pause*” command.

- ```
$ docker pause <container name>
```



---

Stopping a running Container means to stop all the processes running in that Container. Stopping does not mean killing or ending the process.

```
$ docker stop <container name>
```

- **Delete Container**

Removing or deleting the container means destroying all the processes running inside the container and then deleting the Container.

It's preferred to destroy the container, only if present in the stopped state instead of forcefully destroying the running container.

```
$ docker rm <container name>
```

### 3) Graded Lab Tasks

*Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.*

#### Lab Task 1

Implement different Docker Management Commands

---

## Lab 04

### Cloud Deployment and Kubernetes

#### Objective:

In this lab, We will go through Kubernetes and cloud deployment. Kuberneze is actually docker orchestration tool.

#### Activity Outcomes:

The activities provide hands - on practice with the following topics

- Kubernetes
- Kubernetes in Cloud
- Configuration

#### Instructor Note:

## 1. Architecture of Kubernetes

The architecture of Kubernetes is so simple, it consists of Master nodes and Worker nodes, the Master communicates with the Worker using API-server. Multiple Master nodes may also exist in order to provide High Availability, which is indeed one of the most important aspects of application deployment and also an advantage of Kubernetes.



Figure 12 Kubernetes logo

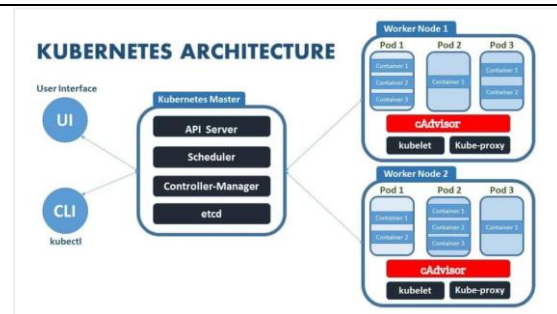


Figure 13 Architecture of Kubernetes

### 1. How to Create Azure Kubernetes AKS Cluster: A Complete Guideline.

Before the starting of this lab, let to discuss about *Kubernets*.

#### What is Kubernetes

“**Kubernetes**” is a Greek word, which means helmsman or pilot which also gives us an idea of how the logo was made. Now, let us come to the technical part of it since Docker has its own limitations, Kubernetes comes into the picture to fill the gaps in the Docker containerization process. K8s is a complete containerization orchestration, which provides the ability to run dynamically scaling, containerized applications, and utilizing an API for management. By doing a comparison of **Docker & Kubernetes** one can deduce the advantages of Kubernetes over other containerization orchestration.

#### Kubernetes Objects

Before one wants to work on K8s, they should know about the basic components in Kubernetes and also about objects in API. Basic objects and several higher-level abstractions are known as controllers. These are the building block of the application lifecycle.

Basic objects include:

- 
- **Pod.** A group of one or more containers, a simple deployable unit of K8S.
  - **Service.** An abstraction that defines a logical set of pods as well as the policy for accessing them.
  - **Volume.** An abstraction that lets us persist data. This is necessary because containers are ephemeral—meaning data is deleted when the container is deleted.
  - **Namespace.** A segment of the cluster dedicated to a certain purpose, for example, a certain project or team of devs.

## Azure Kubernetes Service (AKS)

Azure Kubernetes Service (AKS) is a managed Kubernetes service in which the master node is managed by Azure and end-users manages worker nodes. Users can use AKS to deploy, scale, and manage Docker containers and container-based applications across a cluster of container hosts. One of the best parts about AKS is that you only pay for the worker nodes within your clusters, not for the masters. You can create a cluster in the Azure portal, with the Azure CLI, or template-driven deployment options such as Resource Manager templates and Terraform.

The AKS cluster can be accessed from a local machine's terminal to manage Kubernetes components like **deployments and Kubernetes pods**. It can even be used to create a Kubernetes deployment.

## 2. Solved Lab Activities:

### Activity 1:

#### How To Create Azure Kubernetes Cluster

Azure Kubernetes Service (AKS) is a Kubernetes service that lets you quickly deploy and manage master and worker nodes in clusters. Creating an AKS cluster is easy and there are more than enough manuals who will guide you through the process. It is an effortless process to create a cluster by following the steps given below.

There are 4 ways to deploy an Azure Kubernetes Cluster, which are using:

1. Azure Portal
2. Azure CLI
3. Azure PowerShell
4. Using template-driven deployment options, like Azure Resource Manager templates and Terraform

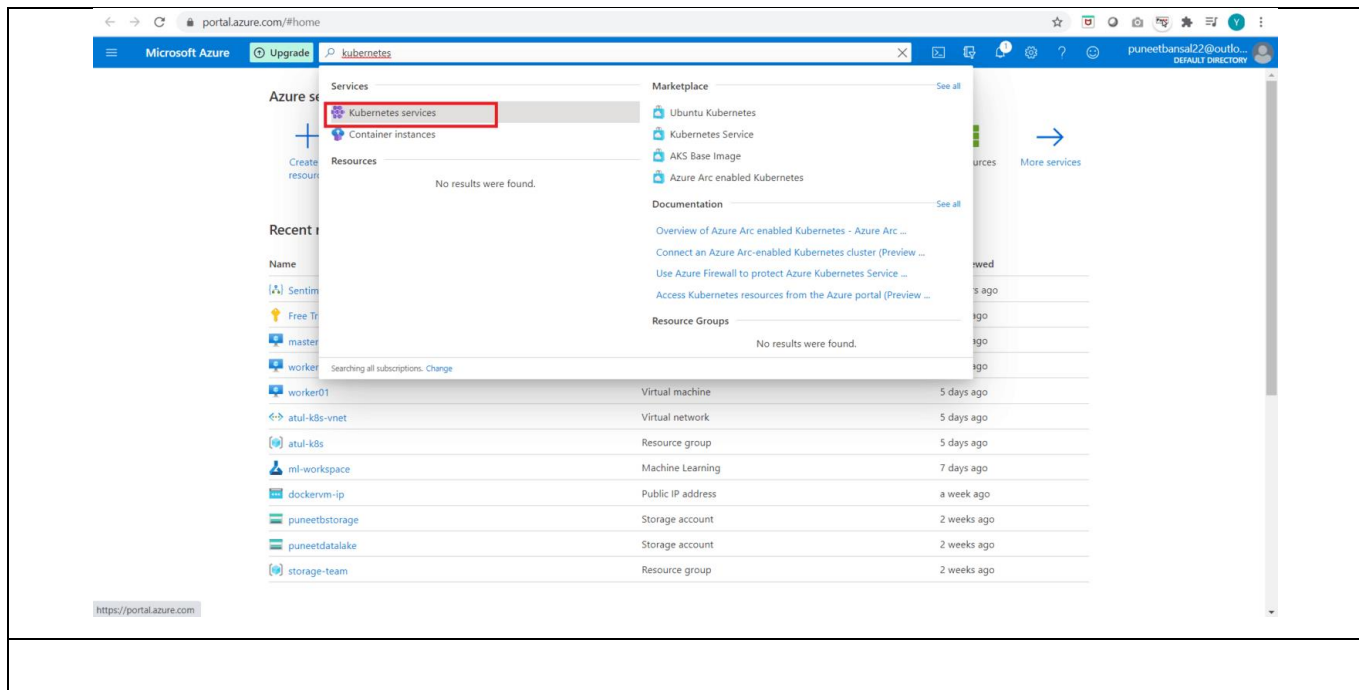
Now let us look at how do we create a cluster using Azure Portal. But before all that you should an Azure Cloud account to create a cluster. Create a **FREE TRIAL AZURE ACCOUNT** to get **\$200 free credit** on your Azure portal.

#### Steps To Create AKS Cluster

Once you have created an Azure account, or if you already have one, please follow the steps given below in order to create an Azure Kubernetes Cluster easily.

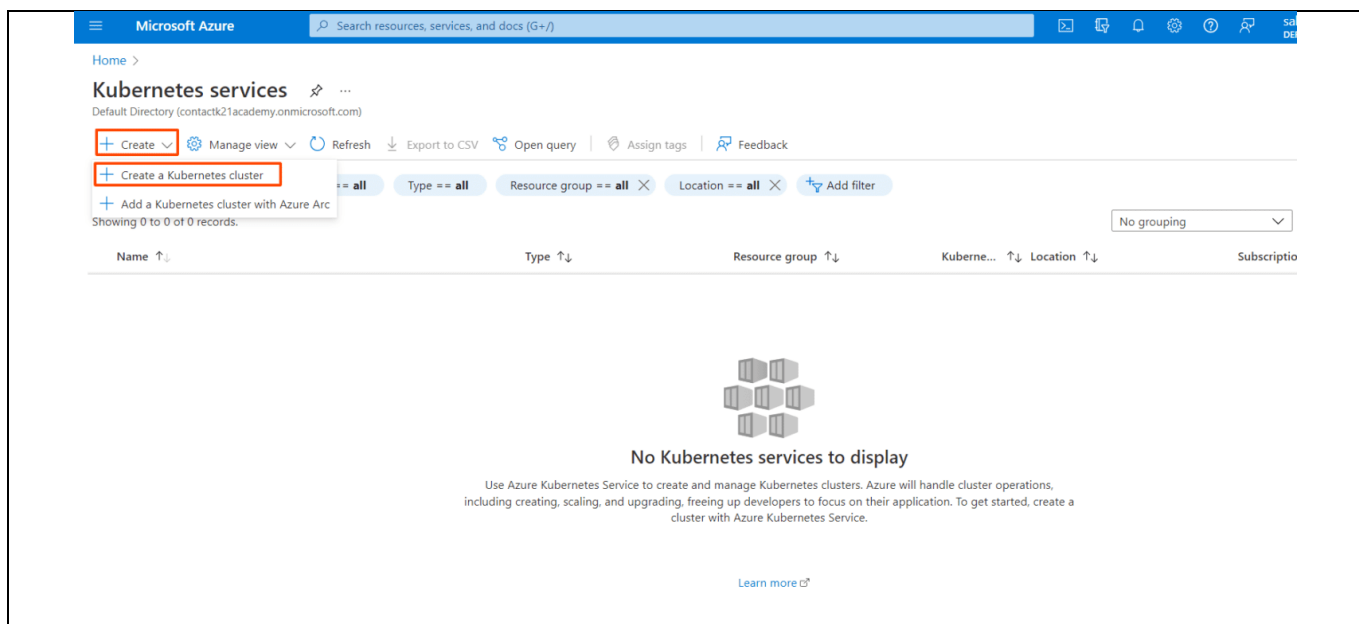
## Step 1 [Kubernetes Services]:

Go to the Azure Portal and search for **Kubernetes Service** in the search bar and click on it.



## Step 2 [Add Kubernetes Cluster]:

Once you reach the Kubernetes Services page, click on **Create** and then **Create Kubernetes Cluster**.



---

## Step 3 [Basics]:

Once you click on add Kubernetes cluster, the next step is to update the specifications of the cluster. So, click on **Basics**.

- Give the Resource Group name as per your requirement.
- Specify a name to your cluster in the **Kubernetes cluster name** field.
- Choose a **Region** in which you want to create your AKS cluster. In the specified region, our master node will be created.
- Based on the region the select the availability zones.
- Select the Kubernetes Version. Here I am choosing the default, i.e., 1.20.9

The screenshot shows the 'Create Kubernetes cluster' wizard in the Azure portal. The 'Basics' tab is selected, and the following fields are filled out:

- Subscription:** Pay-As-You-Go
- Resource group:** (New) atulaks
- Cluster preset configuration:** Standard (\$\$)
- Kubernetes cluster name:** k2tatulclusterdev
- Region:** (US) South Central US
- Availability zones:** Zones 1,2,3
- Kubernetes version:** 1.20.9 (default)

Navigation buttons at the bottom include 'Review + create', '< Previous', and 'Next: Node pools >'.

Next, comes the size and count of the nodes of the AKS cluster that we are gonna create. These can be updated as per the requirements.

- Select the **Node Size**. We are choosing **Standard Ds2 v2** which has the following configuration: 2 vCPUs, 7 GiB RAM, 8 Data Disks, 14 GiB Temp Storage.
- Give the **Node Count** value which specifies how many Worker Nodes we want.

Microsoft Azure

Search resources, services, and docs (G+)

Home > Kubernetes Service >

### Create Kubernetes cluster

Region \* ⓘ (US) South Central US

Availability zones ⓘ Zones 1,2,3  
High availability is recommended for standard configuration.

Kubernetes version \* ⓘ 1.20.9 (default)

**Primary node pool**

The number and size of nodes in the primary node pool in your cluster. For production workloads, at least 3 nodes are recommended for resiliency. For development or test workloads, only one node is required. If you would like to add additional node pools or to see additional configuration options for this node pool, go to the 'Node pools' tab above. You will be able to add additional node pools after creating your cluster. [Learn more about node pools in Azure Kubernetes Service](#)

Node size \* ⓘ **Standard DS2 v2**  
Standard DS2\_v2 is recommended for standard configuration.  
[Change size](#)

Scale method \* ⓘ  
☐ Manual  
☒ Autoscale  
Autoscaling is recommended for standard configuration.

Node count range \* ⓘ 1 2

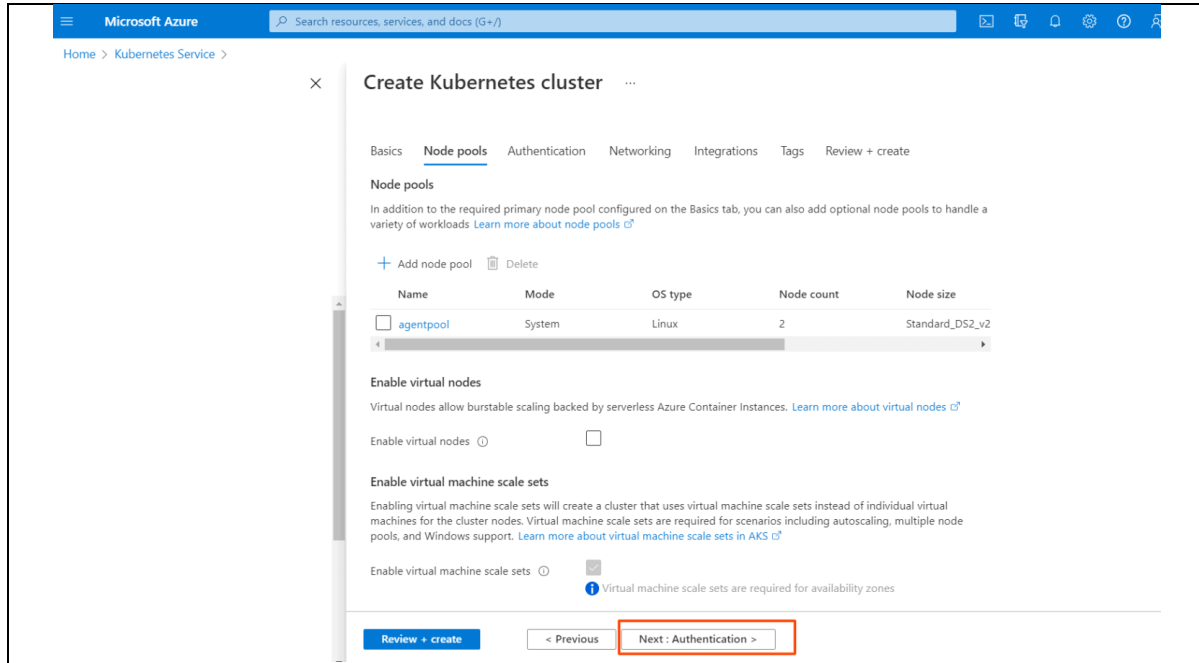
[Review & create](#) < Previous Next : Node pools >

## Activity 2:

### Step 4 [Node Pools]:

Next comes the **Node Pools**, follow the steps given below:

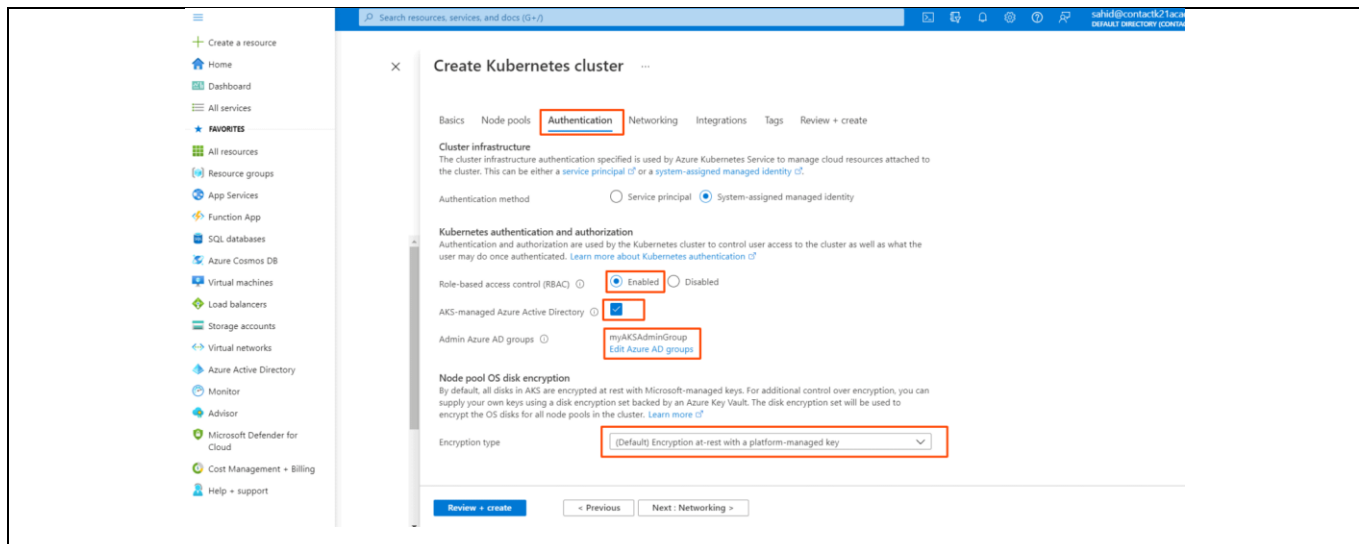
- In Azure Kubernetes Service (AKS), nodes of the same configuration are grouped together into **node pools**. **Node pools** contain the underlying VMs that run your applications.
- The **Virtual nodes** are a type of Serverless container instance. As we want to create the Worker nodes as Virtual Machines, so we **won't enable** this option.



## Step 5 [Authentication]:

Next is to click on Authentication.

- Choose the System-assigned managed identity.
- If you want to go for Role-based Access Control (RBAC) then select Enabled.
- We can use both RBAC & Azure Active Directory for Authentication.
- Note: Here in this step-by-step guide, we are using both RBAC & Azure AD for authentication.
- Check the Azure Active Directory option.
- Create a group in Active Directory and add it.
- Choose the Encryption Type of your choice, I will use the Default one.





---

## Step 6 [Networking]:

Next is the Networking part.

- Select the **Network Configuration**. I will be choosing **Azure CNI**
- The **Cluster Subnet** option is to choose which Subnet you want the Nodes and Containers to be placed in.
- **Kubernetes service address range** is the CIDR notation IP range from which to assign server cluster IPs.
- **Docker Bridge address** is the IP address assigned to Docker Bridge. The Bridge Network is for the container to container communication.
- In **Private Cluster**, the communication between the nodes and the API server happens internally.
- So, I am **Disabling** the Private Cluster.
- Keep the Network Policy to **Azure**.
- Do not enable **HTTP** application routing.

Microsoft Azure

Search resources, services, and docs (G+/I)

Home > Kubernetes services >

### Create Kubernetes cluster

- The **kubernetes** networking plug-in creates a new VNet for your cluster using default values.
- The **Azure CNI** networking plug-in allows clusters to use a new or existing VNet with customizable addresses. Application pods are connected directly to the VNet, which allows for native integration with VNet features.

[Learn more about networking in Azure Kubernetes Service](#)

Network configuration ⓘ

☐ Kubenet

☒ **Azure CNI**

ⓘ The Azure CNI plugin requires an IP address from the subnet below for each pod on a node, which can more quickly exhaust available IP addresses if a high value is set for pods per node. Consider modifying the default values for pods per node for each node pool on the "Node pools" tab. [Learn more](#)

Virtual network \* ⓘ

(New) AKS-K21academy-vnet

[Create new](#)

Cluster subnet \* ⓘ

(new) default (10.240.0.0/16)

Kubernetes service address range \* ⓘ

10.0.0.0/16

Kubernetes DNS service IP address \* ⓘ

10.0.0.10

Docker Bridge address \* ⓘ

172.17.0.1/16

DNS name prefix \* ⓘ

AKS-K21academy-dns

Traffic routing

Load balancer ⓘ

Standard

[Review + create](#) [< Previous](#) [Next: Integrations >](#)

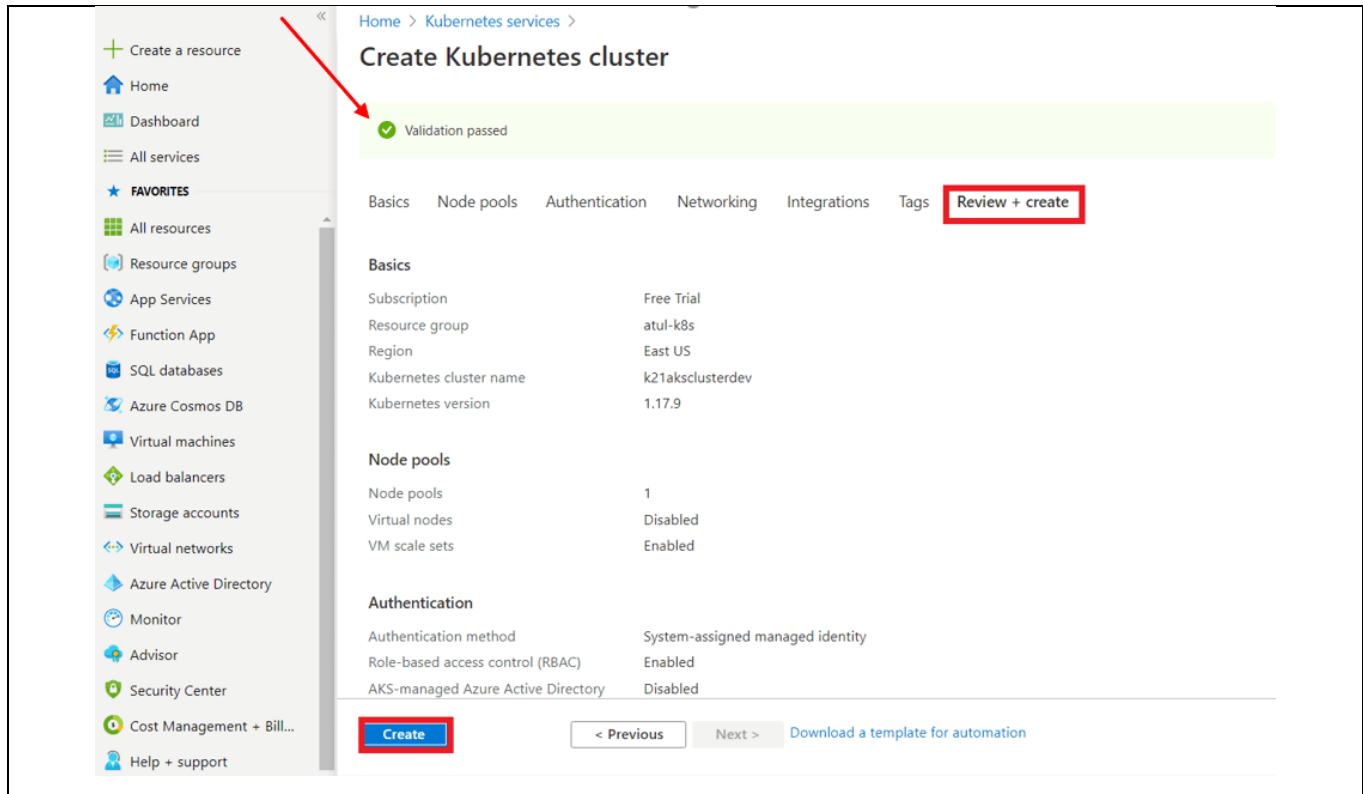
## Step 7 [Integration]:

Next is the Integration.

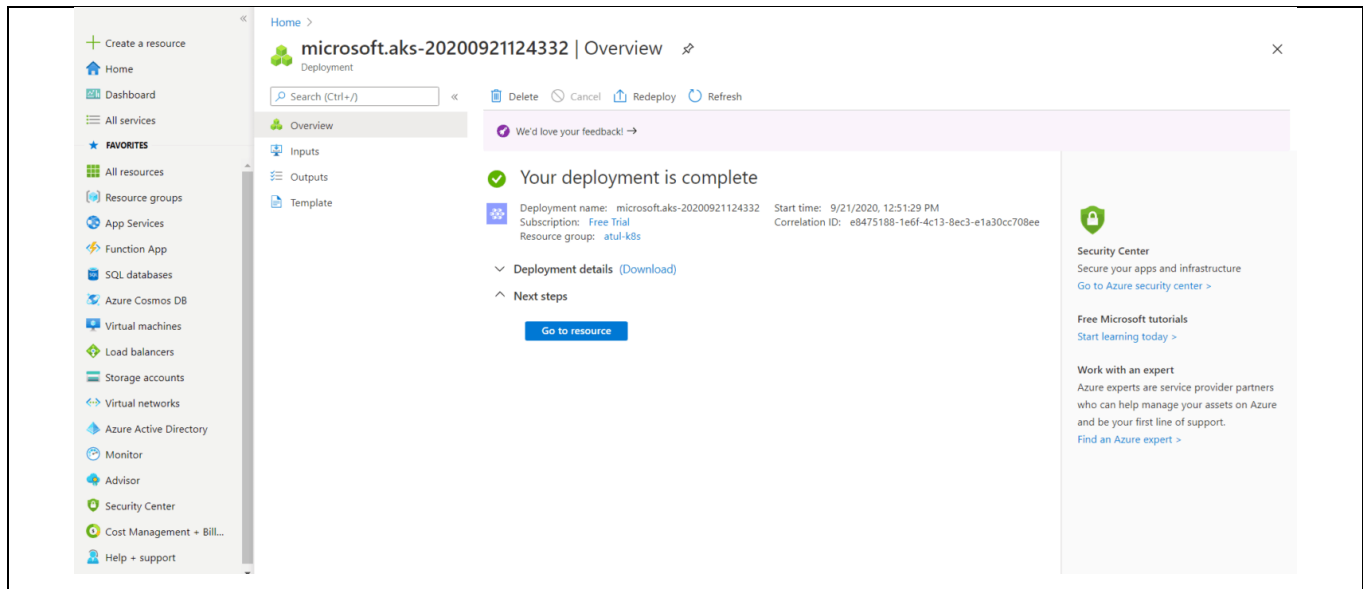
Here we keep all settings to default and move to the next step.

## Step 8 [Review & Create]:

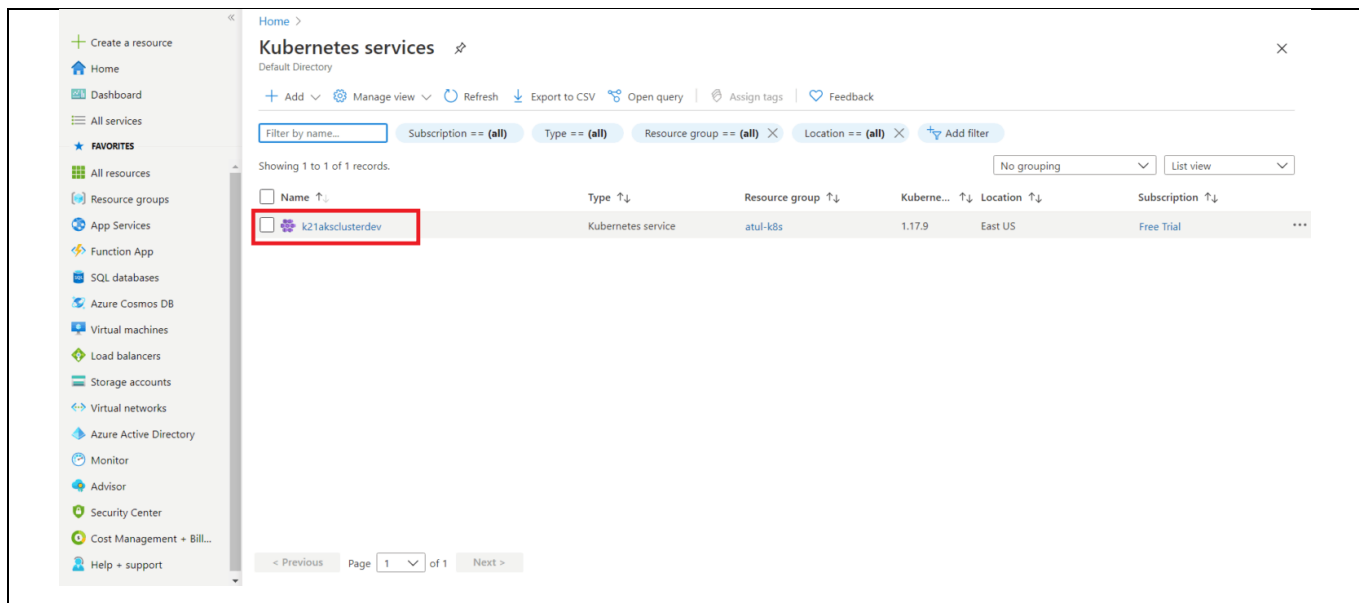
The final step is to click on **Review & Create**. If you click on **Create**, it will first Validate your AKS Cluster and if everything is fine then the cluster will be created.



You can see that our new Azure Kubernetes cluster has been successfully created. Yes! But hold on, we will have to connect this cluster, which I am going to cover next.



To view the cluster, go to **Kubernetes services** and there you can access the AKS cluster.



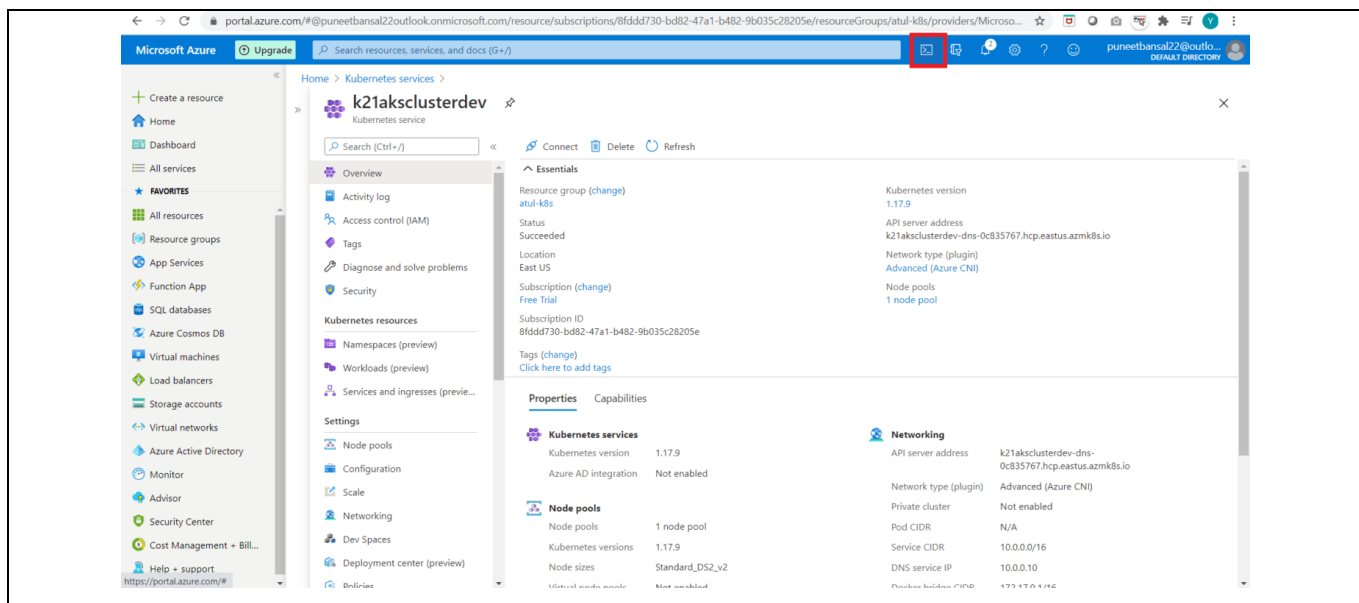
### Activity 3:

## Connect to the Azure Kubernetes Cluster

There are two ways to connect the AKS cluster:

1. Using Cloud Shell
2. Using Azure CLI

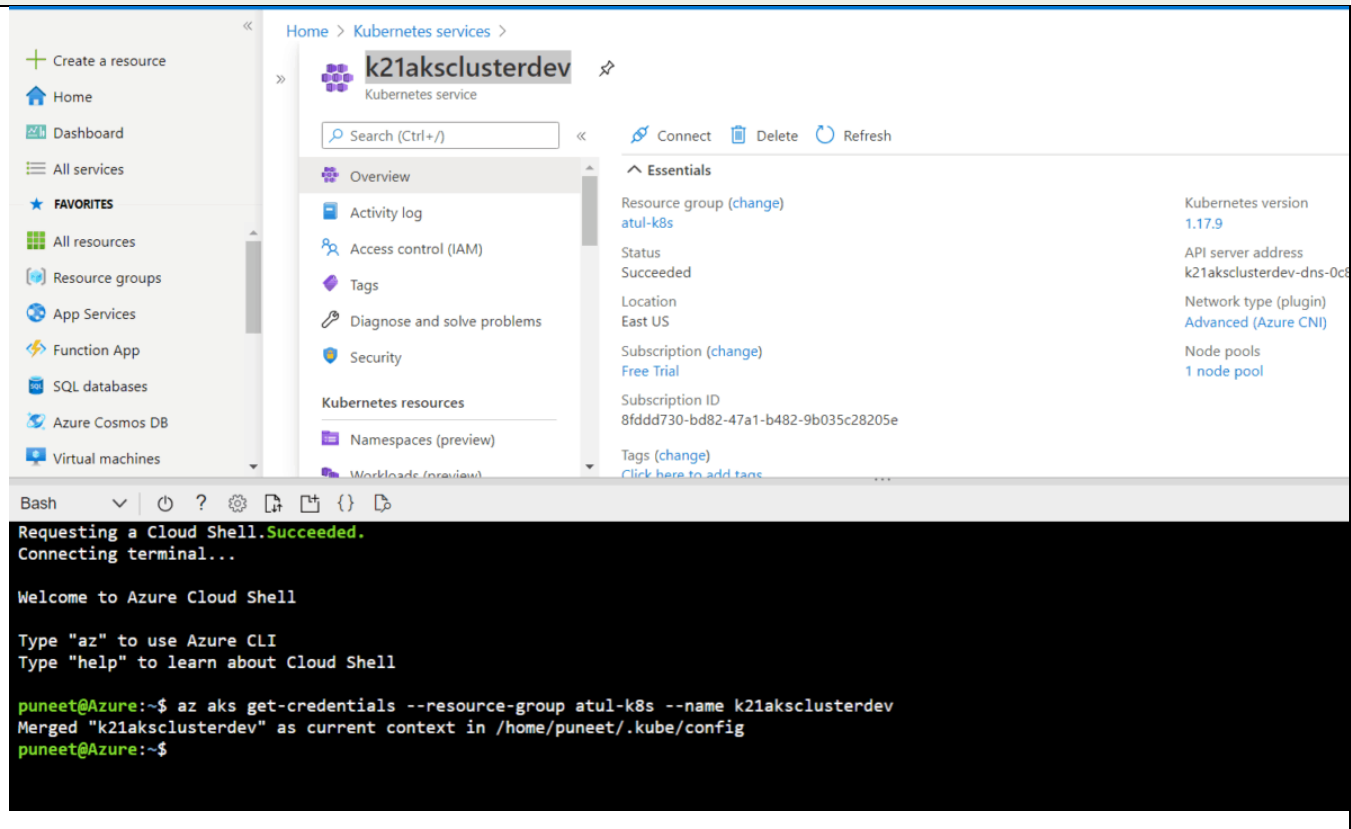
I am going to cover how do we connect using **Cloud Shell**. We can see the option on top of our screen (marked red in the below image).



## Step 1:

Run the following command, on the Azure bash shell:

```
$ az aks get-credentials --resource-groups <name of resource group> --name <name of cluster>
```

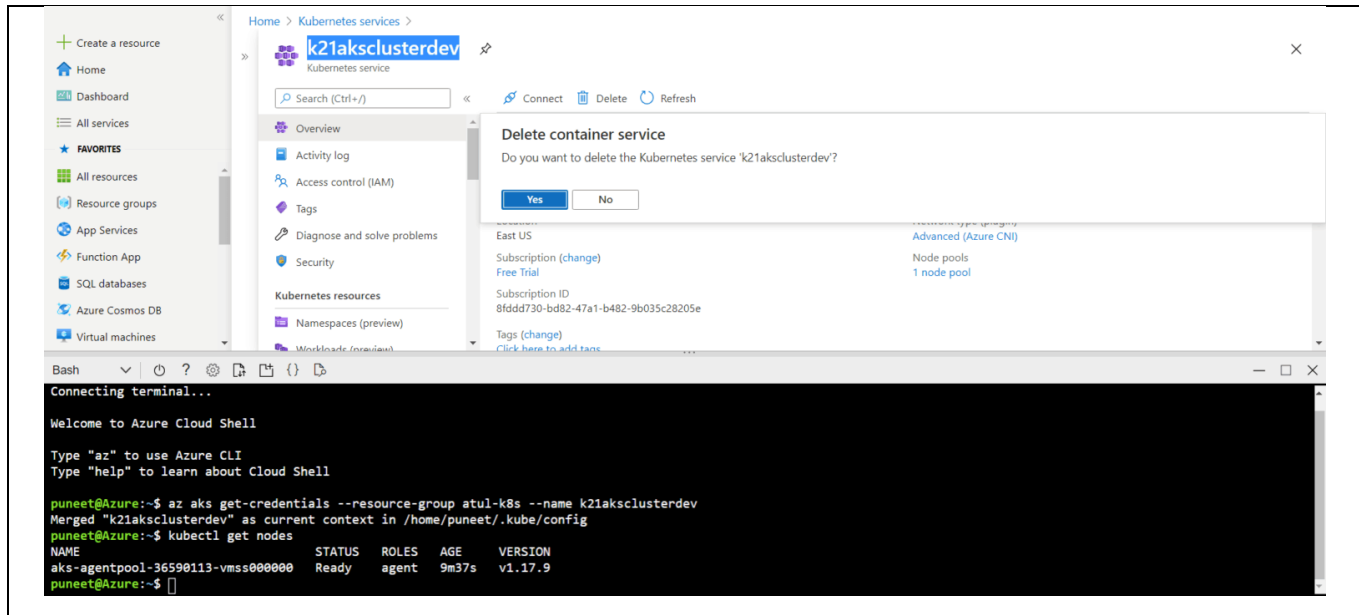


The screenshot displays the Azure portal interface for a Kubernetes service named 'k21aksclusterdev'. The left sidebar shows navigation options like 'Create a resource', 'Home', 'Dashboard', and 'All services'. The main pane shows the 'Overview' tab for the cluster, with details such as 'Resource group (change) atul-k8s', 'Status Succeeded', 'Location East US', and 'Subscription ID 8fddd730-bd82-47a1-b482-9b035c28205e'. Below the portal, a terminal window shows the execution of the 'az aks get-credentials' command, which successfully merges the kubeconfig into the local context.

## Step 2:

To get the Nodes running in our cluster, run the following command, and you will see all the nodes in your AKS cluster.

```
$ kubectl get nodes
```



### 3) Graded Lab Tasks

*Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.*

#### Lab Task 1:

Repeat all above steps for:

1. To Create Azure Kubernetes AKS Cluster.

---

## Lab 05

### Use of GIT

#### Objective:

After completing this lab, students will be able to install and utilize the commands of git

#### Activity Outcomes:

The activities provide hands - on practice with the following topics

- GIT Add
- Commit
- Pull
- Push

#### Instructor Note:

---

# 1. GIT

Git is a free and open-source distributed version control system designed to handle everything from small to very large projects with speed and efficiency. Git is easy to learn and has a tiny footprint with lightning-fast performance. It outclasses SCM tools like Subversion, CVS, Perforce, and ClearCase with features like cheap local branching, convenient staging areas, and multiple workflows.

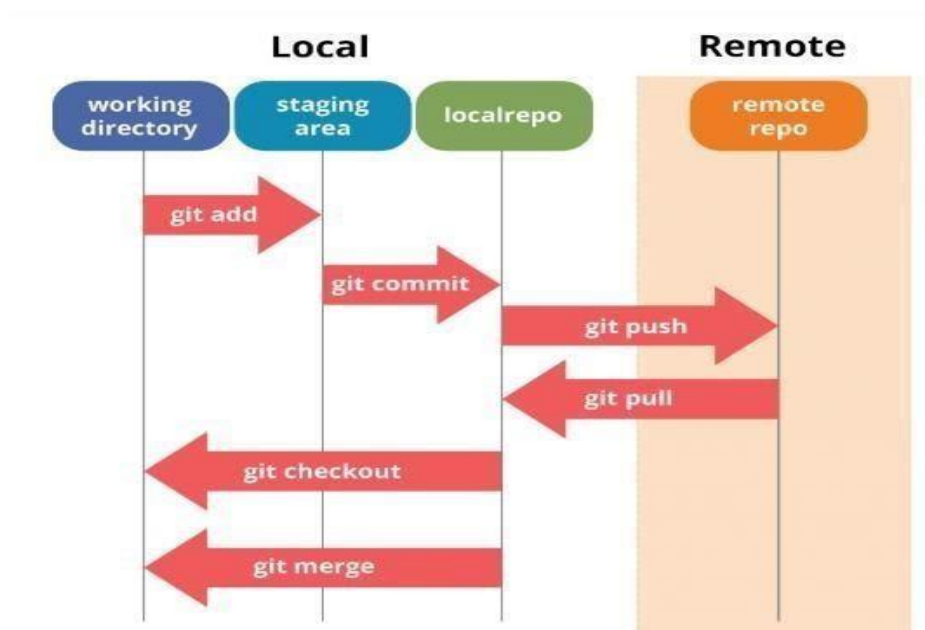
Some of the basic operations in Git are:

1. Initialize
2. Add
3. Commit
4. Pull
5. Push

Some advanced Git operations are:

1. Branching
2. Merging
3. Rebasing

The following diagram depicts all supported operations in GIT





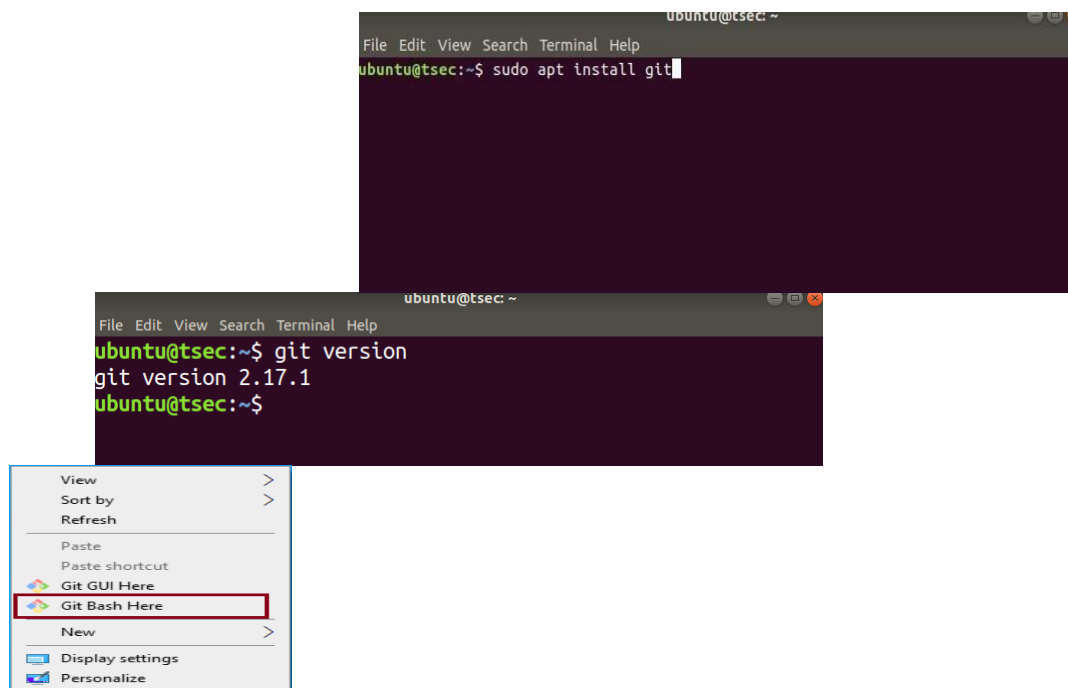
---

## 1. Solved Lab Activities

### Installation of GIT

1. In windows, download GIT from <https://git-scm.com/> and perform the straightforward installation.
2. In Ubuntu, install GIT using `$sudo apt install git`, Confirm the version after installation using command `$git --version`

Once installation is done, open the terminal in Ubuntu and perform the following steps or in windows Right click and select Git bash here.



The output of GIT Bash in windows and GIT shell in Ubuntu is shown below

```
MINGW64/c/Users/ADMIN/Desktop
ADMIN@DESKTOP-J582V2L MINGW64 ~/Desktop
$ |
```

GIT Bash in Windows

```
ubuntu@tsec:~$ git version
git version 2.17.1
ubuntu@tsec:~$ git
usage: git [--version] [--help] [-C <path>] [-c <name>=<value>]
 [--exec-path[=<path>]] [--html-path] [--man-path]
 [--info-path]
 [-p | --paginate | --no-pager] [--no-replace-objects] [--bare]
 [--git-dir=<path>] [--work-tree=<path>] [--name-prefix=<name>]
 <command> [<args>]
```

GIT Shell in Ubuntu

## Activity 2:

In this activity, we will implement different commands

of GIT in our terminal.

Commands are:

1. Init
2. Config
3. Status
4. Commit
5. Branch

etc

```
Lenovo@Lenovo-PC MINGW64 ~
$ mkdir new-repo

Lenovo@Lenovo-PC MINGW64 ~
$ cd new-repo

Lenovo@Lenovo-PC MINGW64 ~/new-repo
$ git config --global user.name "USER"

Lenovo@Lenovo-PC MINGW64 ~/new-repo
$ git config --global user.email "zindagikeupat@gmail.com"

Lenovo@Lenovo-PC MINGW64 ~/new-repo
$ git init
Initialized empty Git repository in C:/Users/Lenovo/new-repo/.git/

Lenovo@Lenovo-PC MINGW64 ~/new-repo (master)
$ touch index.html

Lenovo@Lenovo-PC MINGW64 ~/new-repo (master)
$ touch readme.txt

Lenovo@Lenovo-PC MINGW64 ~/new-repo (master)
$ touch style.py

Lenovo@Lenovo-PC MINGW64 ~/new-repo (master)
$ git status
On branch master

No commits yet

Untracked files:
 (use "git add <file>..." to include in what will be committed)
 index.html
 readme.txt
 style.py

nothing added to commit but untracked files present (use "git add" to track)

Lenovo@Lenovo-PC MINGW64 ~/new-repo (master)
$ git add
Nothing specified, nothing added.
Maybe you wanted to say 'git add .'

Lenovo@Lenovo-PC MINGW64 ~/new-repo (master)
$ git add .
```

```

Lenovo@Lenovo-PC MINGW64 ~/new-repo/new-repo (master)
$ git commit -m "First Commit"
[master (root-commit) d66c9f8] First Commit
3 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 index.html
create mode 100644 readme.txt
create mode 100644 style.py

Lenovo@Lenovo-PC MINGW64 ~/new-repo/new-repo (master)
$ git status
On branch master
nothing to commit, working tree clean

Lenovo@Lenovo-PC MINGW64 ~/new-repo/new-repo (master)
$ git status
On branch master
nothing to commit, working tree clean

Lenovo@Lenovo-PC MINGW64 ~/new-repo/new-repo (master)
$ git add readme.txt

Lenovo@Lenovo-PC MINGW64 ~/new-repo/new-repo (master)
$ git commit -m "changed readme.txt"
On branch master
nothing to commit, working tree clean

Lenovo@Lenovo-PC MINGW64 ~/new-repo/new-repo (master)
$ git log
commit d66c9f88f7d3c8924027e53ea1fbace5f611beda (HEAD -> master)
Author: USER <zindagikeupat@gmail.com>
Date: Thu May 20 01:50:12 2021 +0530

 First Commit

Lenovo@Lenovo-PC MINGW64 ~/new-repo/new-repo (master)
$ git log --oneline readme.txt
d66c9f8 (HEAD -> master) First Commit

Lenovo@Lenovo-PC MINGW64 ~/new-repo/new-repo (master)
$ git branch login

Lenovo@Lenovo-PC MINGW64 ~/new-repo/new-repo (master)
$ git checkout login

```

### Activity 3:

Creating new repository and selecting public option followed by create repository

#### Quick setup — if you've done this kind of thing before

 Set up in Desktop or **HTTPS** **SSH** `https://github.com/shwetashakapnor1999/myrepo` 

Get started by [creating a new file](#) or [uploading an existing file](#). We recommend every repository include a [README](#), [LICENSE](#), and [.gitignore](#).

#### ...or create a new repository on the command line

```
echo "# myrepo" >> README.md
git init
git add README.md
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/shwetashakapnor1999/myrepo.git
git push -u origin main
```



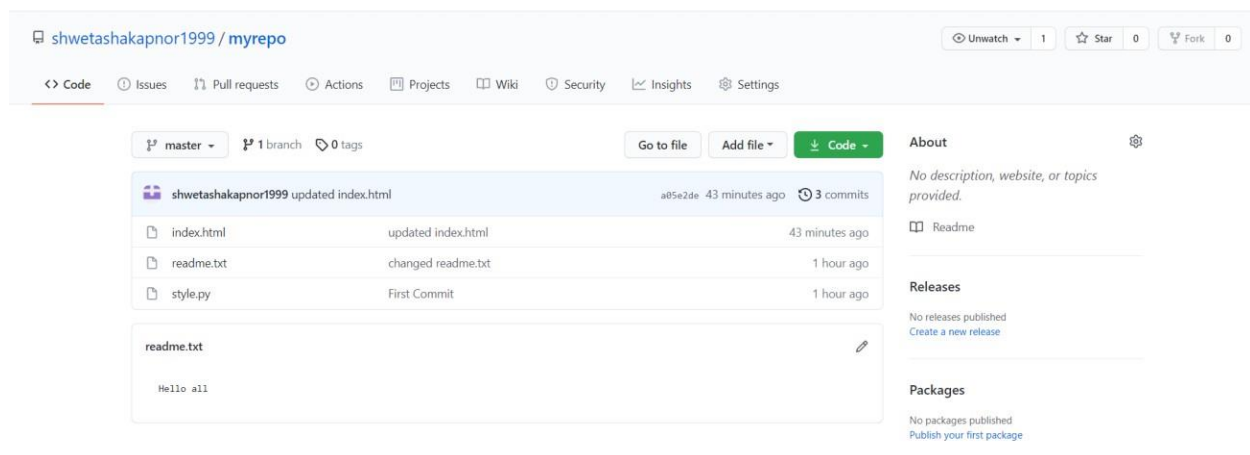
#### ...or push an existing repository from the command line

```
git remote add origin https://github.com/shwetashakapnor1999/myrepo.git
git branch -M main
git push -u origin main
```



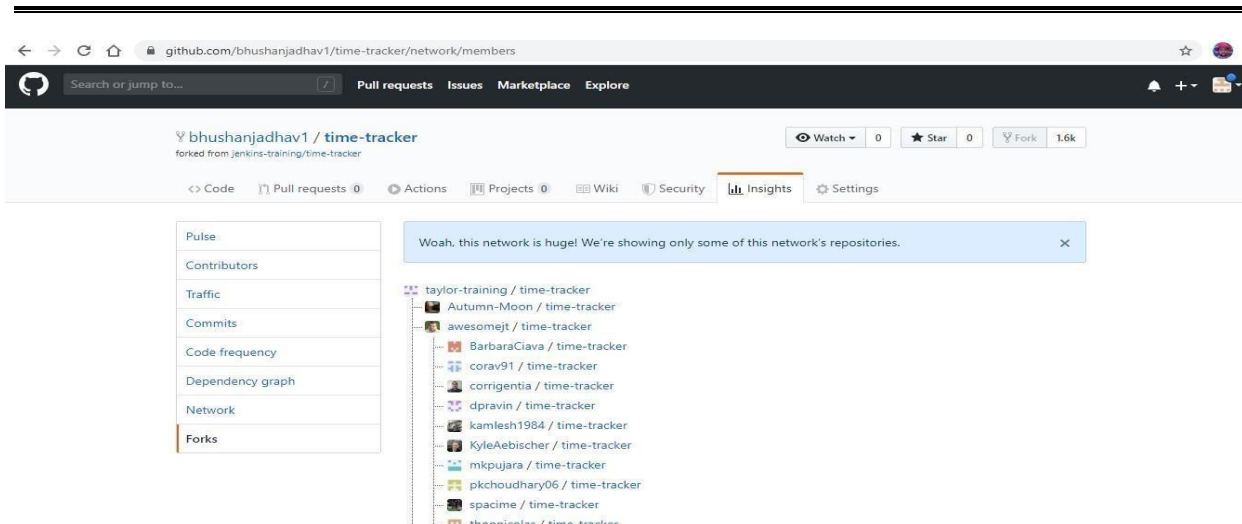
By default, we can create public repository in GitHub. So, we can copy the entire public repository of any other users in to own account using “FORK” Operation. Now fork the repository (Sharing with other users who wants to contribute).

Login with another account→Copy and Paste URL of repository→then just click on fork to clone to others account. Suppose we want to fork public repository “timetracker”. So, search for “timetracker” github repository on google and once its opened clicked on “Fork button” from the top of the github web page as shown below.



The screenshot shows the GitHub interface for a repository named 'myrepo' by user 'shwetashakapnor1999'. The repository is public and has 1 star and 0 forks. The main branch is 'master'. The repository contains three files: 'index.html', 'readme.txt', and 'style.py'. The 'readme.txt' file is selected, showing its content: 'Hello all!'. The right sidebar contains sections for 'About', 'Releases', and 'Packages'.

After fork it will be added in your local repository. Example, it will list all the users who has forked the code.



To delete the repository, open the desired repository you want to delete and go to the settings option. There you will see delete repository button to delete it.

### Danger Zone

Change repository visibility  
This repository is currently public.

Change visibility

Transfer ownership  
Transfer this repository to another user or to an organization where you have the ability to create repositories.

Transfer

Archive this repository  
Mark this repository as archived and read-only.

Archive this repository

Delete this repository  
Once you delete a repository, there is no going back. Please be certain.

Delete this repository

Are you absolutely sure?

×

Unexpected bad things will happen if you don't read this!

This action **cannot** be undone. This will permanently delete the **zindagikeupat/Zindagikeupat** repository, wiki, issues, comments, packages, secrets, workflow runs, and remove all collaborator associations.

Please type **zindagikeupat/Zindagikeupat** to confirm.

zindagikeupat/Zindagikeupat

I understand the consequences, delete this repository

Now, if you want to download a repository in local machine, then git clone command is used followed by path to repository. In GitHub the path of repository can be known through clone or download button and it can be downloaded using git clone command as shown below.

zindagikeupat / Zindagikeupat

Unwatch 1 Star 0 Fork 0

<> Code Issues Pull requests Actions Projects Wiki Security Insights Settings

master had recent pushes 2 minutes ago Compare & pull request

master 2 branches 0 tags Go to file Add file Code

This branch is 1 commit ahead, 1 commit behind main. Pull request Compare

| zindagikeupat changed readme.txt 99c823d 12 hours ago 1 commit |                    |              |
|----------------------------------------------------------------|--------------------|--------------|
| index.html                                                     | changed readme.txt | 12 hours ago |
| readme.txt                                                     | changed readme.txt | 12 hours ago |
| style.py                                                       | changed readme.txt | 12 hours ago |

About

No description, website, or topics provided.

Releases

No releases published  
Create a new release

### 3) Graded Lab Tasks

*Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.*

#### Lab Task 1:

Create a new repository and add two new files into git repository

#### Lab Task 2:

Ignore few files into git repository.

---

## Lab 06

### Source Control Management

#### Objective:

To perform version control on websites/software using GIT with push and pull commands.

#### Activity Outcomes:

Examine the different Version Control Strategies. The activities provide hands - on practice with the following topics

- Use of Push and Pull Command
- Version Control

#### Instructor Note:



# 1. perform version control on websites/software's using GIT with push and pull commands.

First open Github.com and create a new account. After verifying account through E-mail, create a Repository on github.com

**Pull and Push Processes** The pull command is used to fetch the repository from github to local while push is used to commit files from local repository to Github.

**Push** → **Push** changes to Web repository

**Pull** → **Pull** changes to Local repository

## 2. Solved Lab Activities:

### Activity 1:

**Push:**

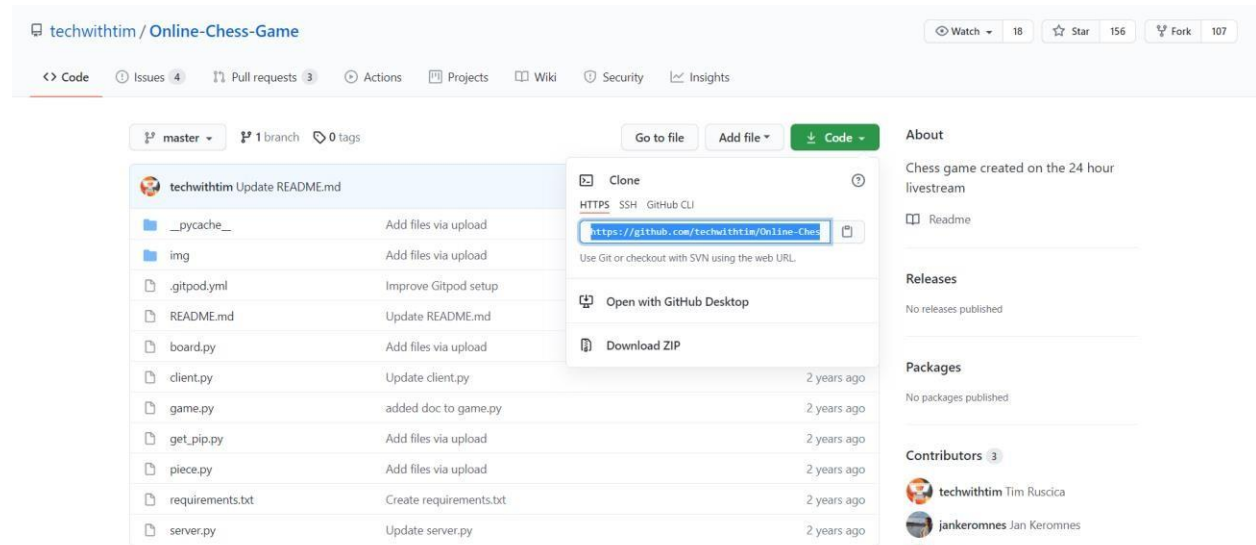
```
Lenovo@Lenovo-PC MINGW64 ~/new-repo (login)
$ git push -u example master
Logon failed, use ctrl+c to cancel basic credential prompt.
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 247 bytes | 123.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0)
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote: https://github.com/zindagikeupat/Zindagikeupat/pull/new/master
remote:
To https://github.com/zindagikeupat/Zindagikeupat
 * [new branch] master -> master
Branch 'master' set up to track remote branch 'master' from 'example'.
```

The screenshot shows the GitHub interface for the repository 'zindagikeupat/Zindagikeupat'. At the top, there's a navigation bar with links for Code, Issues, Pull requests, Actions, Projects, Wiki, Security, Insights, and Settings. Below this, a yellow banner indicates 'master had recent pushes 2 minutes ago' with a 'Compare & pull request' button. The main content area shows the 'master' branch selected, with 2 branches and 0 tags. It states 'This branch is 1 commit ahead, 1 commit behind main.' and provides links for 'Pull request' and 'Compare'. A commit history table is displayed, showing a commit by 'zindagikeupat' that changed 'readme.txt' 12 hours ago. The table lists files: index.html, readme.txt, and style.py, all marked as 'changed readme.txt'. On the right sidebar, there's an 'About' section with a description of the repository and a 'Releases' section indicating no releases have been published.

| File       | Change             | Time         |
|------------|--------------------|--------------|
| index.html | changed readme.txt | 12 hours ago |
| readme.txt | changed readme.txt | 12 hours ago |
| style.py   | changed readme.txt | 12 hours ago |

## Activity 2:

### Pull:



The screenshot shows the GitHub interface for the repository 'techwithtim/Online-Chess-Game'. At the top, there are navigation tabs for Code, Issues (4), Pull requests (3), Actions, Projects, Wiki, Security, and Insights. Below the repository name, there are buttons for 'Go to file', 'Add file', and 'Code'. A 'Clone' dropdown menu is open, showing options for cloning via HTTPS, SSH, or GitHub CLI, with the HTTPS URL 'https://github.com/techwithtim/Online-Chess-Game.git' highlighted. Below the clone options, there are links to 'Open with GitHub Desktop' and 'Download ZIP'. The main content area displays a list of files and folders in the repository, including \_\_pycache\_\_, img, .gitpod.yml, README.md, board.py, client.py, game.py, get\_pip.py, piece.py, requirements.txt, and server.py. Each file entry includes a brief description of its purpose and the last commit message. On the right side, there are sections for 'About' (describing the chess game), 'Releases' (no releases published), 'Packages' (no packages published), and 'Contributors' (listing techwithtim and jankeromnes).

```
Lenovo@Lenovo-PC MINGW64 ~
$ mkdir chess

Lenovo@Lenovo-PC MINGW64 ~
$ cd chess

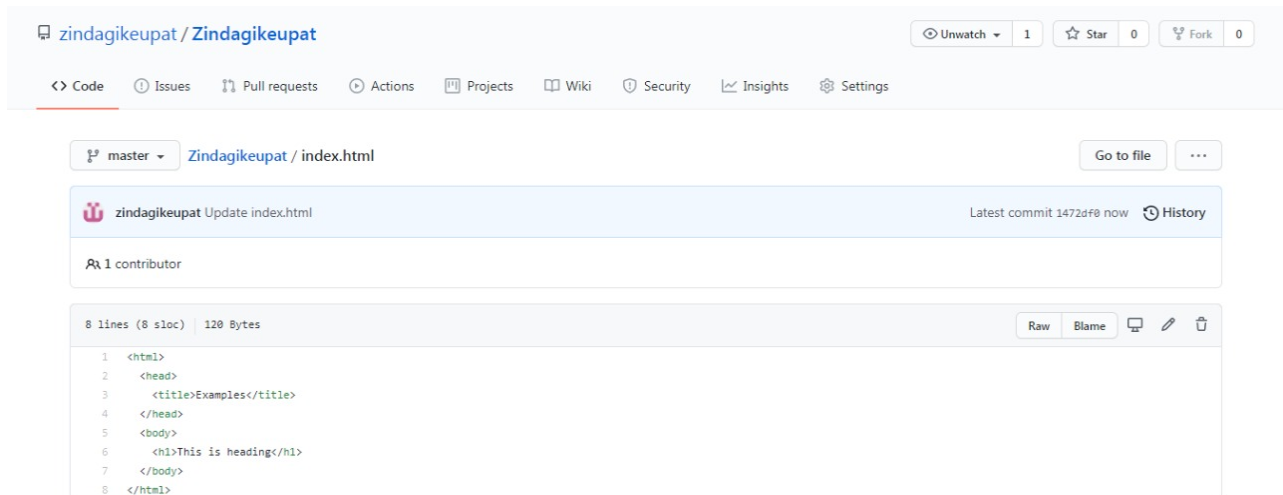
Lenovo@Lenovo-PC MINGW64 ~/chess
$ git init
Initialized empty Git repository in C:/Users/Lenovo/chess/.git/

Lenovo@Lenovo-PC MINGW64 ~/chess (master)
$ git pull https://github.com/techwithtim/Online-chess-Game.git
remote: Enumerating objects: 160, done.
remote: Total 160 (delta 0), reused 0 (delta 0), pack-reused 160
Receiving objects: 100% (160/160), 2.09 MiB | 692.00 KiB/s, done.
Resolving deltas: 100% (80/80), done.
From https://github.com/techwithtim/Online-chess-Game
* branch HEAD -> FETCH_HEAD
```



## Activity 3:

## Fetch:



zindagikeupat / Zindagikeupat

Unwatch 1 Star 0 Fork 0

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

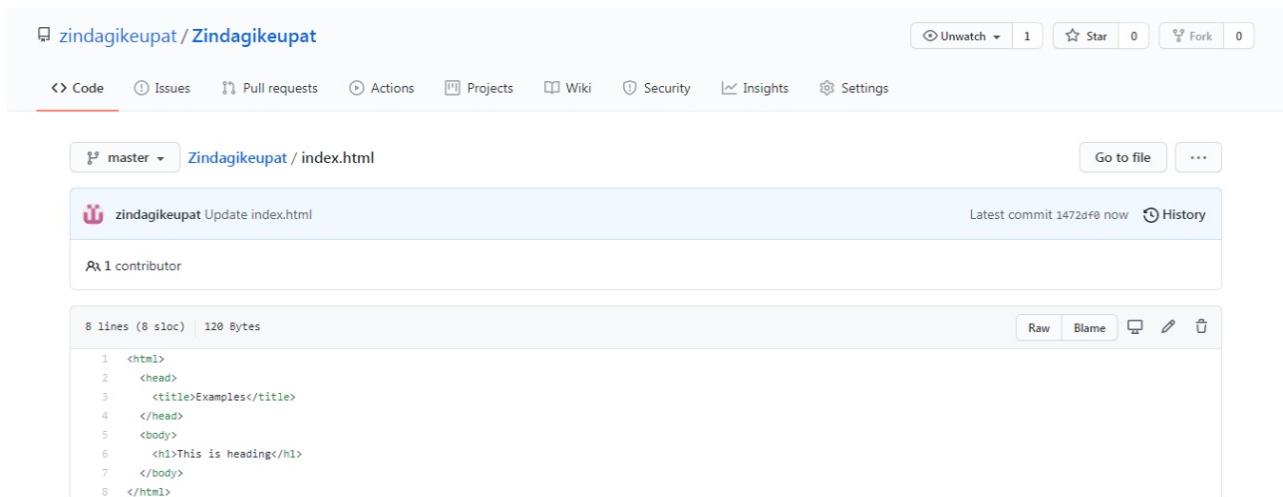
master Zindagikeupat / index.html Go to file

zindagikeupat Update index.html Latest commit 1472df0 now History

1 contributor

8 lines (8 sloc) 120 Bytes Raw Blame

```
1 <html>
2 <head>
3 <title>Examples</title>
4 </head>
5 <body>
6 <h1>This is heading</h1>
7 </body>
8 </html>
```



zindagikeupat / Zindagikeupat

Unwatch 1 Star 0 Fork 0

Code Issues Pull requests Actions Projects Wiki Security Insights Settings

master Zindagikeupat / index.html Go to file

zindagikeupat Update index.html Latest commit 1472df0 now History

1 contributor

8 lines (8 sloc) 120 Bytes Raw Blame

```
1 <html>
2 <head>
3 <title>Examples</title>
4 </head>
5 <body>
6 <h1>This is heading</h1>
7 </body>
8 </html>
```

Before committing changes in index.html

```
Lenovo@Lenovo-PC MINGW64 ~/new-repo (master)
$ git log --oneline example/master
1472df0 (example/master) Update index.html
99c823d (HEAD -> master, login) changed readme.txt
```

```
Lenovo@Lenovo-PC MINGW64 ~/new-repo (master)
$ git log --oneline example/master
1472df0 (example/master) Update index.html
99c823d (HEAD -> master, login) changed readme.txt
```

### 3) Graded Lab Tasks

**Note:** The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.

#### Lab Task 1:

Create new Git repository and upload it on Github using push.

---

## Lab 07

### **Performing Docker commands with push and pull commands**

#### **Objective:**

In this lab, We will go through Docker commands with push and pull commands

#### **Activity Outcomes:**

The activities provide hands - on practice with the following topics

- Git in Cloud
- Docker with GIT
- Push & Pull with Docket

#### **Instructor Note:**

---

# 1. To Build your own Docker Image

## What is DockerHub?

People all over the world create Docker images. You can find these images by browsing the Docker Hub. In this next section, you'll search for and find the image you'll use in the rest of this getting started.

## What is Docker Store?

The Docker Store contains images from individuals like you and official images from organisations like RedHat, IBM, Google, Microsoft, and a whole lot more. Each image repository contains information about an image.

# 2. Solved Lab Activities:

## Activity 1:

### Write a docker file:

```
#mkdir mydockerbuild (create a directory named mydockerbuild)#cd mydockerbuild (go to directory created above)#vi Dockerfile (create a new docker file using vi editor)
```

(copy below mentioned contents into your Dockerfile)

```
FROM docker/whalesay:latest
RUN apt-get -y update && apt-get install -y fortunes
CMD /usr/games/fortune -a | cowsay
```

### 1. Build docker image

```
#docker build -t docker-whale .
```

If you see '**no permission**' error, try to prefix above command with '**sudo**' as shown in above snapshot.

### 2. Now if you can see your latest build using 'docker images' command

```
#docker images
```

### 3. Run your new docker-whale build

```
#docker run docker-whale
```

If you are programmer or in some way related to coding or software world, I assume that you must be aware of [GitHub](#). Similarly, there exists [DockerHub](#).

## Activity 2:

### Create a Docker Hub account and repository

1. Sign up for a docker hub account: In your web browser, go to [the Docker Hub](#)

---

signup page.

2. This account is also valid for docker store.
3. Verify your email address and add a new repository:
4. 4. Learn to 'tag' and 'push' image to your docker hub account.

We will use our docker-whale image to push to docker hub account which you created above.

#### 4.1 Run 'docker images' command to list all images

Third column of above output shows you docker Image ID We

will use this Image ID to tag our docker-whaleimage

4.2 Tag the '**docker-whale**' image using the '**docker tag**' command and the image ID.  
`#docker tag <image-id of docker-whale> <your dockerhub username>/docker-whale:latest`

Now if your use 'docker images' command to list images, you shall see your dockerhub username against docker-whale image as shown in above snapshot.

5. Push your tagged image to Docker Hub, using the 'docker push' command
6. Go back to the Docker Hub website to see the newly-pushed image.
7. Pull your new image from docker hub

#### 7.1 Use 'docker images' to list all your local images

#### 7.2 Let's remove all versions of docker-whale image on our local system.

`#docker rmi -f <Image ID of docker-whale>`

Use 'docker images' command to confirm if all instances of 'docker-whale' has been removed.

---

## 7.3 Now let's run 'docker-whale' and see what happens.

When you use 'docker run' it automatically downloads (pulls) images that don't yet exist locally, creates a container, and starts it. Use the following command to pull and run the 'docker-whale' image, substituting your Docker Hub username.

### Task 3:

**Step 1.** Create a docker Hub account

**Step2:** Build a demo application my-python-app by running the following commands in command prompt

```
C:\python-docker>pip install flask
Collecting flask
 Downloading https://files.pythonhosted.org/packages/f2/28/2a03252dfb9ebf377f40fba6a7841b47083260bf8bd8e737b0c6952df83f/Flask-1.1.2-py2.py3-none-any.whl (94kB)
 100% |#####| 102kB 1.4MB/s
Collecting Werkzeug>=0.15 (from flask)
 Downloading https://files.pythonhosted.org/packages/cc/94/5f7079a0e00bd6863ef8f1da638721e9da21e5bacee597595b318f71d62e/Werkzeug-1.0.1-py2.py3-none-any.whl (298kB)
 100% |#####| 307kB 3.7MB/s
Collecting click>=5.1 (from flask)
 Downloading https://files.pythonhosted.org/packages/d2/3d/fa76db83bf75c4f8d338c2fd15c8d33fdd7ad23a9b5e57eb6c5de26b430e/click-7.1.2-py2.py3-none-any.whl (82kB)
 100% |#####| 92kB 1.3MB/s
Collecting itsdangerous>=0.24 (from flask)
 Downloading https://files.pythonhosted.org/packages/76/ae/44b03b253d6fade317f32c24d100b3b35c2239807046a4c953c7b89fa49e/itsdangerous-1.1.0-py2.py3-none-any.whl
Collecting Jinja2>=2.10.1 (from flask)
 Downloading https://files.pythonhosted.org/packages/7e/c2/1eece8c95ddbc9b1aeb64f5783a9e07a286de42191b7204d67b7496ddf35/Jinja2-2.11.3-py2.py3-none-any.whl (125kB)
 100% |#####| 133kB 3.4MB/s
Collecting MarkupSafe>=0.23 (from Jinja2>=2.10.1->flask)
 Downloading https://files.pythonhosted.org/packages/65/c6/2399700d236d1dd681af8aebff172558cddf6e43d7a5184a675f4711f5/MarkupSafe-1.1.1-cp37-cp37m-win_amd64.whl
Installing collected packages: Werkzeug, click, itsdangerous, MarkupSafe, Jinja2, flask
Successfully installed Jinja2-2.11.3 MarkupSafe-1.1.1 Werkzeug-1.0.1 click-7.1.2 flask-1.1.2 itsdangerous-1.1.0
You are using pip version 18.1, however version 21.1 is available.
You should consider upgrading via the 'python -m pip install --upgrade pip' command.
```

```
C:\python-docker>docker build --tag my-python-app .
[+] Building 74.9s (10/10) FINISHED
=> [internal] load .dockerignore
=> [internal] load build definition from Dockerfile
=> [internal] load dockerfile: 150B
=> [internal] load metadata for docker.io/library/python:alpine3.7
=> [auth] library/python-pull token for registry-1.docker.io
=> [internal] load build context
=> [internal] transferring context: 412B
=> [1/4] FROM docker.io/library/python:alpine3.7@sha256:25f6183ab08f90c727dbcf45373de3b3174a48b4571cc91910bae480dcda847
=> resolve docker.io/library/python:alpine3.7@sha256:25f6183ab08f90c727dbcf45373de3b3174a48b4571cc91910bae480dcda847
=> sha256:01d4f32b0e7a83fcd41201bf4dc400120c83170c11d7daa0d41ba33a0c3f9a00 1.17kB / 1.17kB
=> sha256:33f6f83ab08f90c727dbcf45373de3b3174a48b4571cc91910bae480dcda847 2.04kB / 2.04kB
=> sha256:100ba2573e0479750b17954ba7c310a5f70c23b0f30b72175e27e9e00d6740377e 0.13kB / 0.13kB
=> sha256:6430810450a10d1aa092501f3333722113a0a46d5090c44e6d40c82bb20a07 25.90MB / 25.90MB
=> sha256:40ecb0b270eb081c0d72a50b312de294ec729e1900e92d725f1120c37ce01b 2.12MB / 2.12MB
=> sha256:002729ee08fa0bab4a0a10d0b0ad44e207e5ff3c7e1d579dc3a0c55004d 300.40kB / 300.40kB
=> sha256:3c70e240f76fb19ec575d0547632a0f20b05ee0904cc94fe717d0047ad061159 230B / 230B
=> extracting sha256:40ecb0b270eb081c0d72a50b312de294ec729e1900e92d725f1120c37ce01b
=> sha256:cad3a00ff0ed01a5977edf0f100dec92703a1ae2e91952c0c9f3b54ae51a503 1.81MB / 1.81MB
=> extracting sha256:002729ee08fa0bab4a0a10d0b0ad44e207e5ff3c7e1d579dc3a0c55004d
=> extracting sha256:6430810450a10d1aa092501f3333722113a0a46d5090c44e6d40c82bb20a07
=> extracting sha256:3c70e240f76fb19ec575d0547632a0f20b05ee0904cc94fe717d0047ad061159
=> extracting sha256:cad3a00ff0ed01a5977edf0f100dec92703a1ae2e91952c0c9f3b54ae51a503
=> [2/4] COPY ./app
=> [2/4] WORKDIR /app
=> [3/4] RUN pip install -r requirements.txt
=> exporting to image
=> exporting layers
=> writing image sha256:00150ba5a1a000ba130c30a0bfc70071430f100c7553d130b4401001a719e1
=> naming to docker.io/python-app
```



```
C:\python-docker>docker run --name python-app -p 5000:5000 my-python-app
* Running on http://0.0.0.0:5000/ (Press CTRL+C to quit)
* Restarting with stat
* Serving Flask app "index" (lazy loading)
* Environment: production
 WARNING: This is a development server. Do not use it in a production deployment.
 Use a production WSGI server instead.
* Debug mode: on
* Debugger is active!
* Debugger PIN: 281-336-181
172.17.0.1 - - [10/May/2021 18:10:37] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [10/May/2021 18:10:38] "GET /favicon.ico HTTP/1.1" 404 -
```

Till now we have create the my-python-app file

See if the file is working in the browser using port number



**Step3:** See all the images present in the docker. Here you will see that my-python-app is also present.

```
C:\>docker images
```

| REPOSITORY                              | TAG    | IMAGE ID     | CREATED       | SIZE   |
|-----------------------------------------|--------|--------------|---------------|--------|
| my-python-app                           | latest | 00150bea5a3a | 2 minutes ago | 91.6MB |
| demo-spring-example                     | latest | 5f7f949046b6 | 3 days ago    | 531MB  |
| shwetashakapnor1999/demo-spring-example | latest | 5f7f949046b6 | 3 days ago    | 531MB  |
| alpine/git                              | latest | c99c7d810bc1 | 2 weeks ago   | 25.1MB |
| hello-world                             | latest | d1165f221234 | 2 months ago  | 13.3kB |

**Step4:** Login to the docker hub account

```
C:\>docker login
Authenticating with existing credentials...
Login Succeeded
```

**Step5:** To push the image with your repository name

```
C:\>docker tag my-python-app:latest shwetashakapnor1999/my-python-app
```

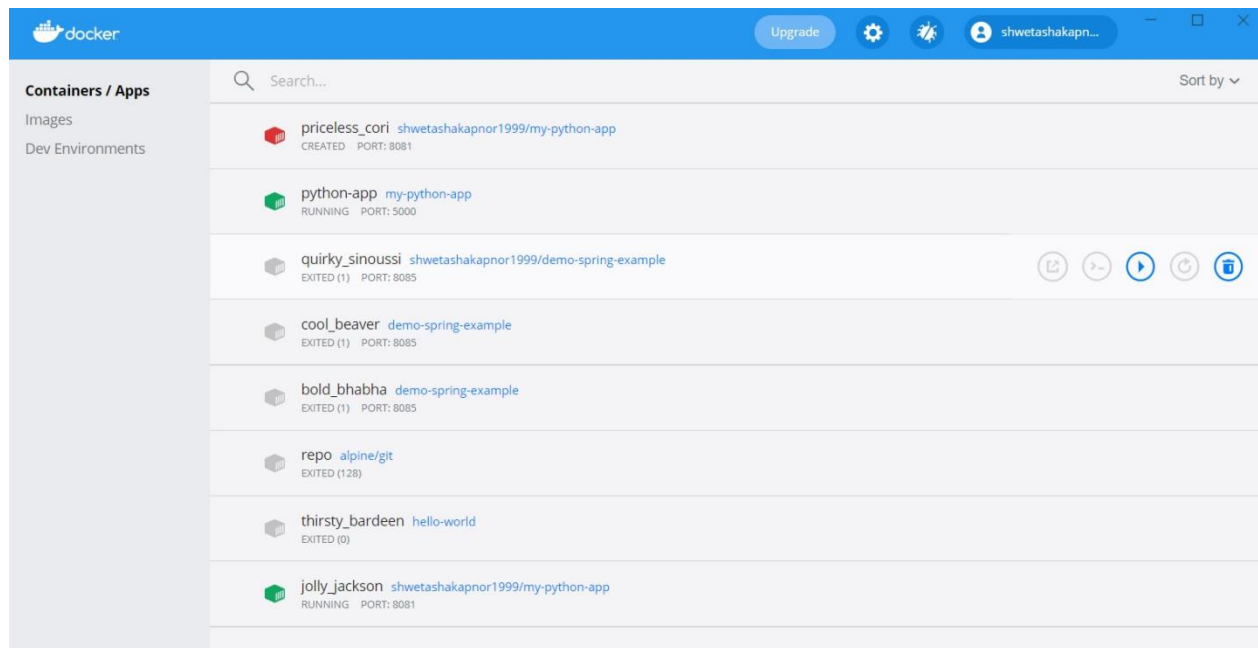
**Step6:** Check the images present, this should include the image name with repository

```
C:\>docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
my-python-app latest 00150bea5a3a 5 minutes ago 91.6MB
shwetashakapnor1999/my-python-app latest 00150bea5a3a 5 minutes ago 91.6MB
demo-spring-example latest 5f7f949046b6 3 days ago 531MB
shwetashakapnor1999/demo-spring-example latest 5f7f949046b6 3 days ago 531MB
alpine/git latest c99c7d810bc1 2 weeks ago 25.1MB
hello-world latest d1165f221234 2 months ago 13.3kB
```

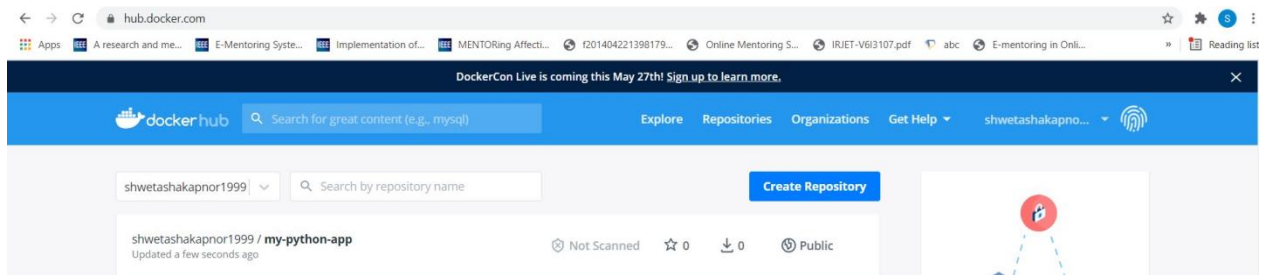
### Step7: Push the image into docker hub account

```
C:\>docker push shwetashakapnor1999/my-python-app
Using default tag: latest
The push refers to repository [docker.io/shwetashakapnor1999/my-python-app]
19593e83d549: Pushed
5f70bf18a086: Pushed
608fd516c32b: Pushed
5fa31f02caa8: Mounted from library/python
88e61e328a3c: Mounted from library/python
9b77965e1d3f: Mounted from library/python
50f8b07e9421: Mounted from library/python
629164d914fc: Mounted from library/python
latest: digest: sha256:06394790b114211f71b2b8b8eec142c4a1e30663cdc5547baf6b0288ea445b54 size: 1992
```

You will see the application generated with the repository name in the docker container



See if build pack demo is listed in docker hub



**Step8:** Check that the images with the repository is now present in the list

```
C:\>docker run -p 8081:5000 shwetashakapnor1999/my-python-app
* Serving Flask app "index" (lazy loading)
* Environment: production
 WARNING: This is a development server. Do not use it in a production deployment.
 Use a production WSGI server instead.
* Debug mode: on
* Running on http://0.0.0.0:5000/ (Press CTRL+C to quit)
* Restarting with stat
* Debugger is active!
* Debugger PIN: 303-727-519
```

**Step9:** Pull the image

```
C:\>docker run -p 8081:5000 shwetashakapnor1999/my-python-app
* Serving Flask app "index" (lazy loading)
* Environment: production
 WARNING: This is a development server. Do not use it in a production deployment.
 Use a production WSGI server instead.
* Debug mode: on
* Running on http://0.0.0.0:5000/ (Press CTRL+C to quit)
* Restarting with stat
* Debugger is active!
* Debugger PIN: 303-727-519
```

Check it in the browser using the port number assigned.



### 3) Graded Lab Tasks

**Note:** The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.

Lab Task 1:

---

Built docker image with the help of python and then learn about the usage of docker image push to share our images to the Docker Hub registry and also how to pull an image.

## **Lab 08**

### **DevOps Tools for CI/CD**

#### **Objective:**

In this lab, We will go through different devops tools such as docker, Jenkins to implement docker images for different operating system.

#### **Activity Outcomes:**

The activities provide hands - on practice with the following topics

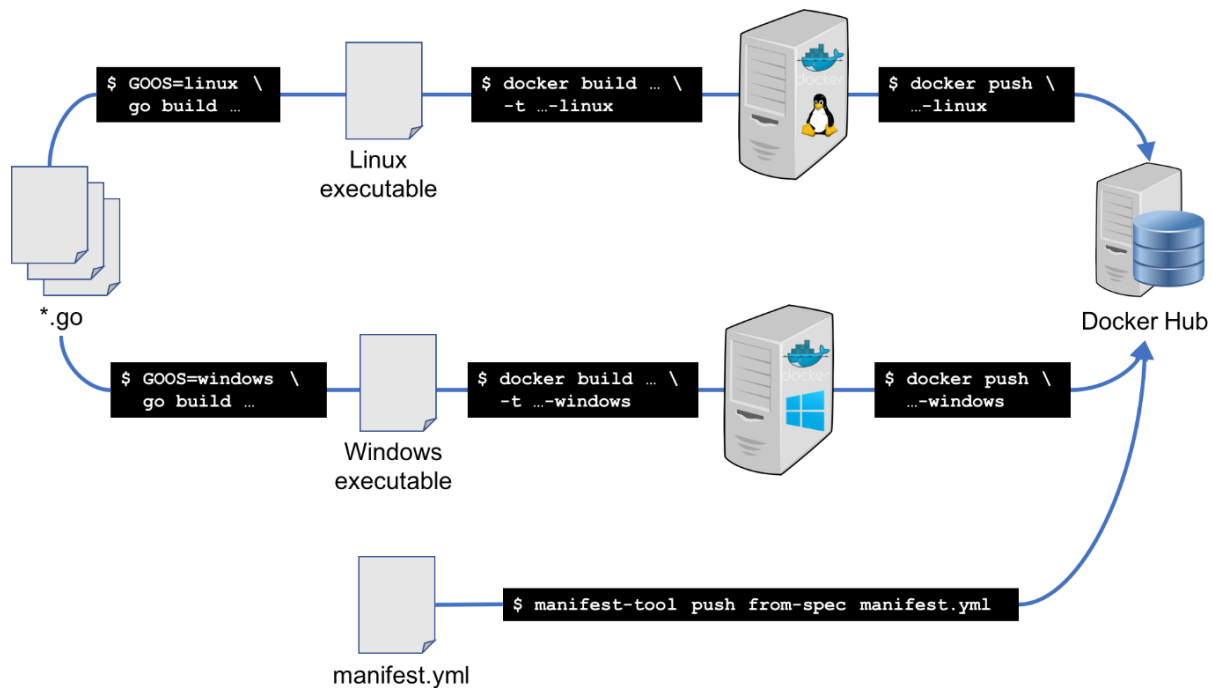
- Containerization
- Orchestration
- DevOps CI/CD Building
- Docker Image building

#### **Instructor Note:**

## 1. Creating Multiple Docker Images:

To create a multi-platform Docker image we need to:

2. Create deployment artifacts (e.g. executable files) for each target platform
3. Create a Dockerfile per target platform
4. Build a Docker image for each target platform
5. Push the target platform specific Docker image to a Docker registry, e.g. DockerHub
6. Create a multi-platform Docker image in the Docker registry based on the individual target platform Docker images



## 2. Solved Lab Activities:

### Activity 1:

#### GET THE CODE

```
git clone https://github.com/callistaenterprise/cadec-2017-service-discovery
```

```
cd cadec-2017-service-discovery\go-quotes\
```

#### BUILD A DOCKER IMAGE FOR WINDOWS

Compile the Go source code to an executable for Windows:

---

```
set GOOS=windows
```

```
go build -o quotes-windows-amd64.exe
```

Try out the Windows executable without using Docker:

```
./quotes-windows-amd64.exe
```

It should startup and say something like:

```
Starting ML Go version of quote-service on port 8080
```

```
2017/12/19 16:23:17 Starting ML HTTP service at 8080
```

Open another “*Windows PowerShell*” terminal window and try out the service using `curl`:

```
curl http://localhost:8080/api/quote -UseBasicParsing
```

**Expected response:**

```
StatusCode : 200
```

```
StatusDescription : OK
```

```
Content
```

```
{"hardwareArchitecture":"amd64","operatingSystem":"windows","ipAddress":"4d131b511ab9/fe80::9846:5be3:c0bb:2d91%Ethernet172.24.224.172","quote":"In Go, the code does exactly what it says on the page ."}...
```

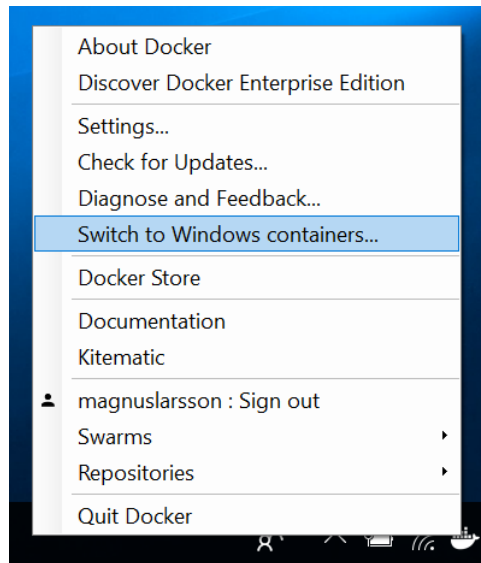
Stop the program with `CTRL/C`.

Ensure that your “Docker for Windows” runs Windows containers. enter the command:

```
docker info
```

Look for the field `OSType`, it should say `windows`.

If not, switch to Windows containers using the Docker menu:



The Dockerfile for building the Windows Docker image, Dockerfile-windows-amd64, looks like:

```
FROM microsoft/nanoserver
```

```
EXPOSE 8080
```

```
ADD quotes-windows-amd64.exe /
```

```
ENTRYPOINT ["/quotes-windows-amd64.exe"]
```

## Activity 2:

### Build the Docker image and push it to DockerHub:

```
docker build -f Dockerfile-windows-amd64 -t magnuslarsson/quotes:24-go-windows-amd64 .
```

```
docker push magnuslarsson/quotes:24-go-windows-amd64
```

You can start the Windows specific Docker image with:

```
docker run -d -p 8080:8080 --name quotes magnuslarsson/quotes:24-go-windows-amd64
```

The current version of “Docker for Windows” has a limitation that prevents access ports published by containers using localhost, e.g. `curl http://localhost:8080/api/quote` does not work. For details, see <https://blog.sixeyed.com/published-ports-on-windows-containers-dont-do-loopback/>.

Instead, we can use the PC’s IP address. You can use `ipconfig` to get the IP address.

---

E.g.:

```
curl http://192.168.1.224:8080/api/quote -UseBasicParsing
```

Expect a similar response as from the `curl` command above. Verify that `operatingSystem` field has the value `windows`!

Stop the Windows container with the command:

```
docker rm -f quotes
```

### Activity 3:

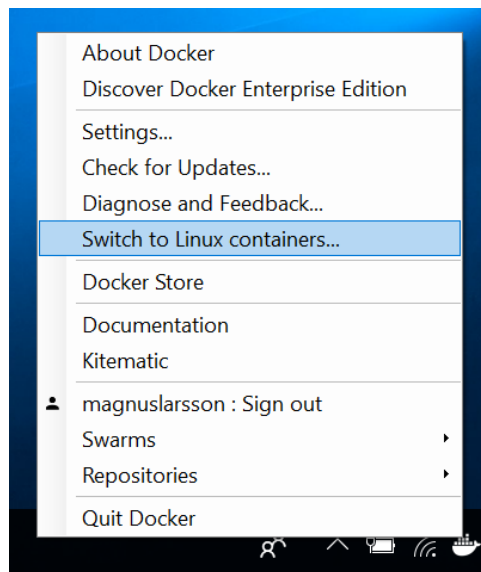
#### BUILD A DOCKER IMAGE FOR LINUX

Compile the Go source to an executable for Linux:

```
set GOOS=linux
```

```
go build -o quotes-linux-amd64
```

Switch to Linux containers using the Docker menu:



The Dockerfile for building the Linux Docker image, `Dockerfile-linux-amd64`, looks like:

```
FROM scratch
```

```
EXPOSE 8080
```

```
ADD quotes-linux-amd64 /
```



---

```
ENTRYPOINT ["/quotes-linux-amd64"]
```

#### **Activity 4:**

##### **Build the Docker image and push it to DockerHub:**

```
docker build -f Dockerfile-linux-amd64 -t magnuslarsson/quotes:24-go-linux-amd64 .
```

```
docker push magnuslarsson/quotes:24-go-linux-amd64
```

You can start the Linux specific Docker image with:

```
docker run -d -p 8080:8080 --name quotes magnuslarsson/quotes:24-go-linux-amd64
```

Now, try out the service using curl:

```
curl http://localhost:8080/api/quote -UseBasicParsing
```

##### **Expected response:**

```
StatusCodes : 200
```

```
StatusDescription : OK
```

```
Content
```

```
:{"hardwareArchitecture":"amd64","operatingSystem":"linux","ipAddress":"0c4e0824f479/172.17.0.2","quote":"I like a lot of the design decisions they made in the [Go] language. Basically, I like all oft...
```

Verify that the operatingSystem field now has the value linux!

Stop the Linux container with the command:

```
docker rm -f quotes
```

#### **Activity 5:**

##### **BUILD A MULTI-PLATFORM DOCKER IMAGE**

Now, it's time to combine the two platform specific Docker images into one common Docker image.

We will use the standalone tool `manifest-tool`. Executables can be downloaded from: <https://github.com/estesp/manifest-tool/releases>.

---

I used version v0.7.0 of the tool compiled for macOS (I got stuck on some strange authentication problem when trying out the Windows version).

The manifest file, `manifest-quotes-multi-platform.yml`, looks like:

```
image: magnuslarsson/quotes:24-go
manifests:
-
 image: magnuslarsson/quotes:24-go-linux-amd64
 platform:
 architecture: amd64
 os: linux
-
 image: magnuslarsson/quotes:24-go-windows-amd64
 platform:
 architecture: amd64
 os: windows
```

The multi-platform Docker image is create with the command:

```
./manifest-tool-darwin-amd64 --username=magnuslarsson --password=xxx push from-spec
manifest-quotes-multi-platform.yml > **Note:** `./manifest-tool-darwin-amd64` requires
macOS, you can however try out the Windows version (or a Linux version) on your own.
```

Verify that we now have a Docker image for our Go service that supports both Linux and Windows:

```
docker run --rm mplatform/mquery magnuslarsson/quotes:24-go
```

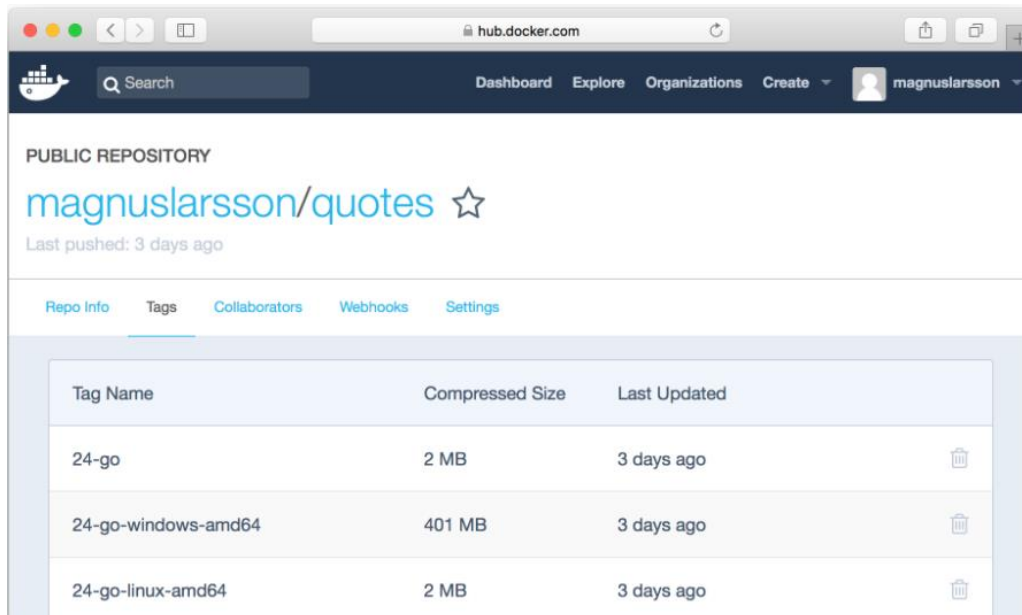
**Expected response:**

```
Image: magnuslarsson/quotes:24-go
* Manifest List: Yes
* Supported platforms:
- linux/amd64
- windows/amd64:10.0.14393.1944
```

---

You can also take a look into DockerHub to see the resulting three Docker images, e.g. in my case: <https://hub.docker.com/r/magnuslarsson/quotes/tags/>

### Expected result:



### TRY OUT THE MULTI-PLATFORM DOCKER IMAGE

Now we should be able to run our Go service in both Windows and Linux containers using one and the same Docker image: `magnuslarsson/quotes:24-go`!

Since we currently have “Docker for Windows” configured to run Linux containers, let’s start with trying it out on Linux:

```
docker run -d -p 8080:8080 --name quotes magnuslarsson/quotes:24-go
```

Test it using curl:

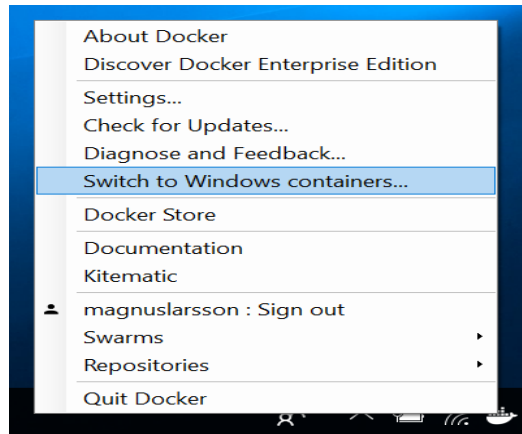
```
curl http://localhost:8080/api/quote -UseBasicParsing
```

Verify that the `operatingSystem` field in the response has the value `linux`!

Kill the Linux container:

```
docker rm -f quotes
```

Switch “Docker for Windows” to use Windows containers:



Start a Windows container:

```
docker run -d -p 8080:8080 --name quotes magnuslarsson/quotes:24-go
```

Test it using curl:

```
curl http://192.168.1.224:8080/api/quote -UseBasicParsing
```

Verify that the `operatingSystem` field in the response now has the value `windows`!

Wrap up with killing the Windows container:

```
docker rm -f quotes
```

### 3) Graded Lab Tasks

*Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.*

Lab Task 1:

we have learned how to create a Docker image that works on different hardware architectures and operating systems.

Lab Task 2:

Now create a Docker image based on a service written in Go and package it for use in both Linux and Windows on 64 bit Intel based hardware.

---

## Lab 09

### DevOps Planning: Agile Project Management in Azure

#### Objective:

In this lab, We will go through Azure Platform for agile project management

#### Activity Outcomes:

The activities provide hands - on practice with the following topics

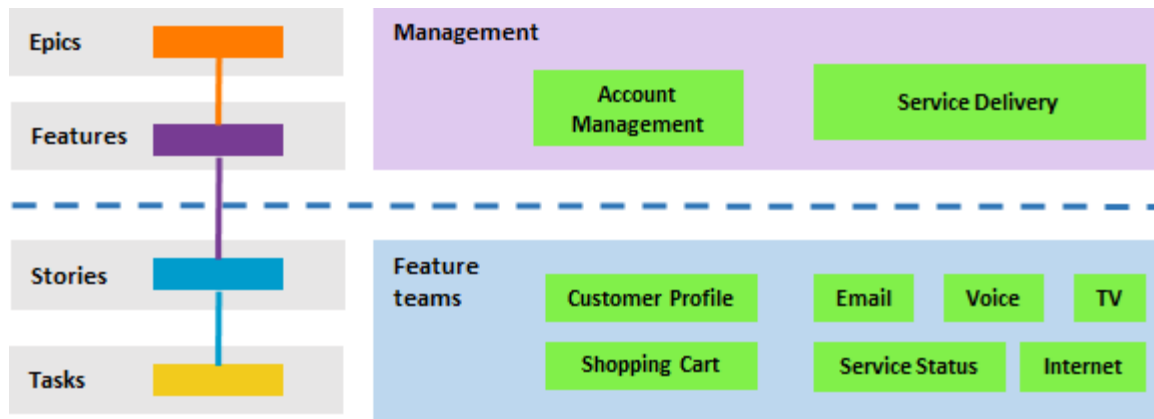
- Agile Project Management
- Cloud DevOps Planning

#### Instructor Note:

---

## 1. Agile Project Management:

Work items play a prominent role in Azure DevOps. Whether describing work to be done, impediments to release, test definitions, or other key items, work items are the workhorse of modern projects. In this task you'll focus on using various work items to set up the plan to extend the Parts Unlimited site with a product training section. While it can be daunting to build out such a substantial part of a company's offering, Azure DevOps and the Scrum process make it very manageable.

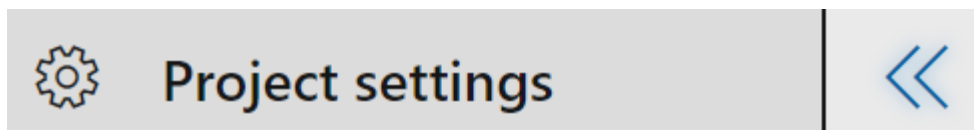


## 2. Solved Lab Activities

### Activity 1:

#### Task 1: Working with teams, areas, and iterations

1. Navigate to your Parts Unlimited project on Azure DevOps. It will be something like <https://dev.azure.com/YOURACCOUNT/Parts%20Unlimited>.
2. Open the settings page using the **Project settings** navigation located at the bottom left of the page.



3. Select the **Teams** tab. There are already a few teams in this project, but you'll make a new one for this lab. Click **New team**.

**Project Settings**  
PartsUnlimited

**General**

- Overview
- Teams**
- Permissions
- Notifications
- Service hooks
- Dashboards

**Boards**

- Project configuration
- Team configuration

**Teams**

Filter teams

Total 3

| Name ↓                        | Description               | Members |
|-------------------------------|---------------------------|---------|
| PT PartsUnlimited_... Default | The default project team. | 1       |
| P PUL                         | Parts Unlimited Web Team  | 0       |
| P PUL-DB                      | PUL DB Team               | 0       |

New Team

- Use **"PUL-Web"** as the **Team name** and click **Create team** (Leave defaults).

**Create a new team**



**Name**

PUL-Web

**Members**

 Add members

**Description**

Add a description to your team this will appear in the team page

Cancel Create

- Select the newly created team to view its details.

| Name ↓                                                                                                                             | Description               | Members                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------|
|  <b>Parts Unlimited Team</b> <span>Default</span> | The default project team. |  1 |
|  <b>PUL</b>                                       | Parts Unlimited Web Team  |  0 |
|  <b>PUL-DB</b>                                    | PUL DB Team               |  0 |
|  <b>PUL-Web</b>                                   |                           |  1 |

- By default, the new team has only you as its member. You can use this view to manage membership, notifications, dashboards, and more. But first you will want to define the schedule and scope of the team. Click **Iterations and Area Paths**.


**PUL-Web**  
 Relevant links: [Notifications](#) | [Dashboards](#) | [Iterations and Area Paths](#)

[Members](#) [Settings](#)

Total 1

☐ Name Type Username or scope

☐  Michelle Admin aad user

[Direct Members](#) [Add](#)

- Select the **Iterations** tab and click **Select iterations**. This team will use the same iteration schedule that's already in place for the other teams, although you can take a different route if that's better for your organization.

Boards
 

This project is currently using the Customized Scrum process. To customize your work item types, [go to the process customization page](#).

[General](#) [Iterations](#) [Areas](#) [Templates](#)

**Iterations**  
 Teams plan and track work at regular time intervals, commonly referred to as an iteration or sprint. Here you can manage the iterations for your team. [Learn more about customizing areas and iterations](#)

**Default iteration** ⓘ  
 @CurrentIteration [Change](#)

**Backlog iteration** ⓘ  
 PartsUnlimited [Change](#)

The iterations selected below will appear in the Sprints hub for your team. Each iteration will have a backlog and board to help your team plan and manage work.

[+ Select iteration\(s\)](#)
[X Remove](#)
[New](#)
[New child](#)



8. Select **Parts Unlimited\Sprint 1** and click **Save and close**. Note that this first sprint has already passed. This is because the demo data generator is designed to build out project history so that this sprint occurs in the past.

## Select iteration(s)

These iterations will be available on your team's backlog

+ Iteration

[Learn more](#)

Save and close

Cancel

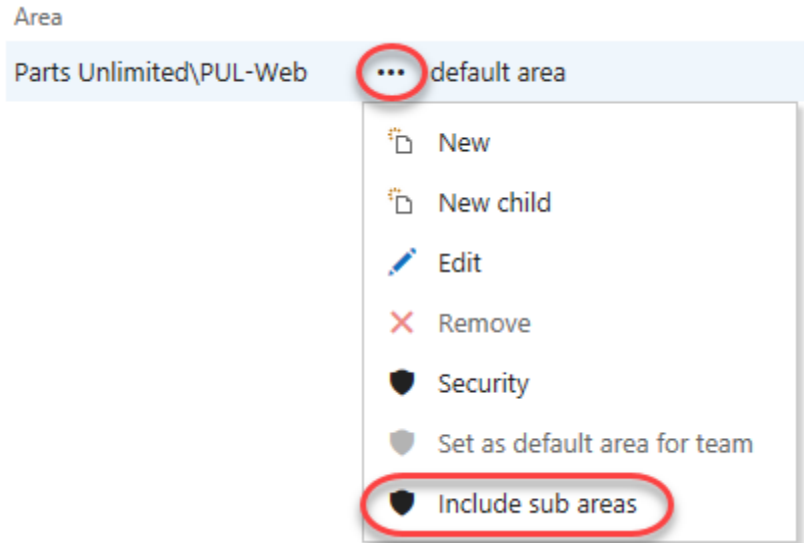
9. Repeat the process to add **Sprint 2** and **Sprint 3** (add them as in previous step). The second sprint is our current iteration, and the third is in the near future.

| + Select iteration(s)   ✕ Remove       New   New child |            |           |
|--------------------------------------------------------|------------|-----------|
| Iteration                                              | Start Date | End Date  |
| Parts Unlimited\Sprint 1                               | 6/30/2019  | 7/21/2019 |
| Parts Unlimited\Sprint 2                               | 7/22/2019  | 8/12/2019 |
| Parts Unlimited\Sprint 3                               | 8/13/2019  | 9/3/2019  |

10. Select the **Areas** tab. By default, there is an area matching the name of the team.

General   Iterations   **Areas**   Templates

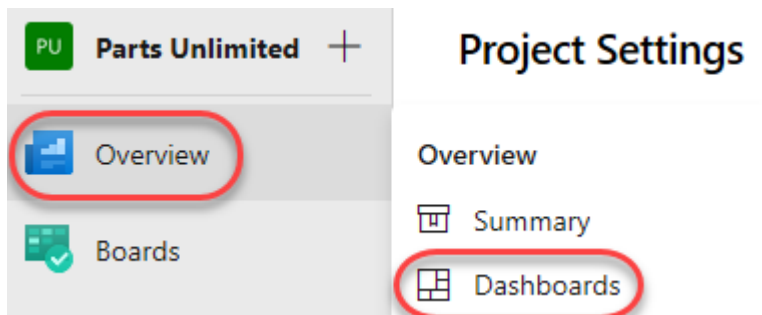
11. From the area dropdown, select **Include sub areas**. The default setting for all teams is to exclude sub-area paths. We will change it to include sub-areas so that the team gets visibility into all of the work items from all teams. Optionally, the management team could also choose to not include sub-areas, thereby removing work items from their view as soon as they are assigned to one of the teams.



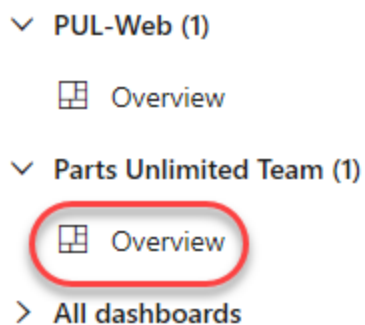
## Activity 2: Working with work items

This task is designed to illustrate a variety of ways you can create different kinds of work items, as well as to demonstrate the breadth of features available on the platform. As a result, these steps should not be viewed as prescriptive guidance for project management. The features are intended to be flexible enough to fit your process needs, so explore and experiment as you go.

1. Navigate to **Overview | Dashboards**.



2. Select the **Overview** dashboard for **Parts Unlimited Team**.



- There are many ways to create work items in Azure DevOps, and we'll explore a few of them. Sometimes it's as simple as firing one off from a dashboard. In the **New Work Item** form, type "**Product training**" and select the **Epic** type. Click **Create**.

The screenshot shows the Azure DevOps dashboard for the 'PartsUnlimited' team. The 'Dashboards' menu is open, and the 'New Work Item' form is visible. The form has a red border and contains the following elements:

- A text input field with the value 'Product training'.
- A dropdown menu with the value 'Epic'.
- A blue 'Create' button.

The dashboard also displays several charts and metrics, including 'User Stories by Assigned To', 'User Stories by State', and 'Average burndown'.

- Assign the new work item to yourself and set the **Area** to **Parts Unlimited\PUL-Web**. Set the **Iteration** to **Parts Unlimited\Sprint 2** and click **Save & Close**. Ordinarily you would want to fill out as much information as possible, but you can run lean here for the purposes of this lab.

NEW EPIC \*

Product training

0 comments Add tag

Save & Close

Statg ● New Area Parts Unlimited\PUL-Web  
Reason New epic Iteration Parts Unlimited\Sprint 2

Details LaunchDarkly

**Description**

B / U A [Link] [Image] [List] [Table] [Code]

**Acceptance Criteria**

B / U A [Link] [Image] [List] [Table] [Code]

**Discussion**

EC Add a comment. Use # to link a work item, ! to link a pull request, or @ to mention a person.

**Status**

Target Date

**Details**

Priority 2

Effort

Business Value

Time Criticality

Value area Business

**Development**

+ Add link

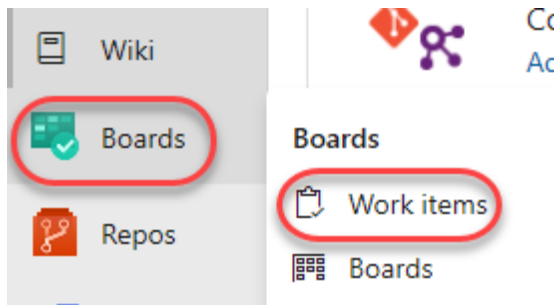
Development hasn't started on this item.

**Related Work**

+ Add link v

There are no links in this group.

5. Navigate to **Boards | Work Items**.



6. Locate the newly created epic for **Product training** and open it.

Work Items

Assigned to me v | + New Work Item v | Open in Queries | Column Options | Recycle Bin

Filter by keyword | Types v | States v | Area v

| ID    | Title                                                                 | State         | Area Path               |
|-------|-----------------------------------------------------------------------|---------------|-------------------------|
| 26637 | <b>Product training</b>                                               | ● New         | Parts Unlimited\PUL-Web |
| 26625 | Parts Unlimited_TestPlan1                                             | ● In Progress | Parts Unlimited         |
| 26636 | 26490 : As a customer, I should be able to put items to shopping c... | ● In Progress | Parts Unlimited         |

7. The work item form includes everything you could ever want to know about a work item. This includes details about who it's assigned to, its status across many parameters, and all the associated

---

information and history for how it has been handled since creation. One of the key areas to focus on is the **Related Work**. One of the ways to add a feature to this epic is to select **Add link | New item**.

### Status

Target Date

### Details

Priority

2

Effort

Business Value

### Development

+ Add link

Development hasn't started on this it

[Create a new branch](#)

### Related Work

+ Add link ▾

🔗 Existing item

📋 New item

8. Set the **Work item type** to **Feature** and set the **Title** to “**Training dashboard**”. Click **OK**.

## Add link

You are adding a link from:

   26637 Product training  
Updated 21 minutes ago, ● New

Link type

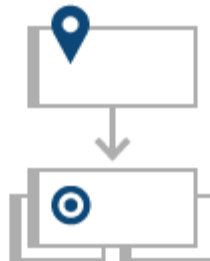
Child

 Work item type

Feature

Title

Training dashboard

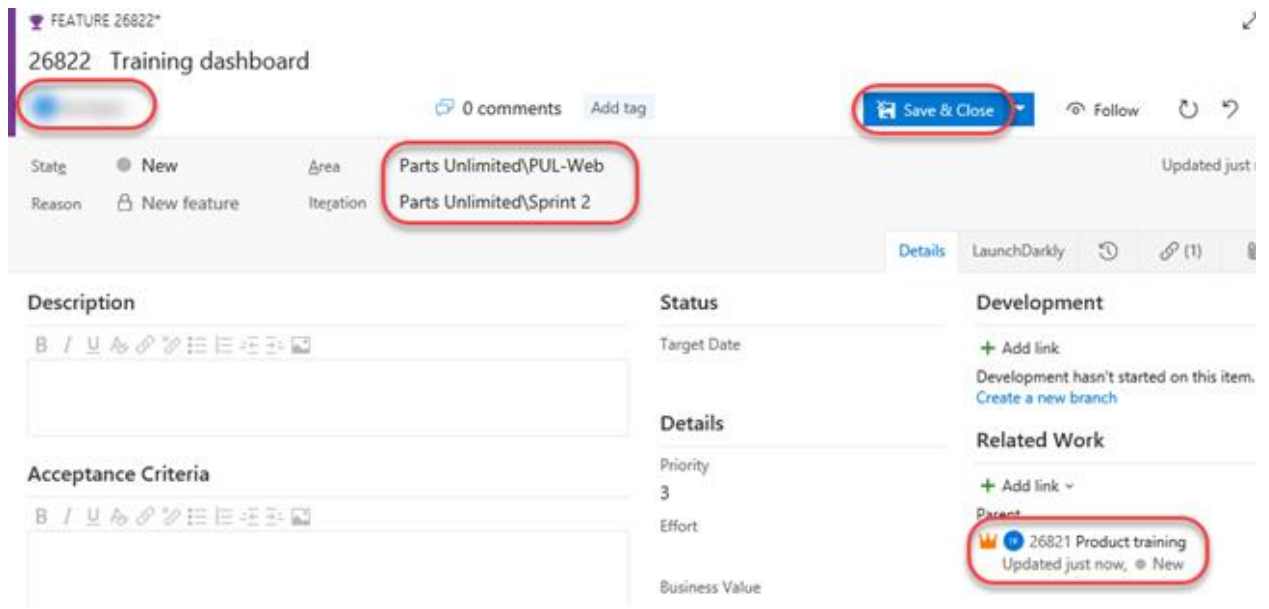


Comment

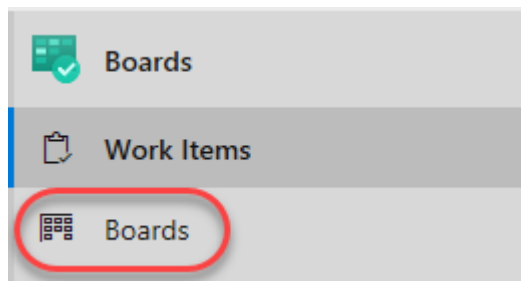
OK

Cancel

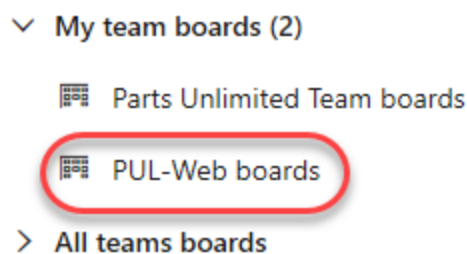
9. That **Assignment**, **Area**, and **Iteration** should already set to the same as the epic, and it's even linked to the parent item it was created from. Click **Save & Close**.



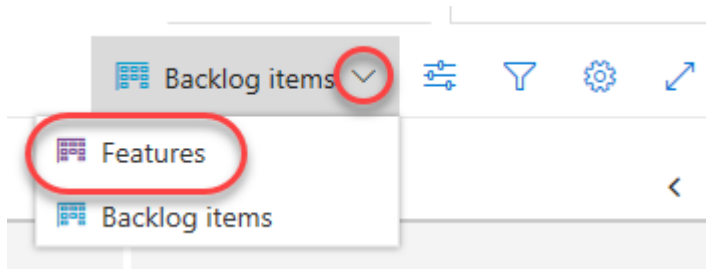
10. Navigate to the **Boards** view.



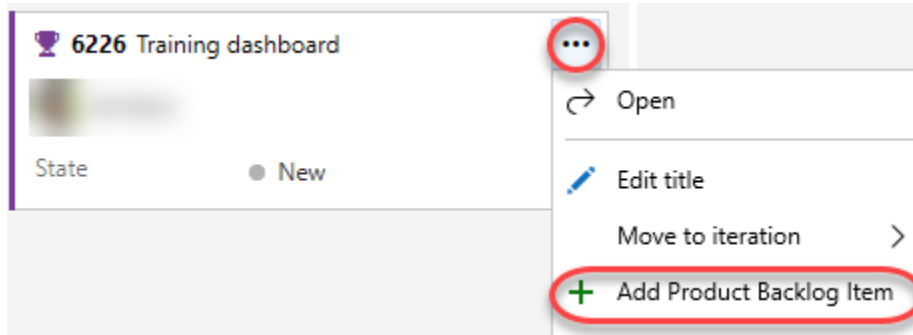
11. Select **PUL-Web Boards**. This will open the board for that particular team.



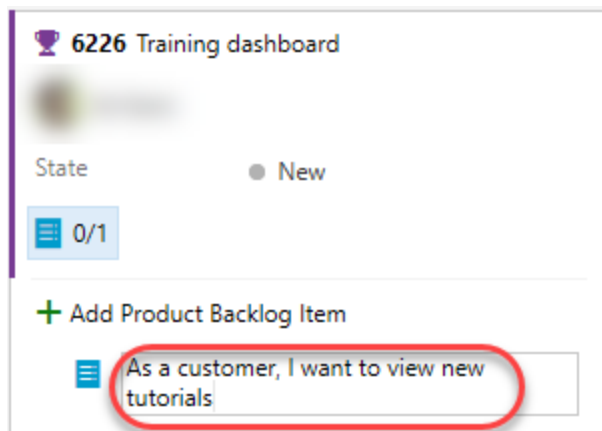
12. Switch the board from showing **Backlog items** to showing **Features**. This will make it easy to add tasks and other work items to the features.



13. From the **Training dashboard** dropdown, select **Add Product Backlog Item**.

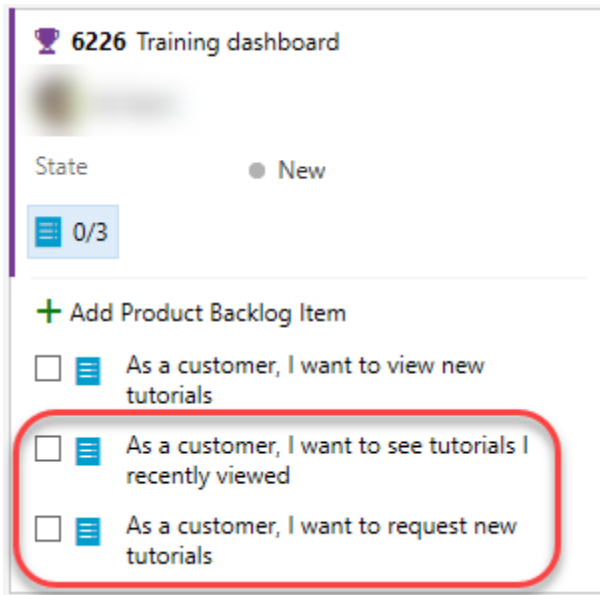


14. Name the first backlog item “**As a customer, I want to view new tutorials**” and press **Enter** to save. This creates a new **Product Backlog Item** (PBI) work item that is a child of the feature and shares its area and iteration.

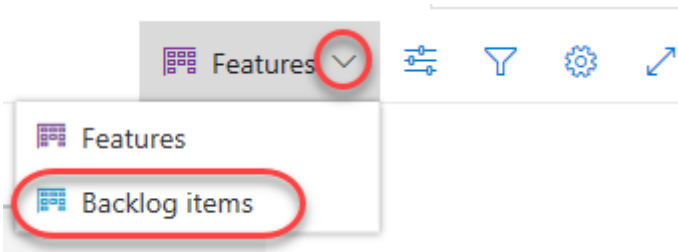


15. Add two more PBIs designed to enable the customer to see their recently viewed tutorials and to request new tutorials.

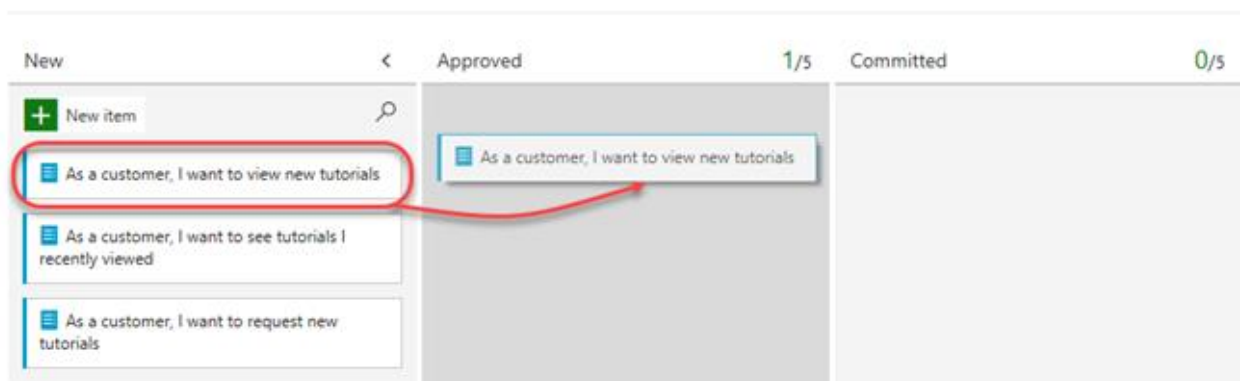




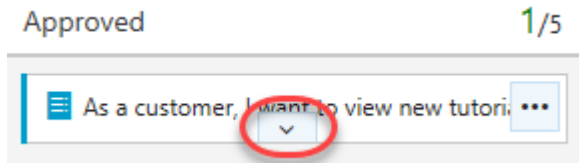
16. Switch the task board view back to **Backlog items**.



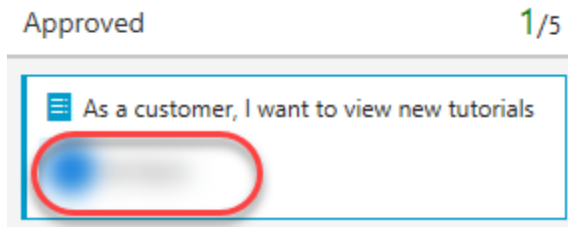
17. Backlog items have a state that defines where they are relative to being completed. While you could open and edit the work item using the form, it's easier to just drag cards on the board. Drag the first work item to **Approved**.



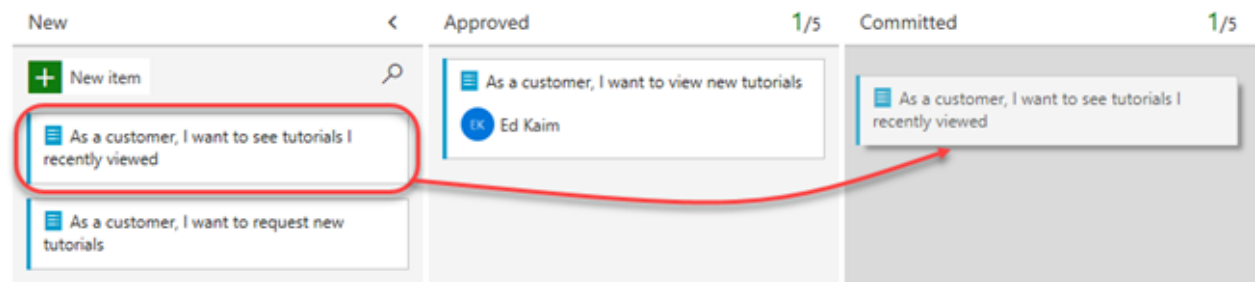
18. You can also expand work item cards to get to conveniently editable details.



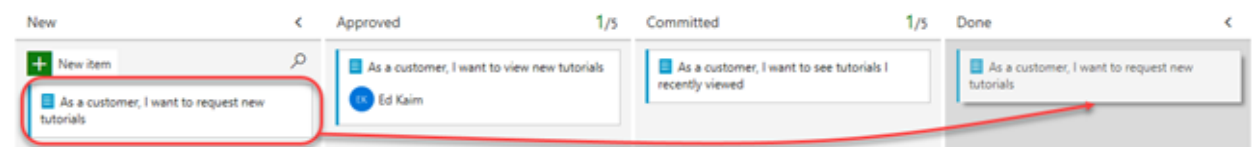
19. Assign the moved PBI to yourself.



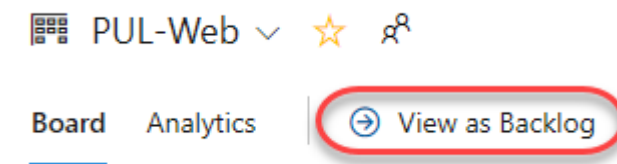
20. Drag the second work item to the **Committed** stage.



21. Drag the final PBI to the **Done** stage.



22. The task board is one view into the backlog. View the tabular form by clicking **View as Backlog**.



23. Click the **Expand** button, which allows you to view nested tasks under these work items. Another easy way to create work items is using the **Add** button on the backlog. Click it to add a new task to the first backlog item.

|  | Order | Work Item Type   | Title                                       |
|-----------------------------------------------------------------------------------|-------|------------------|---------------------------------------------|
|  | 1     | Product Backl... | As a customer, I want to view new tutorials |
|                                                                                   | 2     | Product Backl... | As a customer, I want to second thing       |

24. Set the **Title** to “Add page for most recent tutorials”. Set the **Remaining Work** to “5” and the **Activity** to “Development”. Click **Save & Close**.

**NEW TASK \***

Add page for most recent tutorials

Unassigned 0 comments Add tag

Save & Close

State: To Do Area: Parts Unlimited\PUL-Web Reason: New task Iteration: Parts Unlimited\Sprint 2

**Description**

**Details**

Priority: 2

Remaining Work: 5

Activity: Development

**Development**

+ Add link

Development hasn't started c

**Related Work**

+ Add link v

25. Add another task to “Optimize data query for most recent tutorials”. Set its **Remaining Work** to “3” and its **Activity** to “Design”. Click **Save & Close**.

**NEW TASK \***

Optimize data query for most recent tutorials

Unassigned 0 comments Add tag

Save & Close

State: To Do Area: Parts Unlimited\PUL-Web Reason: New task Iteration: Parts Unlimited\Sprint 2

**Description**

**Details**

Priority: 2

Remaining Work: 3

Activity: Design

**Development**

+ Add link

Development hasn't started i

**Related Work**

+ Add link v

### Activity 3: Managing sprints and capacity

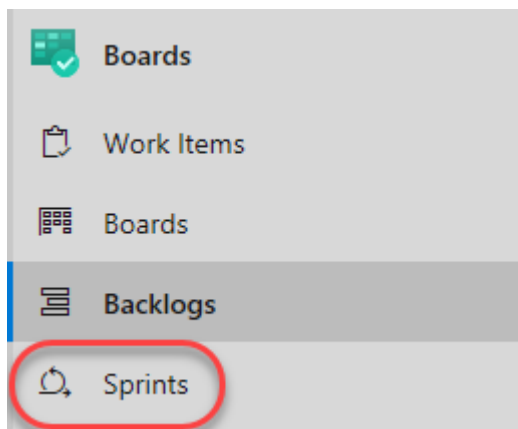
Your team builds the sprint backlog during the sprint planning meeting, typically held on the first day of the sprint. Each sprint corresponds to a time-boxed interval which supports your team’s ability to work using

---

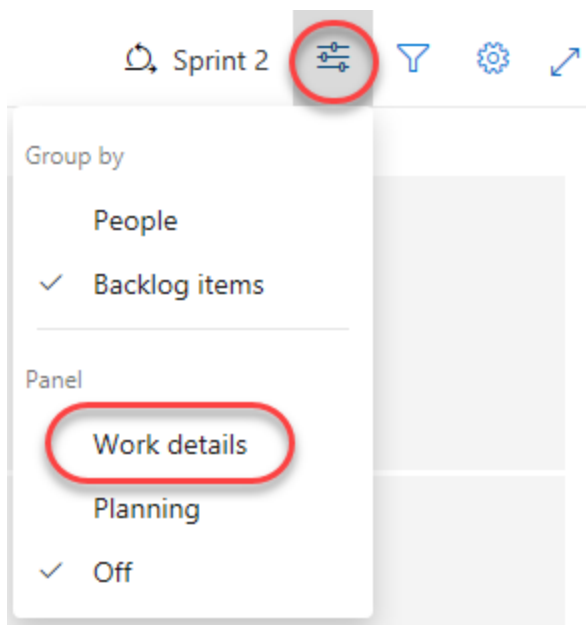
Agile processes and tools. During the planning meeting, your product owner works with your team to identify those stories or backlog items to complete in the sprint.

Planning meetings typically consist of two parts. In the first part, the team and product owner identify the backlog items that the team feels it can commit to completing in the sprint, based on experience with previous sprints. These items get added to the sprint backlog. In the second part, your team determines how it will develop and test each item. They then define and estimate the tasks required to complete each item. Finally, your team commits to implementing some or all the items based on these estimates.

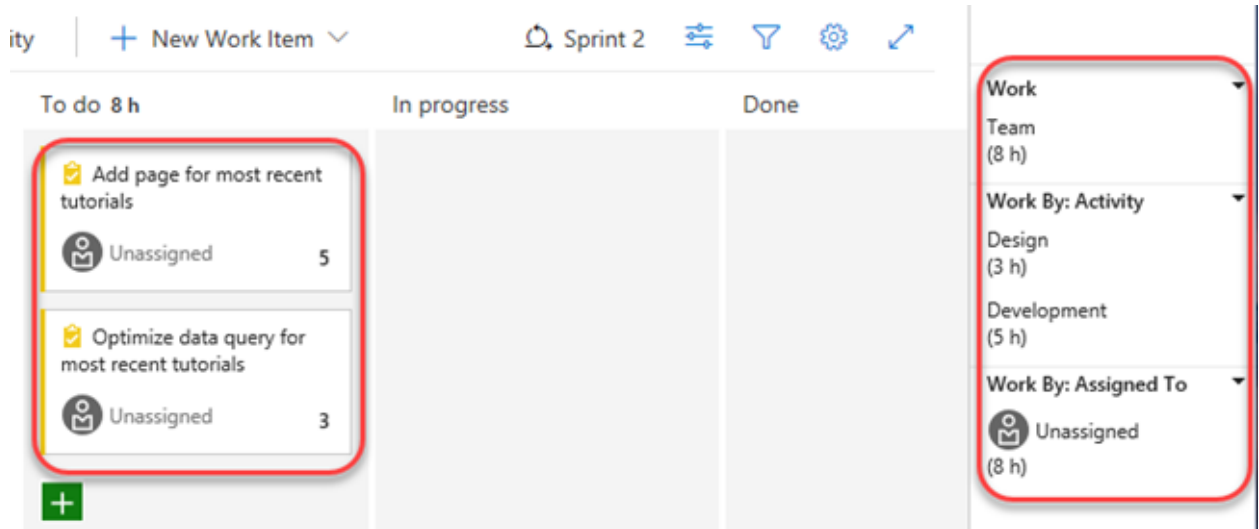
1. Your sprint backlog should contain all the information your team needs to successfully plan and complete work within the time allotted without having to rush at the end. Before you start planning your sprint, you'll want to have created, prioritized, and estimated your backlog and defined your sprints. Navigate to the **Sprints** view using the navigation.



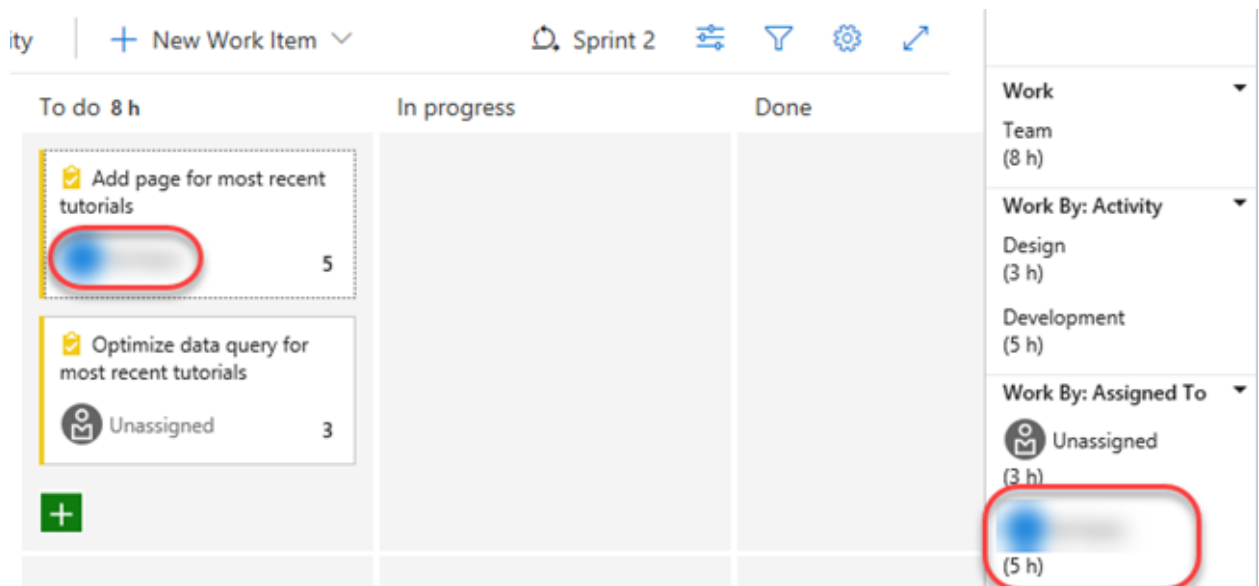
2. From the **View options** dropdown, select the **Work details** panel option.



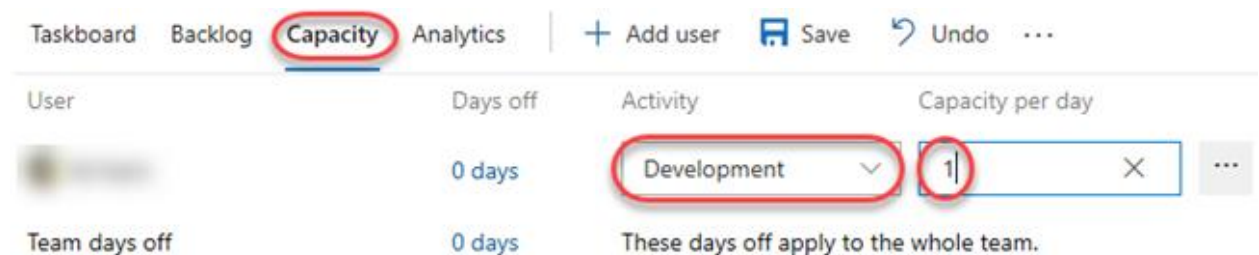
3. The current sprint has a pretty limited scope. There are two tasks in the **To do** stage that combine for 8 hours of estimated work. At this point, neither task has been assigned.



- Assign the **Add page** task to yourself. Note that this updates the capacity view.



- Select the **Capacity** tab. This view enables you to define what activities a user can take on and at what level of capacity. In this case, set your capacity to allow “1” hour of **Development** per day. Note that you can add additional activities per user in the case they do more than just development.



- However, let's assume you're going to take some vacation. Click **0 days** under **Days off**.

---

Days off

0 days

7. Set your vacation to span five work days during the current sprint (within the next few weeks). Click **OK**.

| Start     | End       | Days off |
|-----------|-----------|----------|
| 6/25/2018 | 6/29/2018 | 5        |

+ Add additional days off

**Total 5**

**OK** Cancel

8. Click **Save**.

+ Add user **Save** Undo ...

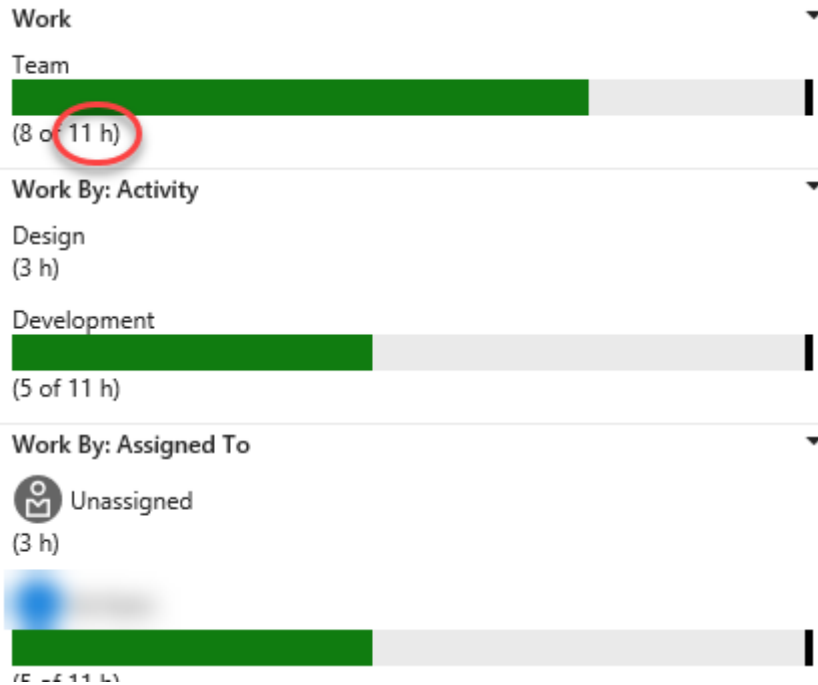
| Days off | Activity    | Capacity per day |
|----------|-------------|------------------|
| 5 days   | Development | 1                |

0 days These days off apply to the whole team.

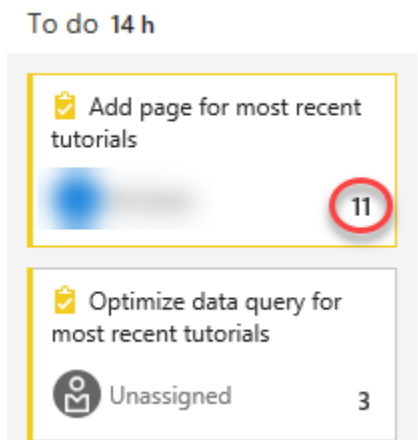
9. Return to the **Taskboard**.

**Taskboard** Backlog Capacity

10. Note that the capacity view has been updated to reflect your available bandwidth. This exact number may vary, but for the screenshots here, that sprint capacity is 11 hours (1 hour per day over 11 working days).



11. One convenient feature of the boards is that you can easily update key data in-line. It's a good practice to regularly update the **Remaining Work** estimate to reflect the amount of time expected for each task. Let's say you've reviewed the work for the **Add page** task and found that it will actually take longer than originally expected. Set it to whatever your total capacity is for this sprint.



12. Note how this expands the **Development** and your personal capacities to their maximum. Since they're large enough to cover the assigned tasks, they stay green. However, the overall **Team** capacity is exceeded due to the additional 3 hours required by the other task.



13. One way to resolve this capacity issue would be to move the task to a future iteration. There are a few ways this could be done. First, you could open the task here and edit it in the dialog. The **Backlog** view, on the other hand, provides an in-line menu option to move it. Don't move it now.

Taskboard Backlog Capacity + New Work Item ...

| Order | Work Item Type   | Title                                                    | State    | Effort | Value Area | Iteration Path           |
|-------|------------------|----------------------------------------------------------|----------|--------|------------|--------------------------|
| 1     | Product Backl... | As a customer, I want to view new tutorials              | Approved |        | Business   | Parts Unlimited\Sprint 2 |
|       | Task             | Add page for most recent tutorials                       | To Do    |        |            | Parts Unlimited\Sprint 2 |
|       | Task             | Optimize data query for most recent tutorials            | To Do    |        |            | Parts Unlimited\Sprint 2 |
| 2     | Product Backl... | As a customer, I want to see tutorials I recently viewed |          |        | business   | Parts Unlimited\Sprint 2 |
| 3     | Product Backl... | As a customer, I want to request new tutorials           |          |        | business   | Parts Unlimited\Sprint 2 |

Menu options for the selected task:

- Edit...
- Assign to
- Copy to clipboard
- Delete
- Templates
- Add link
- Move to iteration
- Move to top
- Change type...
- Move to team project...

Right sidebar:

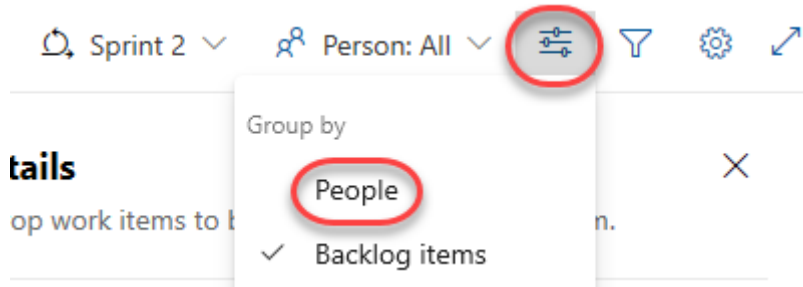
- Backlog
- Current (Sprint 2 - 6/18/2018 - 7/9/2018)
- FUTURE ITERATIONS
- Sprint 3 - 7/10/2018 - 7/31/2018

14. Return to the **Taskboard** view.

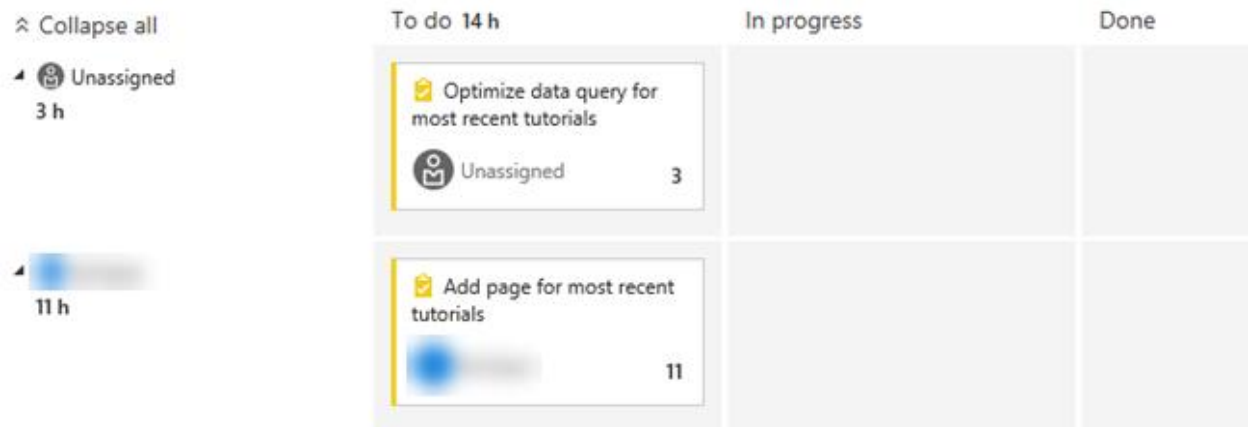




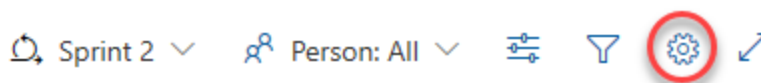
15. Select **People** from the **View options** dropdown.



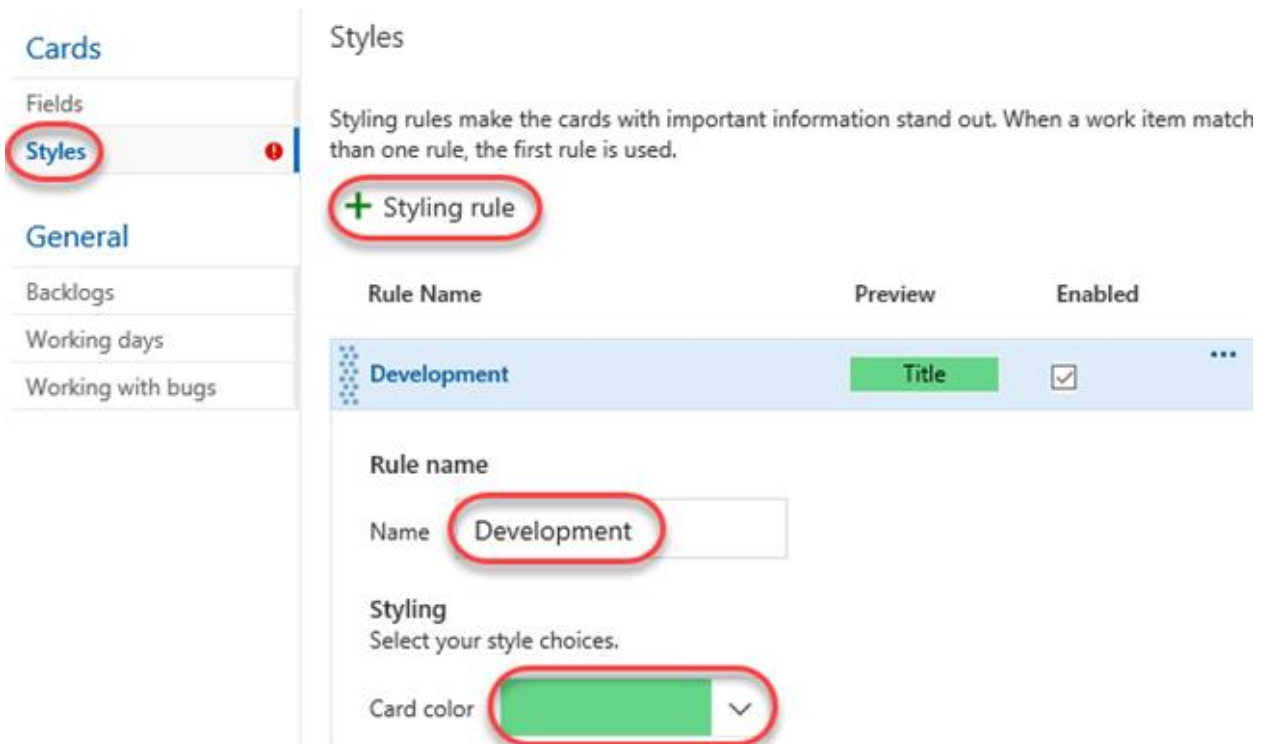
16. This adjusts your view such that you can review the progress of tasks by person instead of by backlog item.



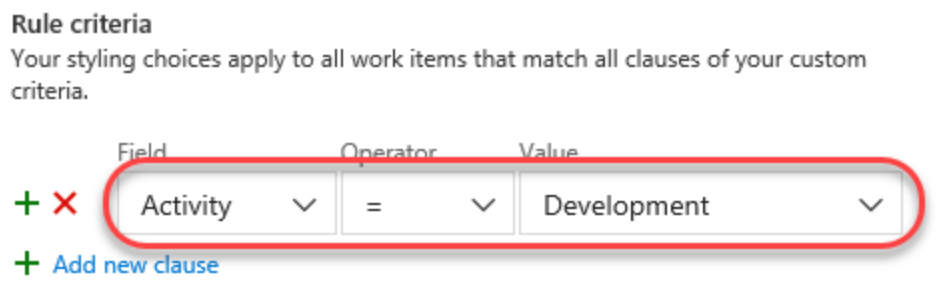
17. There is also a lot of customization available. Click the **Configure team settings** button.



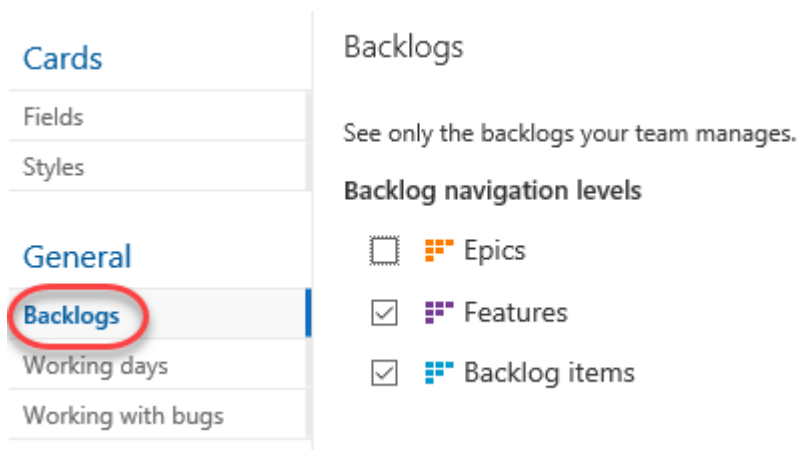
18. On the **Styles** tab, click **Add Styling rule** and set the **Name** to **“Development”**. Choose a green **Card color**. This will color all cards green if they meet the rule criteria set below.



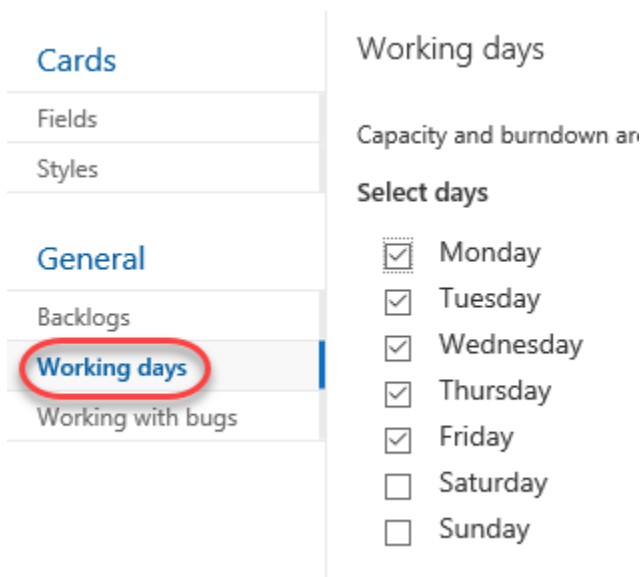
19. Add a rule for **Activity = Development**. This will set all cards assigned to **Development** activities green.



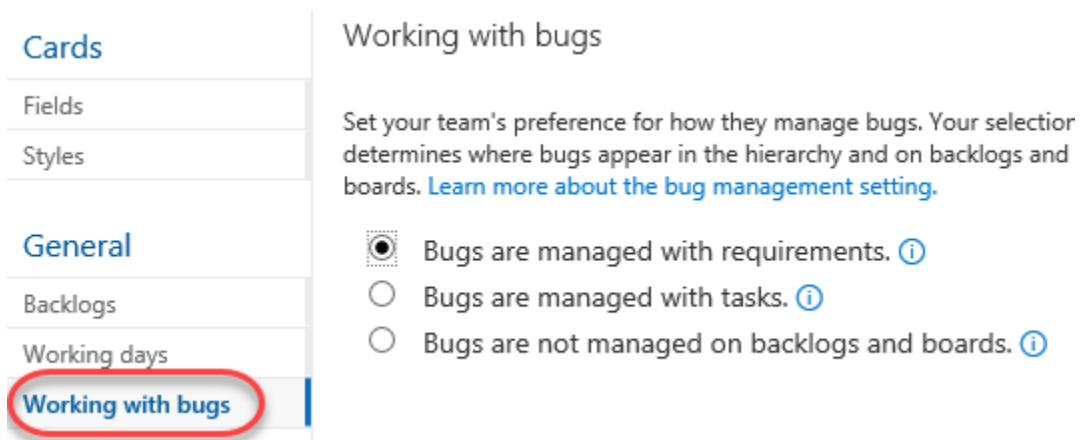
20. The **Backlogs** tab allows you to set the levels available for navigation. Epics are not included by default, but you could change that here.



21. You can also specify the **Working days** the team follows. This applies to capacity and burndown calculations.



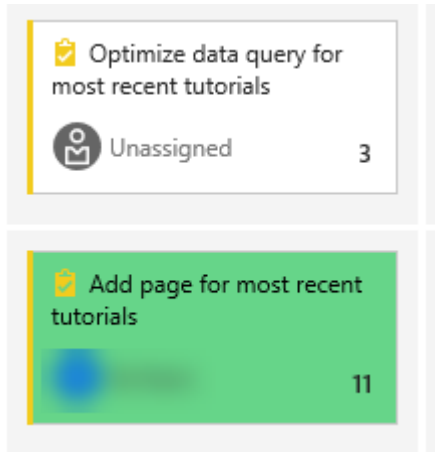
22. The **Working with bugs** tab allows you to specify how bugs are presented on the board.



- 
23. Click **Save and close** to save the styling rule.



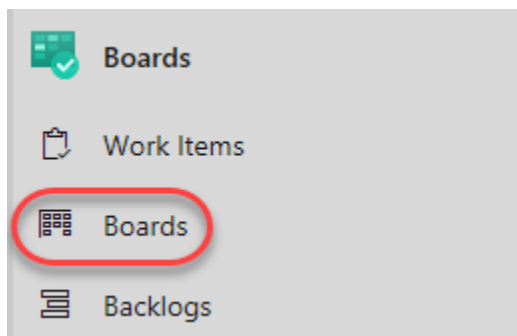
24. The task associated with **Development** is now green and very easy to identify.



#### Activity 4: Customizing Kanban boards

To maximize a team's ability to consistently deliver high quality software, Kanban emphasize two main practices. The first, visualize the flow of work, requires you to map your team's workflow stages and configure your Kanban board to match. The second, constrain the amount of work in progress, requires you to set work-in-progress (WIP) limits. You're then ready to track progress on your Kanban board and monitor key metrics to reduce lead or cycle time. Your Kanban board turns your backlog into an interactive signboard, providing a visual flow of work. As work progresses from idea to completion, you update the items on the board. Each column represents a work stage, and each card represents a user story (blue cards) or a bug (red cards) at that stage of work. However, every team develops its own process over time, so the ability to customize the Kanban board to match the way your team works is crucial.

1. Navigate to **Boards**.



2. Click the **Configure team settings** button.

- The team is emphasizing work done with data, so there is special attention paid to any task associated with accessing or storing data. Select the **Tag colors** tab. Click **Add tag color** enter a tag of “data”. Whenever a backlog item or bug is tagged with **data**, that tag will be highlighted.

Cards
Fields
Styles
Tag colors \*
Annotations
Tests
Board

### Tag colors

Emphasize selected tags using colors. Only one color applies to a tag at a time.

**+ Tag color**

| Tag  | Color                                                                              | Enabled                                                      |
|------|------------------------------------------------------------------------------------|--------------------------------------------------------------|
| data |  | <input checked="" type="checkbox"/> <input type="checkbox"/> |

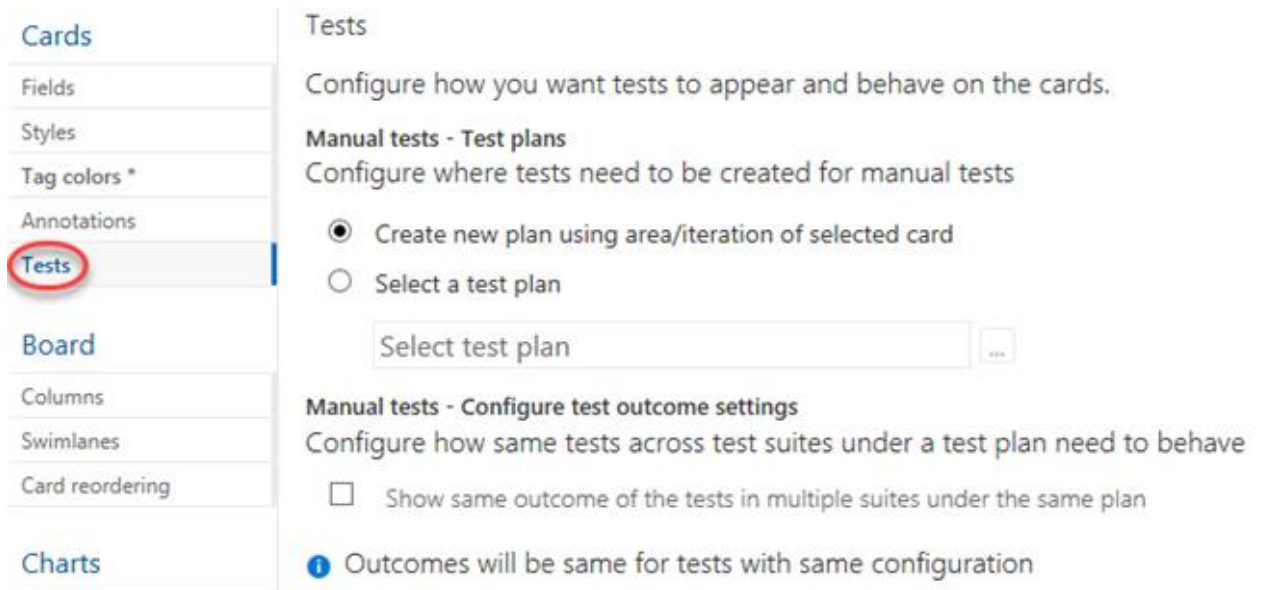
- You can also specify which **Annotations** you would like included on cards to make them easier to read and navigate. When an annotation is enabled, the child work items of that type are easily accessible by clicking the visualization on each card.

Fields
Styles
Tag colors \*
Annotations
Tests
Board

### Show visual cues on cards.

| Annotation | Visualization                                                                         | Enabled                             |
|------------|---------------------------------------------------------------------------------------|-------------------------------------|
| Task       |  | <input checked="" type="checkbox"/> |
| GitHub     |  | <input checked="" type="checkbox"/> |
| Tests      |  | <input checked="" type="checkbox"/> |

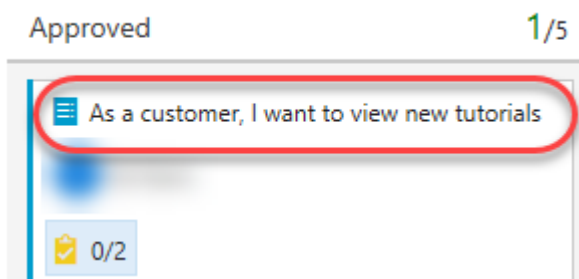
- The **Tests** tab enables you to configure how tests appear and behave on the cards.



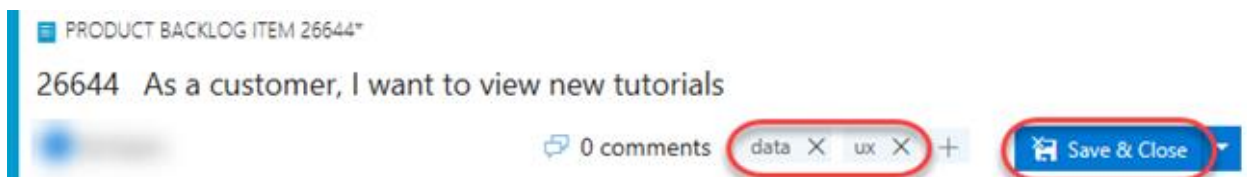
6. Click **Save and close**.



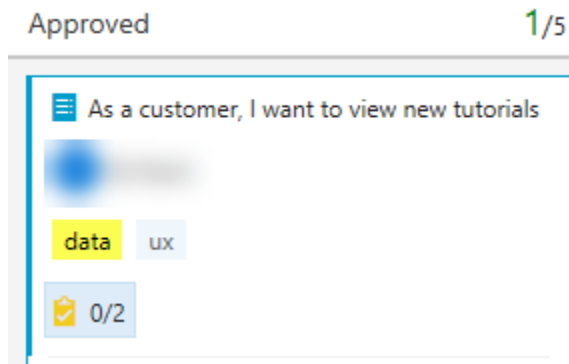
7. Open the **view new tutorials** backlog item.



8. Add tags for **“data”** and **“ux”**. Click **Save & Close**.



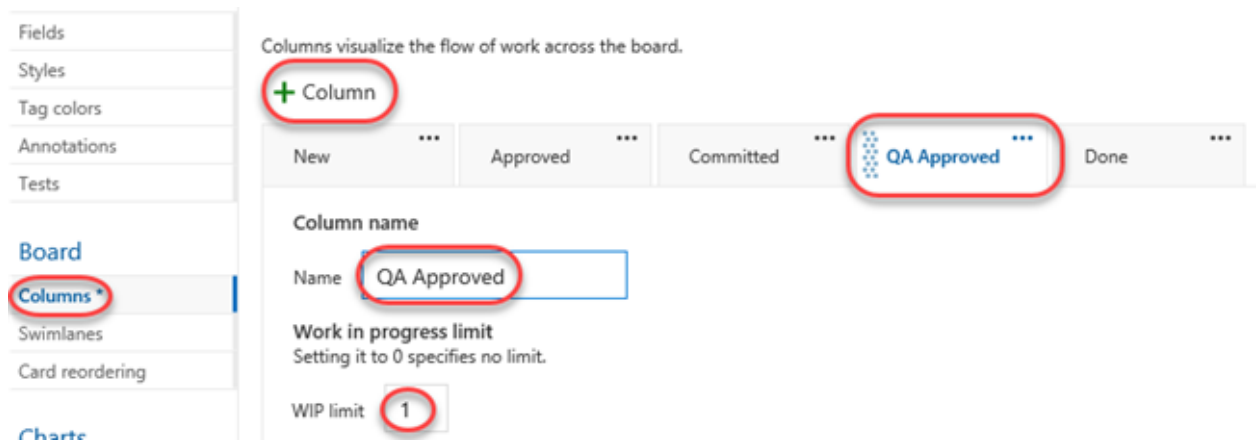
9. Note that the two tags are now visible on the card, although the **data** tag is highlighted yellow as configured.



10. Click the **Configure team settings** button.



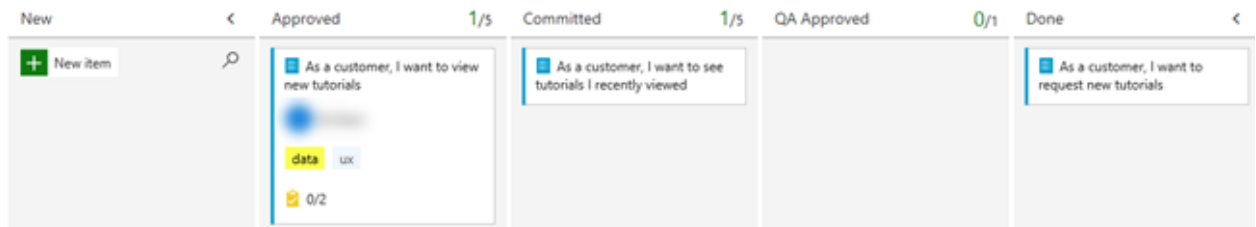
11. Select the **Columns** tab. This section allows you to add new stages to the workflow. Click **Add Column** and set the **Name** to “QA Approved”. Set the **WIP limit** to “1”, which indicates that only one work item should be in this stage at a time. You would ordinarily set this higher, but there are only two work items to demonstrate the feature with here. Move the stage to occur between **Committed** and **Done**.



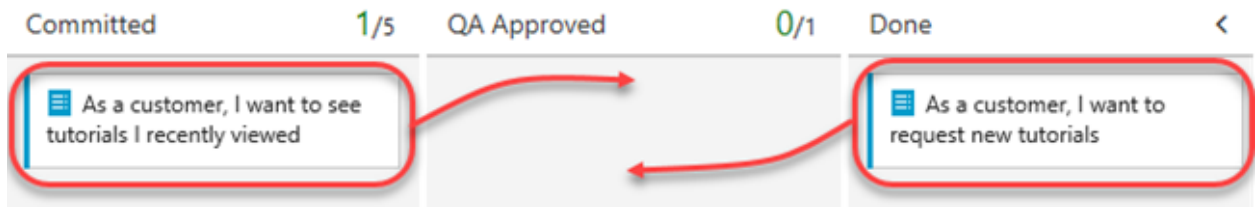
12. Click **Save and close**.



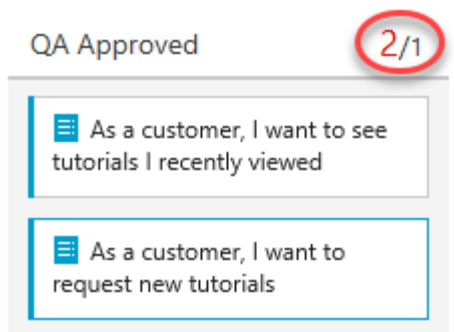
13. You will now see the new stage in the workflow.



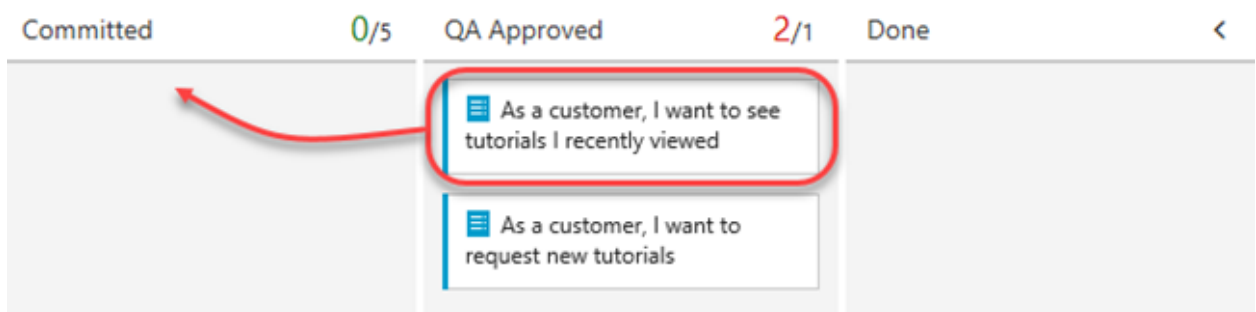
14. Move the work items from **Committed** and **Done** into **QA Approved**.



15. The stage now exceeds its **WIP** limit and is colored red as a warning.



16. Move the **recently viewed** backlog item back to **Committed**.

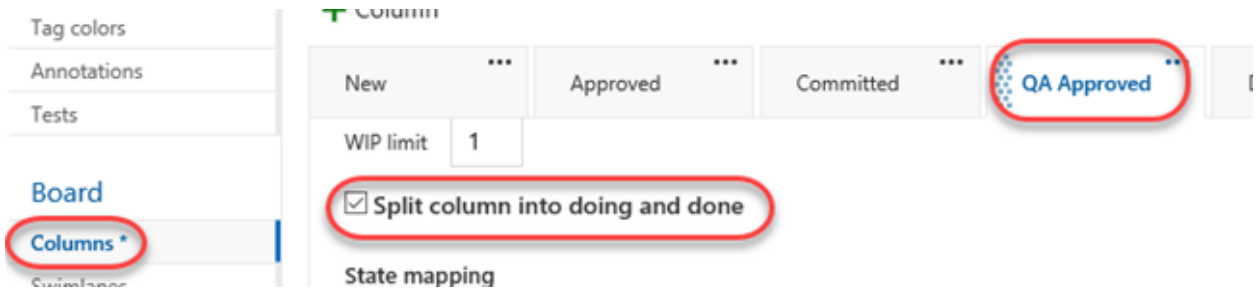


17. Click the **Configure team settings** button.





18. Return to the **Columns** tab and select **QA Approved**. A lag often exists between when work gets moved into a column and when work starts. To counter that lag and reveal the actual state of work in progress, you can turn on split columns. When split, each column contains two sub-columns: **Doing** and **Done**. Split columns let your team implement a pull model. Without split columns, teams push work forward, to signal that they've completed their stage of work. However, pushing it to the next stage doesn't necessarily mean that a team member immediately starts work on that item. Check **Split column into doing and done** to create two separate columns for this.



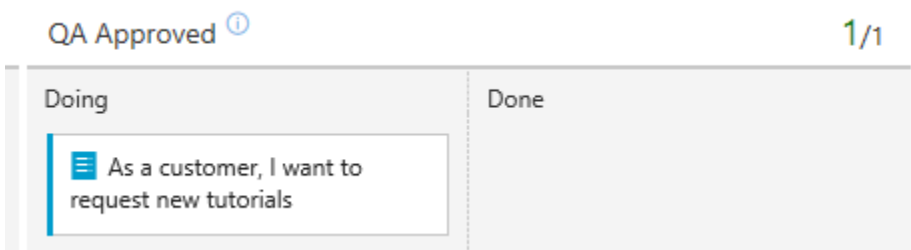
19. As your team updates the status of work as it progresses from one stage to the next, it helps that they agree on what **done** means. By specifying the **Definition of done** criteria for each Kanban column, you help share the essential tasks to complete before moving an item into a downstream stage. Add a **Definition of done** using markdown, such as "**Passes `**all**` tests.**". Click **Save and close**.

**Definition of done**  
Enter plain text or format using markdown.

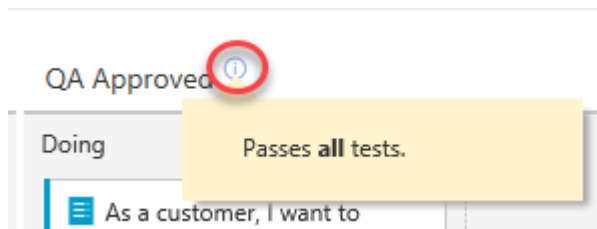
Passes **`**all**`** tests.

**Save and close** Cancel

20. Note that the **QA Approved** stage now has **Doing** and **Done** columns.



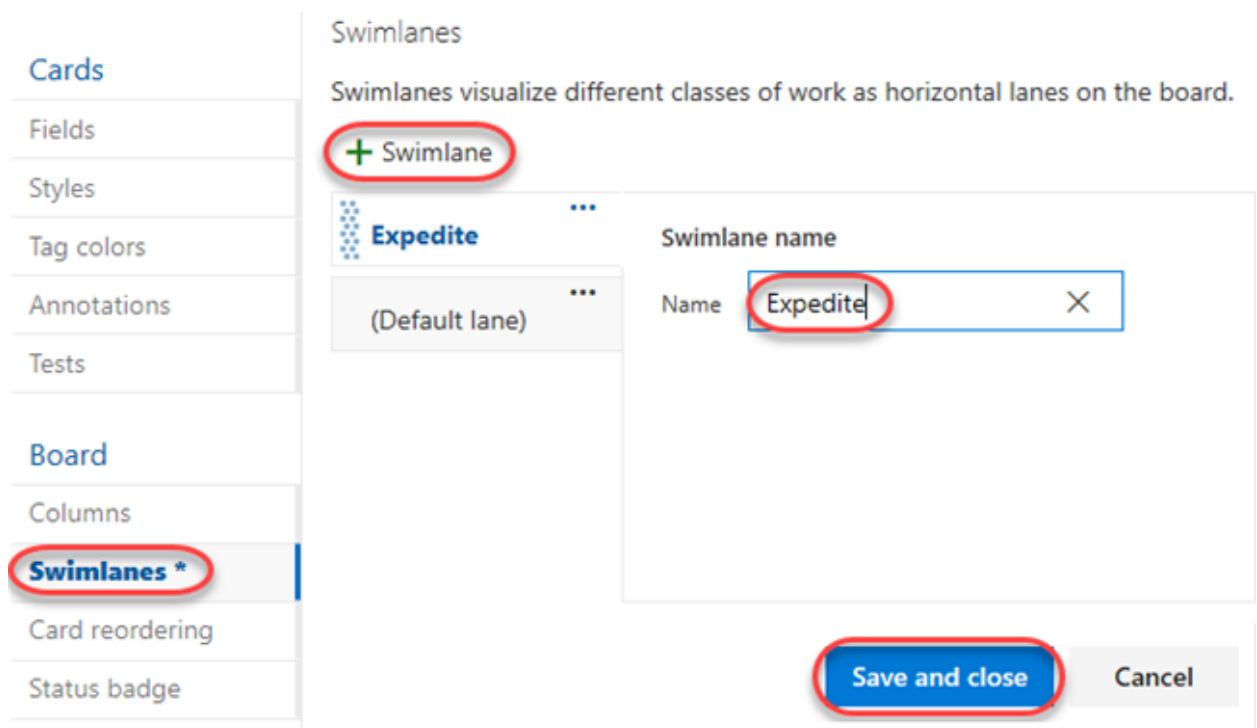
21. You can also click the icon next to the column header to read the **Definition of done**.



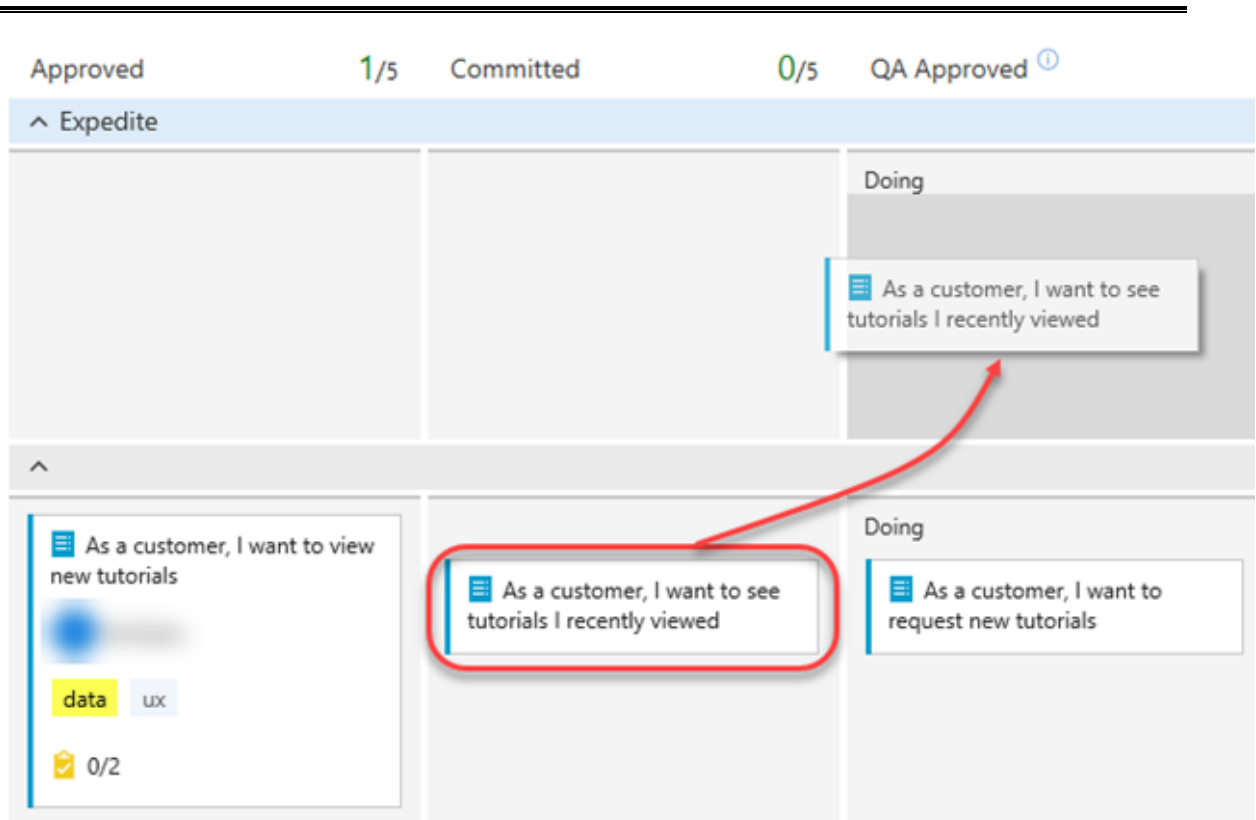
22. Click the **Configure team settings** button.



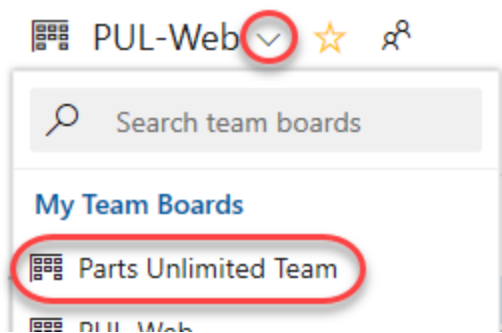
23. Your Kanban board supports your ability to visualize the flow of work as it moves from new to done. When you add **swimlanes**, you can also visualize the status of work that supports different service-level classes. You can create a swimlane to represent any other dimension that supports your tracking needs. From the **Swimlanes** tab, click **Add Swimlane** and set the **Name** to “**Expedite**”. Click **Save and close** and close.



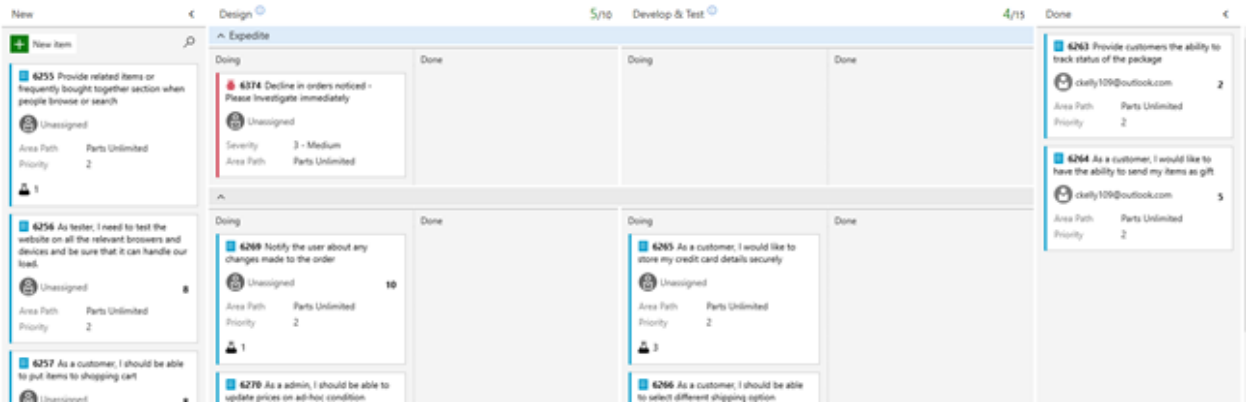
24. Drag and drop the **Committed** work item onto **QA Approved | Doing** so that it gets recognized as having priority when QA bandwidth becomes available.



25. If you would like to review a more sophisticated board with many more work items, select the **Parts Unlimited Team** from the team dropdown.

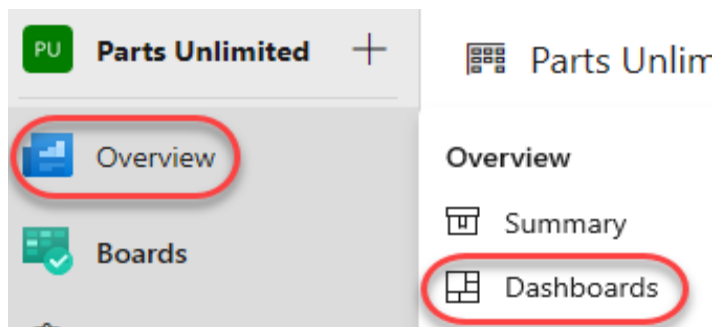


26. This board provides a playground for you to experiment with and review the results.



## Activity 5: Defining dashboards

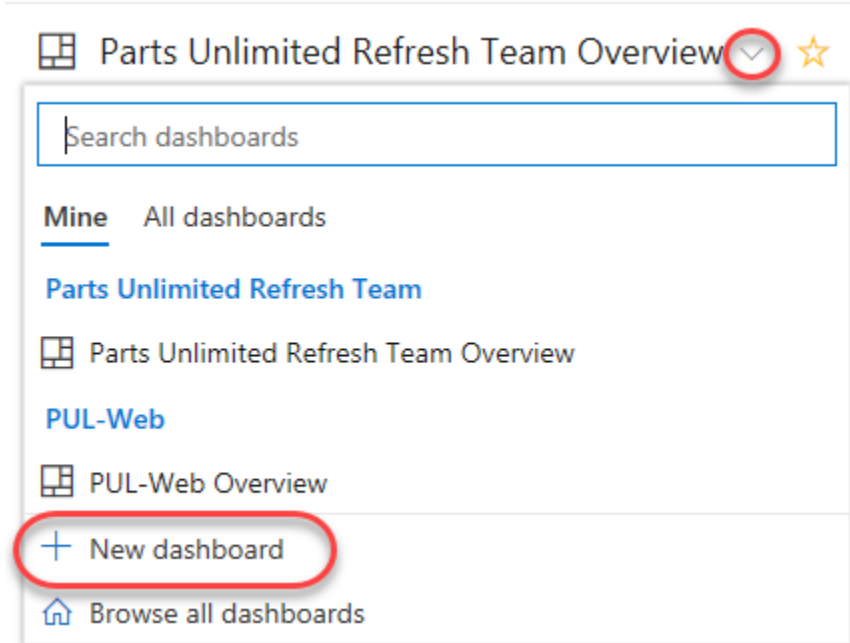
1. Select **Overview | Dashboards**.



2. From the dashboard dropdown, select **Parts Unlimited Team Overview**. Dashboards allow teams to visualize status and monitor progress across the project. At a glance, you can make informed decisions without having to drill down into other parts of your team project site. The Overview page provides access to a default team dashboard which you can customize by adding, removing, or rearranging the tiles. Each tile corresponds to a widget that provides access to one or more features or functions.



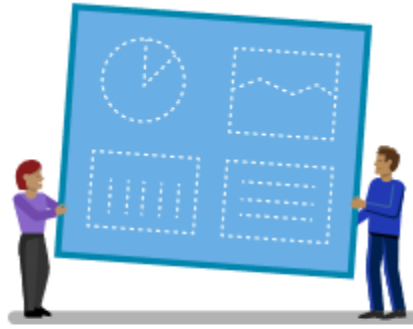
- From the dashboard dropdown, select **New dashboard**.



- Set the **Name** to “**Product training**” and select the **PUL-Web** team. Click **Create**.

A screenshot of a 'Create a dashboard' modal window. The title is 'Create a dashboard' with a close button (X) in the top right. The form has two main sections: 'Name' and 'Description'. The 'Name' field contains the text 'Product Training' and is highlighted with a red box. The 'Description' field is a large text area with the placeholder text 'Enter a description'. Below the description field is a checkbox labeled 'Automatically refresh the dashboard every 5 minutes', which is currently unchecked. Underneath is the 'Dashboard Type' section, where 'Team Dashboard' is selected with a radio button. A small note below says 'The dashboard is associated with a single team. Team admins can'. At the bottom right, there are two buttons: 'Create' (highlighted with a red box) and 'Cancel'.

- Click **Add a widget**.

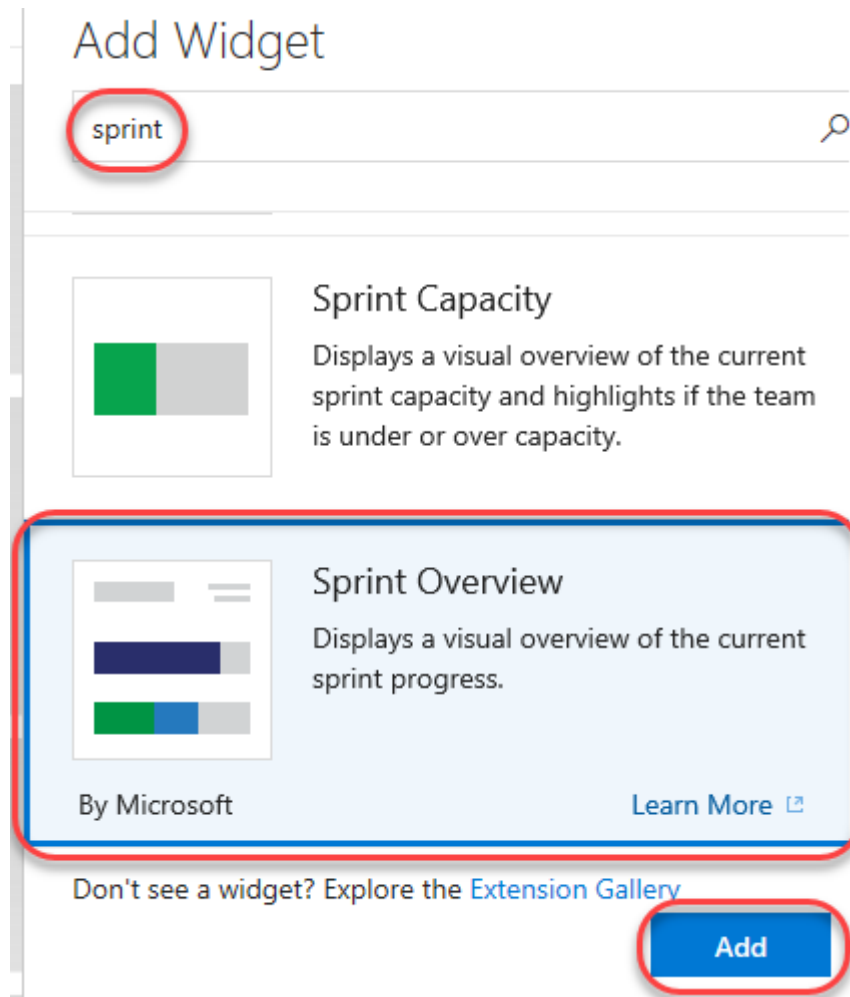


This dashboard doesn't have widgets just yet!

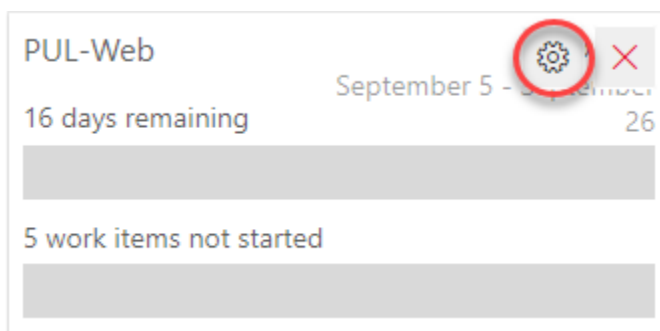
Add one or more widgets to gain visibility into your team's progress.

Add a widget

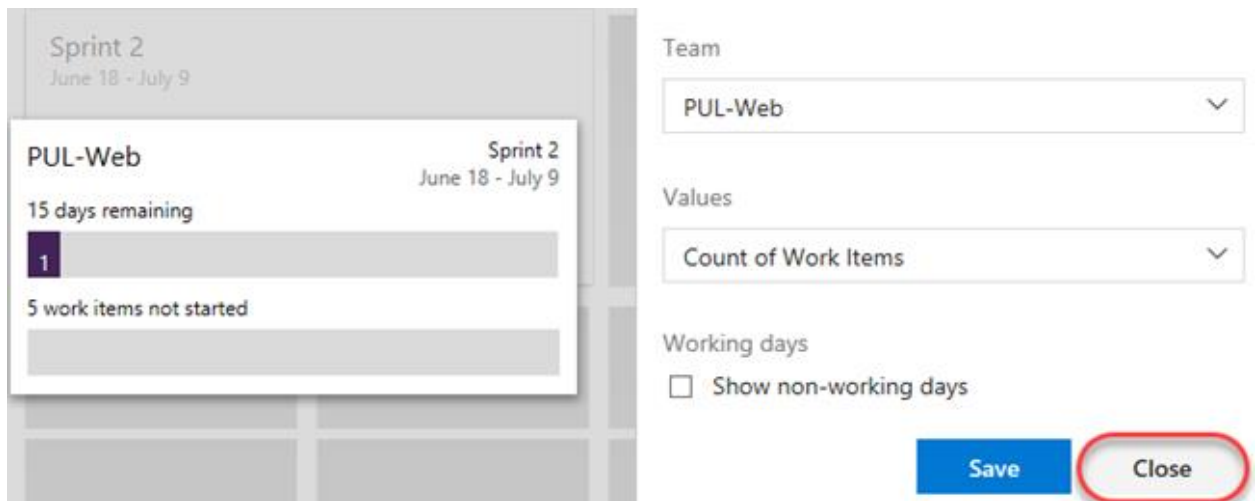
6. In the **Add Widget** panel, search for “**sprint**” to find existing widgets that focus on sprints. Select **Sprint Overview** and click **Add**.



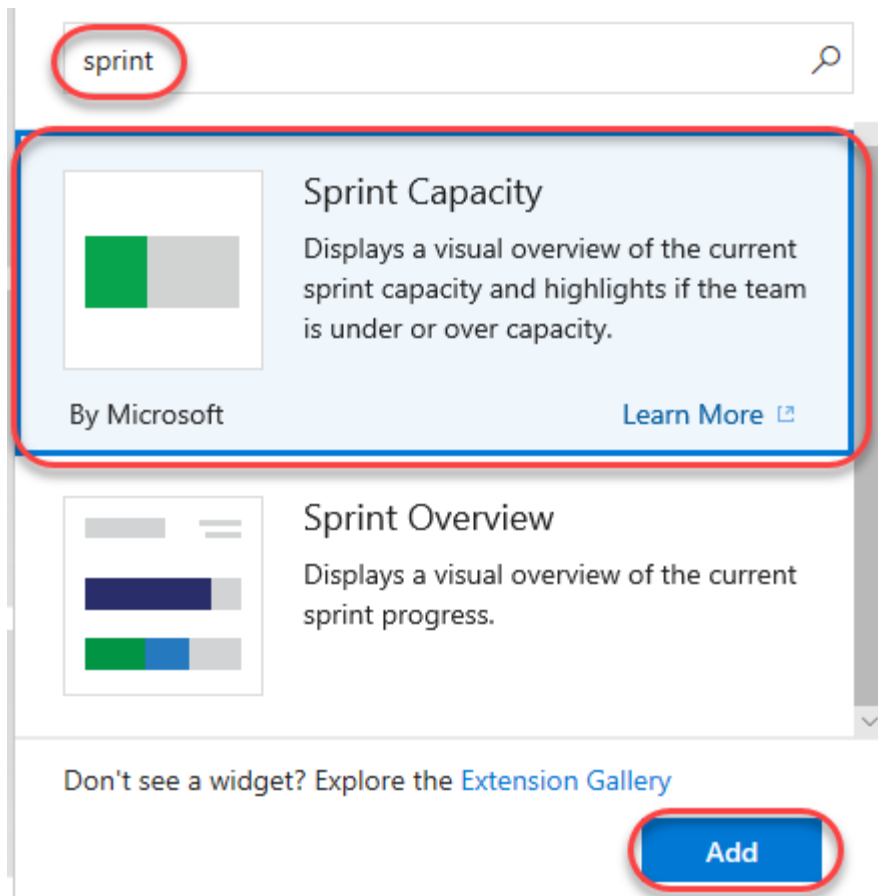
7. Many widgets have options you can configure. Click the **Settings** button.



8. The quantity and depth of settings will vary by widget. Click **Close** to dismiss.

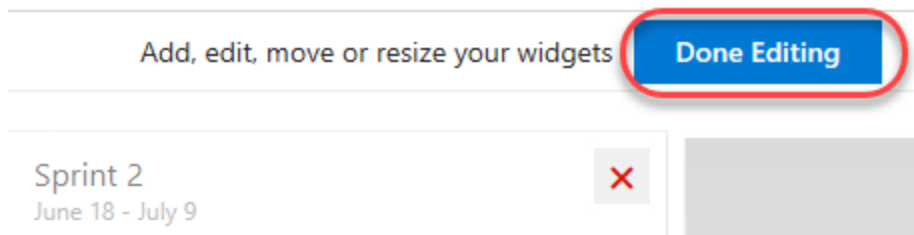


9. Search the widgets again for “sprint” and add the **Sprint Capacity** widget.

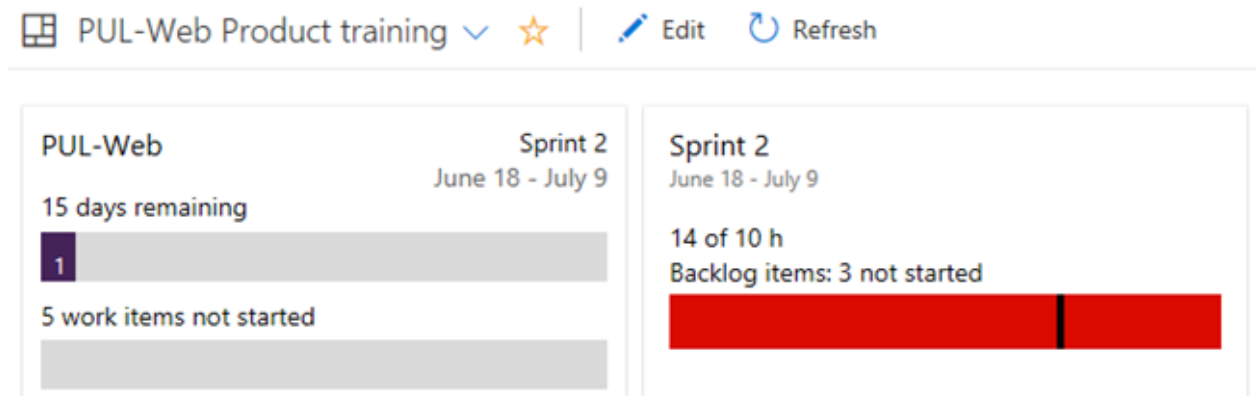


10. Click **Done Editing**.

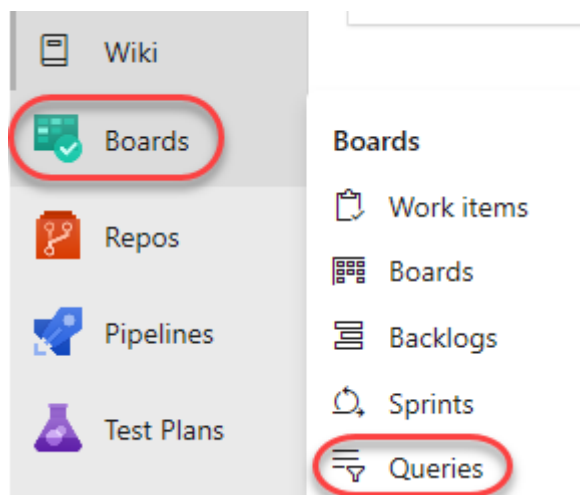




11. You can now review two important aspects of your current sprint on your custom dashboard.



12. Another way of customizing dashboards is to generate charts based on work item queries, which you can share to a dashboard. Select **Boards** | **Queries**.



13. Click **New query**.

## Queries

Favorites

All

+ New query

Title

14. Set the first term to **Work Item Type = Task** and the second term to **Area Path = Parts Unlimited\PUL-Web**.

Filters for top level work items

|                  | And/Or                       | Field          | Operator | Value                   |
|------------------|------------------------------|----------------|----------|-------------------------|
| + X              | <input type="checkbox"/>     | Work Item Type | =        | Task                    |
| + X              | <input type="checkbox"/> And | Area Path      | =        | Parts Unlimited\PUL-Web |
| + Add new clause |                              |                |          |                         |

15. Click **Save query**.

Save query

Revert changes

Column options

16. Set the **Name** to “**Web tasks**” and the **Folder** to **Shared Queries**. Click **OK**.

## New query



Name \*

Web tasks

Folder \*

Shared Queries

OK

Cancel

17. Select the **Charts** tab and click **New chart**.

Results

Editor

Charts



Refresh charts



New chart

18. Click **New chart**.



## Visualize query results with charts

Create several types of charts - such as pie, column, or trend - to quickly view the status of work.

New chart

19. Set the **Name** of the chart to “**Web tasks - By assignment**” and **Group by** to **Assigned To**. Click **OK** to save.

Name

Group by ?

Aggregation ?  
 of

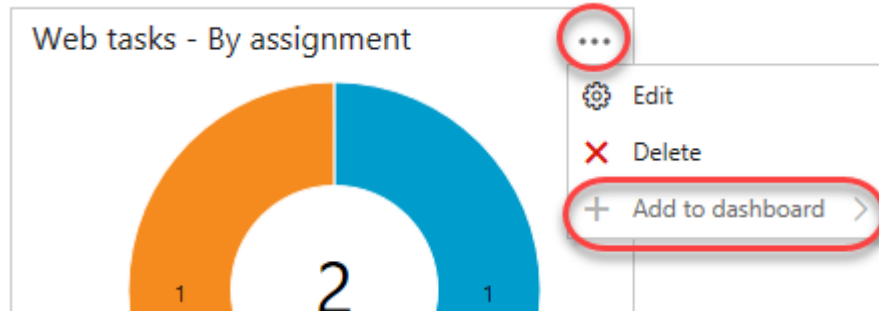
Sort ?

Series ?

☐ ☐ (blank)



20. You can now add this chart to a dashboard.



### 3) Graded Lab Tasks

*Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.*

#### Lab Task 1:

You will create a new project and use a sprint to run the task of mind map. Show the management output of the activity.

#### Lab Task 2:

##### Customizing team process

In Azure DevOps, you customize your work tracking experience through a process. A process defines the building blocks of the work item tracking system as well as other sub-systems you access through Azure DevOps. Whenever you create a team project, you select the process which contains the building blocks you want for your project.

Azure DevOps supports two process types. The first, the core system processes-Scrum, Agile, and CMMI system processes-are locked. You cannot customize these processes. The second type, inherited processes, you create from a core system process. These processes you can customize.

In addition, all processes are shared. That is, one or more team projects can reference a single process. Instead of customizing a single team project, you customize a process. Changes made to the process automatically update all team projects that reference that process.

Once you've created an inherited process, you can customize it, create team projects based on it, and migrate existing team projects to reference it. The Git team project can't be customized until it's migrated to an inherited process.

In this task we'll create a new process that inherits from Scrum. The one change we'll make is to add a backlog item field designed to track to a proprietary PartsUnlimited ticket ID.

---

## **Lab 10**

### **Build using Maven**

#### **Objective:**

In this lab, We will go through Jenkins to install and configure to test and deploy Java or Web Applications using Maven and Jenkins

#### **Activity Outcomes:**

The activities provide hands - on practice with the following topics

- Installation of Jenkins
- Using of Maven for building Java Application

#### **Instructor Note:**

---

## **1. Install and configure Jenkins to test and deploy Java or Web Applications.**

To install Jenkins following software packages are required

- 1) GIT ([git-scm.com](https://git-scm.com))
- 2) Notepad++ (<https://notepad-plus-plus.org/downloads/>)
- 3) Latest Java development kit (JDK)
- 4) Jenkins
- 5) Apache Maven

(Optional) Step 1-: Install

GIT

Step 2 -: Install Notepad++

Step 3 -: Install Java

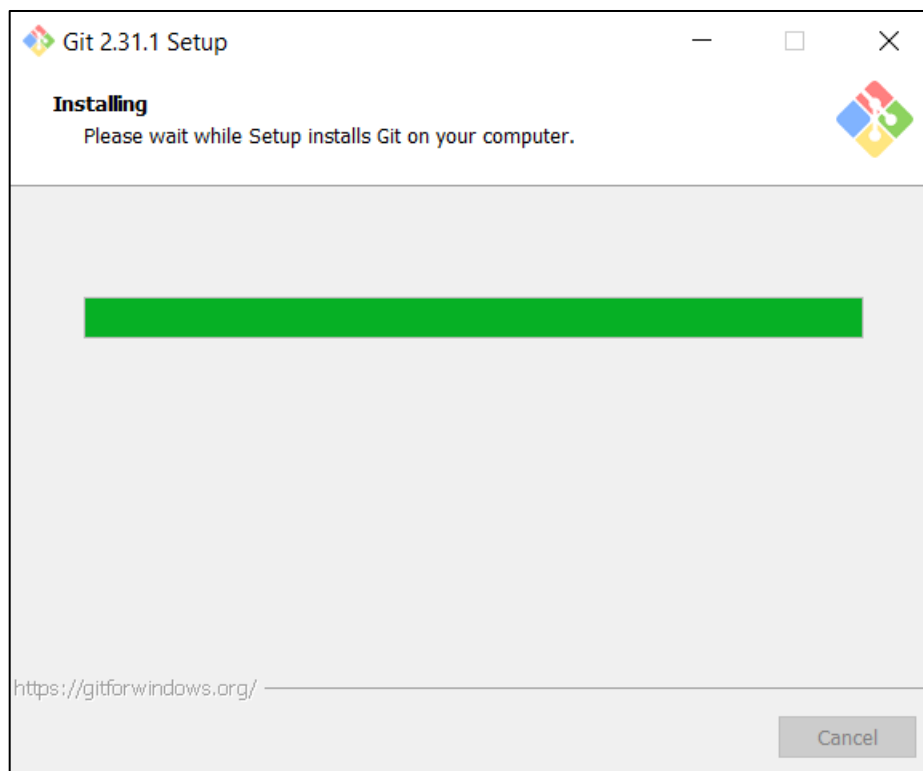
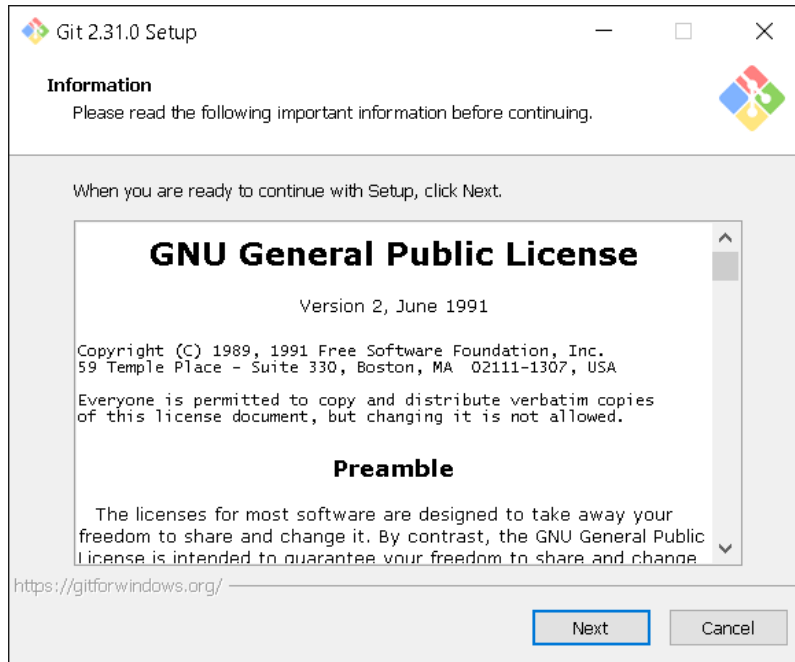
Step 4 -: Install Jenkins

Step 5 -: Install Maven

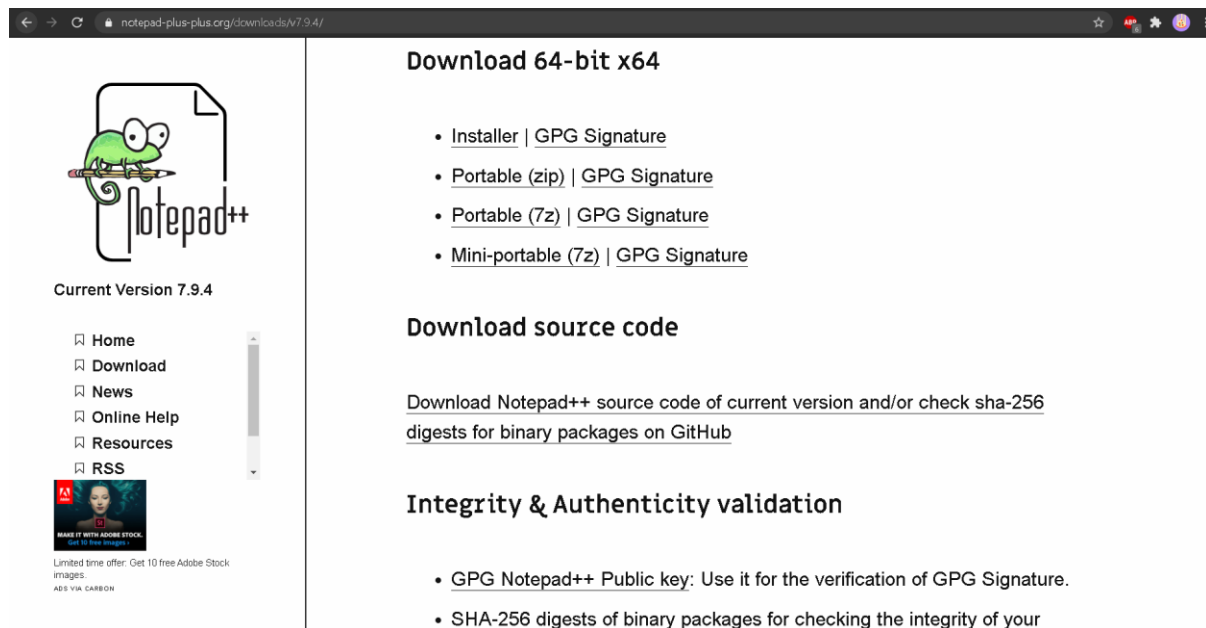
## **2. Solved Lab Activities**

### **Activity 1:**

**Step 1:** Install GIT



## Step 2: Install Notepad++



The screenshot shows the Notepad++ website. On the left is a sidebar with the Notepad++ logo (a green frog), the current version (7.9.4), and a navigation menu with links to Home, Download, News, Online Help, Resources, and RSS. Below the menu is a small advertisement for Adobe Stock. The main content area has three sections: 'Download 64-bit x64' with links for Installer, Portable (zip), Portable (7z), and Mini-portable (7z), each with a GPG Signature link; 'Download source code' with a link to download source code and check sha-256 digests on GitHub; and 'Integrity & Authenticity validation' with links for GPG Notepad++ Public key and SHA-256 digests.

notepad-plus-plus.org/downloads/v7.9.4/

### Download 64-bit x64

- [Installer](#) | [GPG Signature](#)
- [Portable \(zip\)](#) | [GPG Signature](#)
- [Portable \(7z\)](#) | [GPG Signature](#)
- [Mini-portable \(7z\)](#) | [GPG Signature](#)

### Download source code

[Download Notepad++ source code of current version and/or check sha-256 digests for binary packages on GitHub](#)

### Integrity & Authenticity validation

- [GPG Notepad++ Public key](#): Use it for the verification of GPG Signature.
- [SHA-256 digests of binary packages](#) for checking the integrity of your

### Step 3: Install Java



The screenshot shows the 'Java Setup - Welcome' window. It features the Java logo and the text 'Welcome to Java - Updated License Terms'. Below this, it states that the terms have changed and provides a link to the 'Updated License Agreement'. It explains that the Java Runtime is licensed for personal (non-commercial) use and that commercial use requires a separate license. It prompts the user to click 'Install' to accept the license and install Java, or 'Remove' to uninstall it. At the bottom, there is a checkbox for 'Change destination folder' and buttons for 'Install' and 'Remove'.

Java Setup - Welcome

 **Java™**  
ORACLE

### Welcome to Java - Updated License Terms

The terms under which this version of the software is licensed have changed.  
[Updated License Agreement](#)

This version of the Java Runtime is licensed only for your personal (non-commercial) desktop and laptop use.

Commercial use of this software requires a separate license from Oracle or from your software vendor. Click Install to accept the license agreement and install Java now or click Remove to uninstall it from your system.

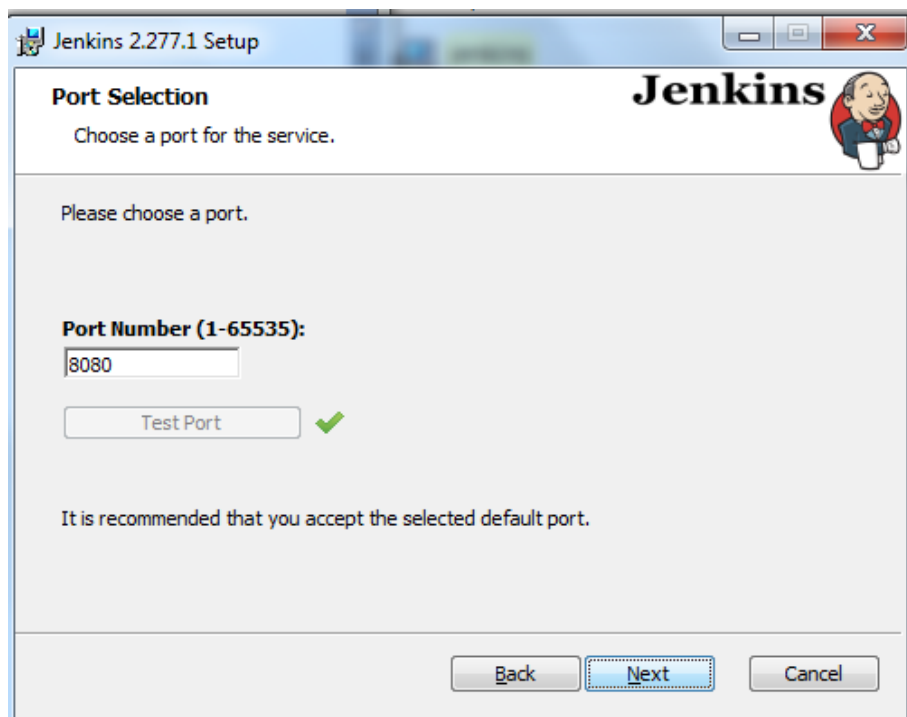
No personal information is gathered as part of our install process. [Details on the information we collect](#)

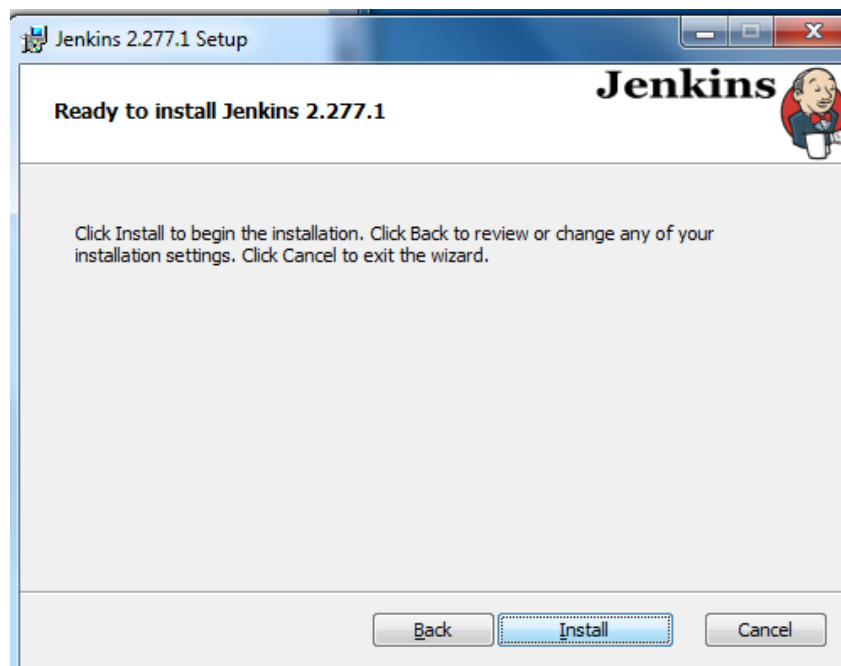
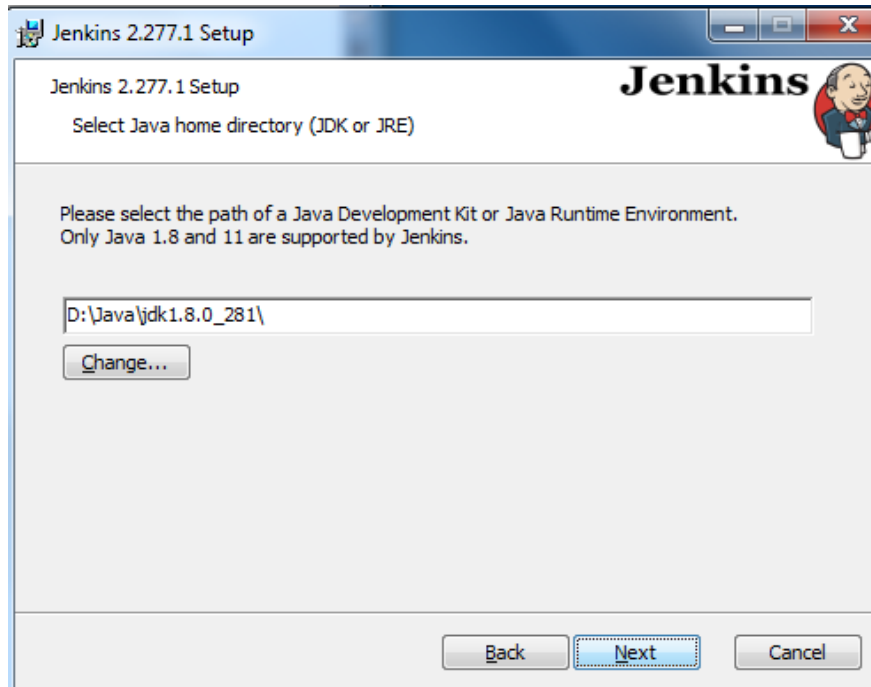
☐ Change destination folder

[Install](#) [Remove](#)

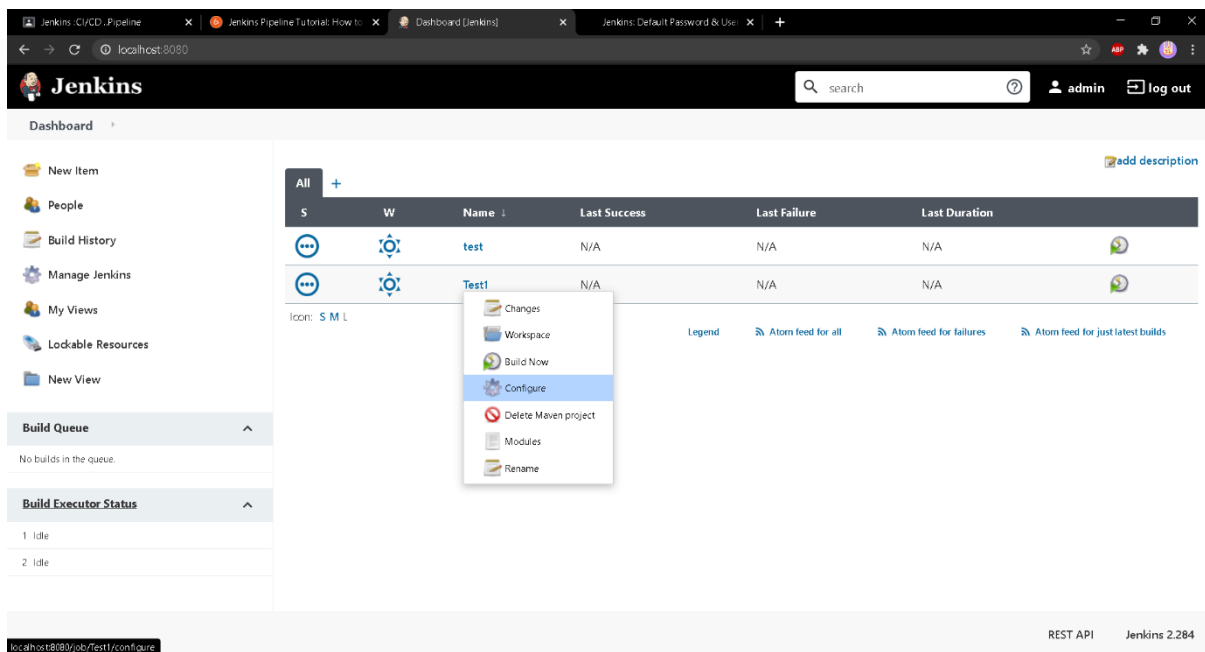
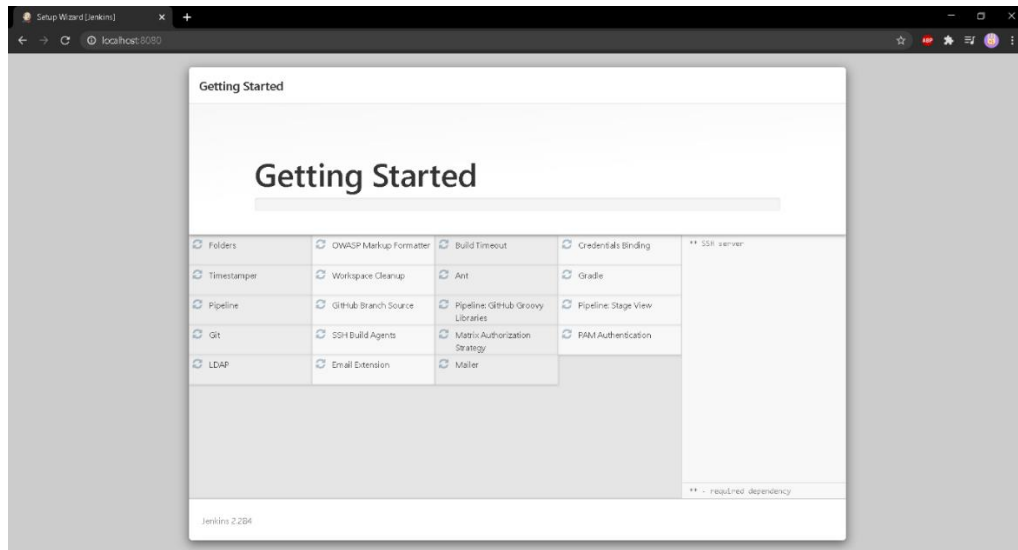
### Step 4: Install Jenkins







## Step 5: Configuration and Jenkins Pipeline

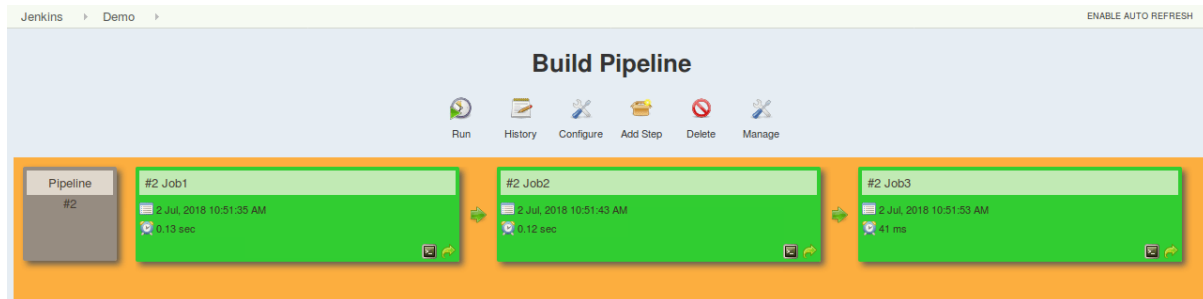


## Build Triggers

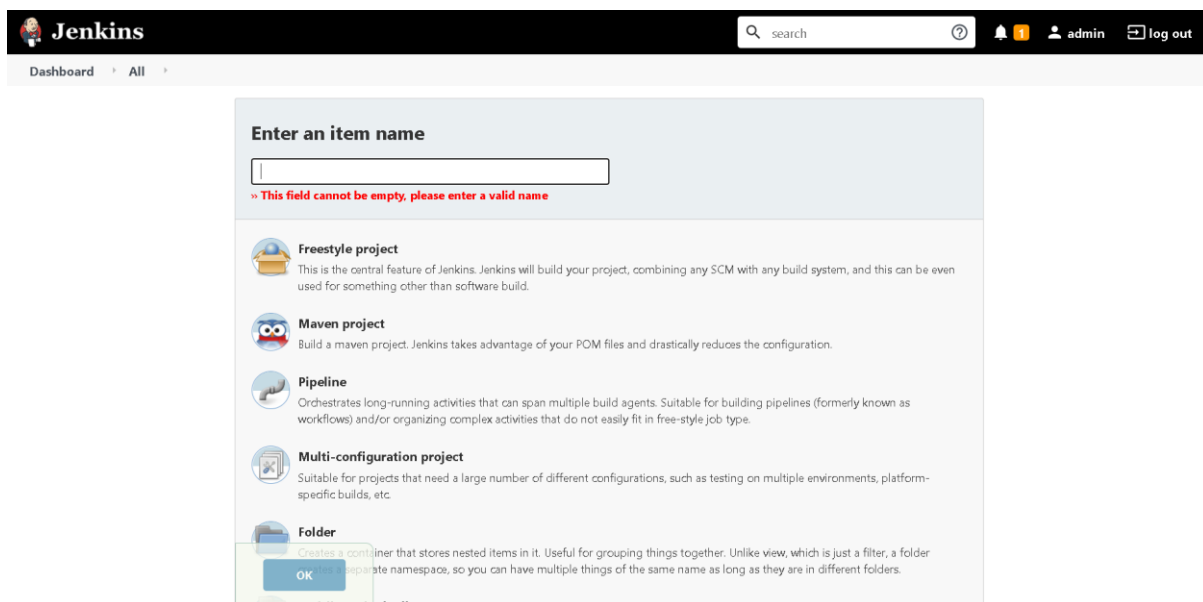
- ☒ Build whenever a SNAPSHOT dependency is built ?
- ☐ Schedule build when some upstream has no successful builds ?
- ☐ Trigger builds remotely (e.g., from scripts) ?
- ☒ Build after other projects are built ?

Projects to watch

Test1,



## Step 6: Creation of job using Jenkins



## Step 7: Build maven project using Jenkins

Maven - Download Apache Ma... x +

maven.apache.org/download.cgi

Apache Maven Project  
http://maven.apache.org/

Download | Get Sources | Last Published: 2021-03-20

## Downloading Apache Maven 3.6.3

Apache Maven 3.6.3 is the latest release and recommended version for all users.

The currently selected download mirror is <https://mirrors.estointernet.in/apache/>. If you encounter a problem with this mirror, please select another mirror. If all mirrors are failing, there are [backup mirrors](#) (at the end of the mirrors list) that should be available. You may also consult the [complete list of mirrors](#).

Other mirrors:

### System Requirements

|                                   |                                                                                                                                                                                                                                                      |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Java Development Kit (JDK)</b> | Maven 3.3+ require JDK 1.7 or above to execute - they still allow you to build against 1.3 and other JDK versions <a href="#">by Using Toolchains</a>                                                                                                |
| <b>Memory</b>                     | No minimum requirement                                                                                                                                                                                                                               |
| <b>Disk</b>                       | Approximately 10MB is required for the Maven installation itself. In addition to that, additional disk space will be used for your local Maven repository. The size of your local repository will vary depending on usage but expect at least 500MB. |
| <b>Operating System</b>           | No minimum requirement. Start up scripts are included as shell scripts and Windows batch files.                                                                                                                                                      |

### Files

Maven is distributed in several formats for your convenience. Simply pick a ready-made binary distribution archive and follow the [installation instructions](#). Use a source archive if you intend to build Maven yourself.

In order to guard against corrupted downloads/installations, it is highly recommended to [verify the signature](#) of the release bundles against the public [KEYS](#) used by the Apache Maven developers.

Edit User Variable

Variable name:

Variable value:

Dashboard ▾ ▸

 New Item

 People

 Build History

 Manage Jenkins

 My Views

 Lockable Resources

 New View

Q maven

Updates Available Installed Advanced

| Install ↑                | Name                                                                                                                                                                                                                                                                                | Version | Released    |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------------|
| <input type="checkbox"/> | <b>Maven Integration</b><br><small>Build Tools</small><br>This plug-in provides, for better and for worse, a deep integration of Jenkins and Maven: Automatic triggers between projects depending on SNAPSHOTS, automated configuration of various Jenkins publishers (JUnit, ...). | 3.10    | 15 days ago |

Javadoc

✓ Success

Maven Integration

✓ Success

Loading plugin extensions

✓ Success

### Enter an item name

Test1

» Required field



#### Freestyle project

This is the central feature of Jenkins. Jenkins will build your project, combining any SCM with any build system, and this can be even used for something other than software build.



#### Maven project

Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.

JDK

(System) ▼

JDK to be used for this project

Advanced...

Dashboard > Test1 >

General
Source Code Management
Build Triggers
Build Environment
Pre Steps
Build
**Post Steps**
Build Settings

Post-build Actions

**Settings file**
Use default maven settings

**Global Settings file**
Use default maven global settings

**Post Steps**

☐ Run only if build succeeds
☐ Run only if build succeeds or is unstable
☒ Run regardless of build result

Should the post-build steps run only for successful builds, etc.

Add post-build step

**Build Settings**
☐ E-mail Notification

**Post-build Actions**
Add post-build action

Save
Apply

**Jenkins**

admin
log out

Dashboard > Test1 >

Back to Dashboard
Status
Changes
Workspace
Build Now

**Maven project Test1**

Workspace
 Recent Changes

add description
Disable Project

### 3) Graded Lab Tasks

*Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.*

#### Lab Task 1:

Create a joava project. Use Maven to build the project. Display the output.

---

## **Lab 11**

### **Cloud Deployment Testing in DevOps: Selenium**

#### **Objective:**

In this lab, We will go through selenium and perform continuous testing of web applications using Selenium.

#### **Activity Outcomes:**

The activities provide hands - on practice with the following topics

- Selenium
- Continuous Testing in Cloud

#### **Instructor Note:**



---

## 1. Selenium as Testing Process automation

Continuous Testing is the process of executing automated tests as part of the software delivery pipeline in order to obtain feedback on the business risks associated with a software release candidate as rapidly as possible. It evolves and extends test automation to address the increased complexity and pace of modern application development and delivery.

Continuous Testing in DevOps is a software testing type that involves testing the software at every stage of the software development life cycle. The goal of Continuous testing is evaluating the quality of software at every step of the Continuous Delivery Process by testing early and testing often. The continuous testing process in DevOps involves stakeholders like Developer, DevOps, QA and

The old way of testing was hand off centric. The software was handed off from one team to another. A project would have definite Development and QA phases. QA teams always wanted

---

more time to ensure quality. The goal was that the quality should prevail over project schedule. However, business wants faster delivery of software to the end user. The newer is the software, the better it can be marketed and increase revenue potential of the company. Hence, a new way of testing was evolved.

Continuous means uninterrupted testing done on a continuous basis. In a Continuous DevOps process, a software change (release candidate) is continuously moving from Development to Testing to Deployment.

## **Selenium**

Selenium is an open-source tool which automates web browser testing. It is mainly used for functional (unit testing & regression testing).

Reasons for considering Selenium (advantages):

1. Programming Language support – Selenium scripts can be written in major programming languages like Java, Python, Perl, Php, Ruby and C#.
2. Web Browser support – Support cross browser testing on chrome, Firefox, internet explorer and safari.
3. Operating Systems support – Selenium can be used on Windows, Mac or Linux.

In this practical, we would use Selenium for creating test cases, TestNG for getting detailed report of those test cases and further we use Jenkins to run our test cases. Jenkins will see if our test cases are getting passed and if those test cases fail's we would receive an email notification. And this is how we complete the process of continuous testing in DevOps.

**Test Case:** Automate amazon.in search and Facebook login. And at the end, receive a mail confirmation if the test result fail. This whole process has to be automated and in continuous manner for continuous testing.

## **2. Solved Lab Activities**

### **Activity 1:**

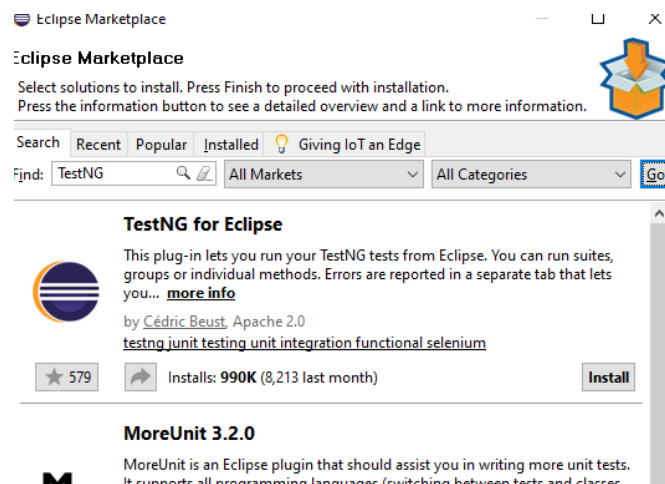
**Step 1: Create Test cases with Selenium on Eclipse IDE.**

Step 1.1: Download Selenium standalone-3.9.0.jar and Selenium server-3.9.0.zip files from the link below and unzip the file.

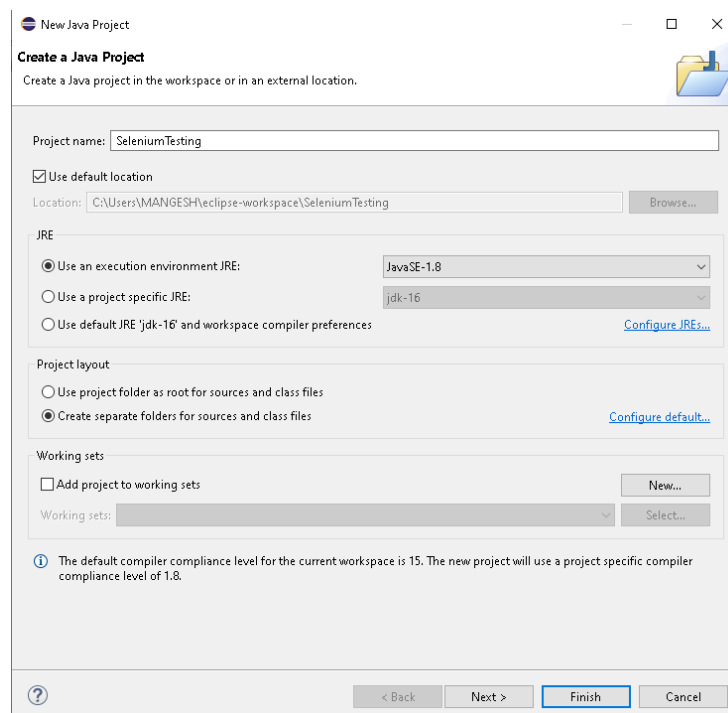
<https://selenium-release.storage.googleapis.com/index.html?path=3.9/>

Step 1.2: Download latest Java as a prerequisite for Eclipse followed by installation of Eclipse.

Step 1.3: Launch the Eclipse IDE and open Eclipse Marketplace to install TestNG plugin.

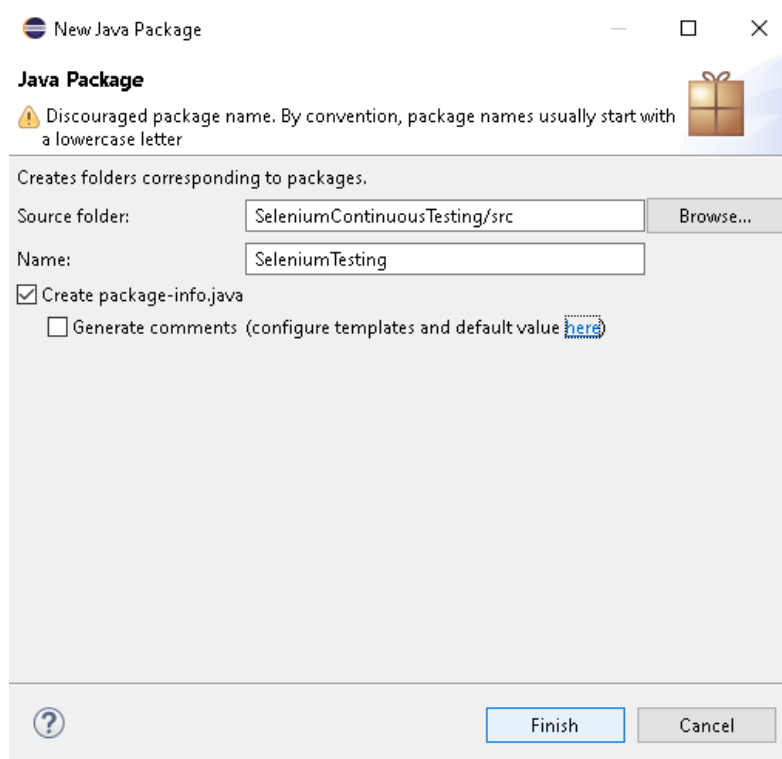


Step 1.4: Restart Eclipse IDE and create a new java project under New  Project  Java Project and Select JRE 1.8 under “Use Execution Environment for JRE”



---

Step 1.5: Now add new package from File Tab.



Step 1.6: Select Project ☐ Click New ☐ Class and Click Finish and enter the below code

package <Package name>;

import java.util.concurrent.TimeUnit;

import org.openqa.selenium.By;

import org.openqa.selenium.WebDriver;

import org.openqa.selenium.chrome.ChromeDriver;

public class <File name> {

    public void JenkinsTest() {

        System.setProperty("webdriver.chrome.driver","file\_location");

        WebDriver driver = new ChromeDriver();

        driver.navigate().to("https://www.amazon.in");

        driver.manage().timeouts().implicitlyWait(5, TimeUnit.SECONDS);

        driver.findElement(By.id("twotabsearchtextbox")).sendKeys("Nike shoes");

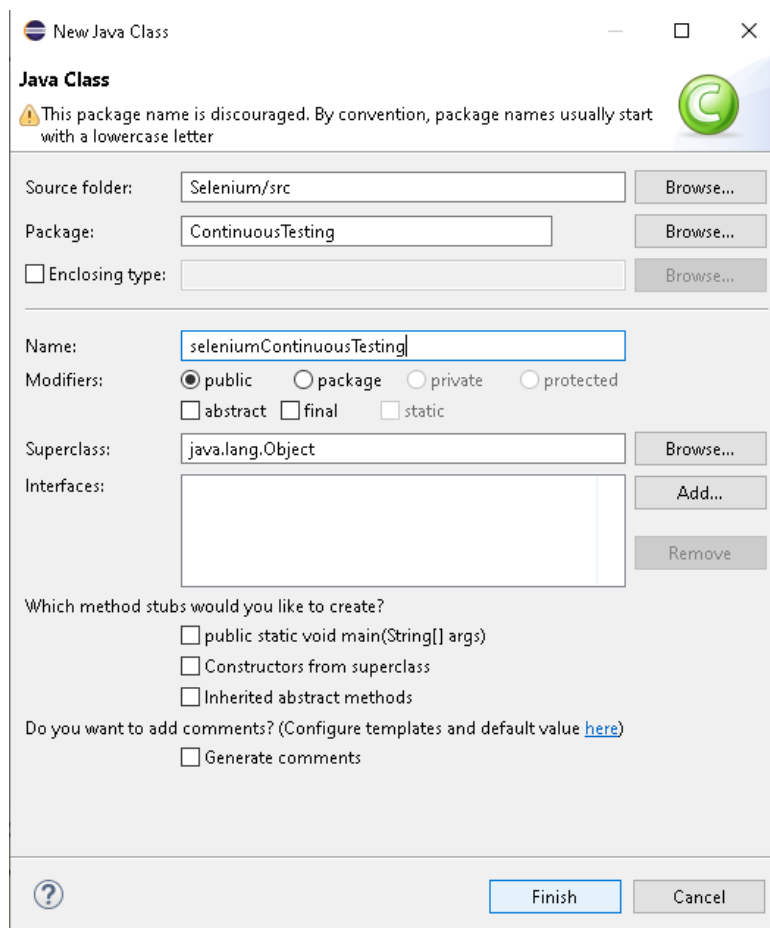
        driver.findElement(By.xpath("/html/body/div[1]/header/div/div[1]/div[3]/div/form/div[2]/div/input")).click();

        driver.manage().timeouts().implicitlyWait(10, TimeUnit.SECONDS);

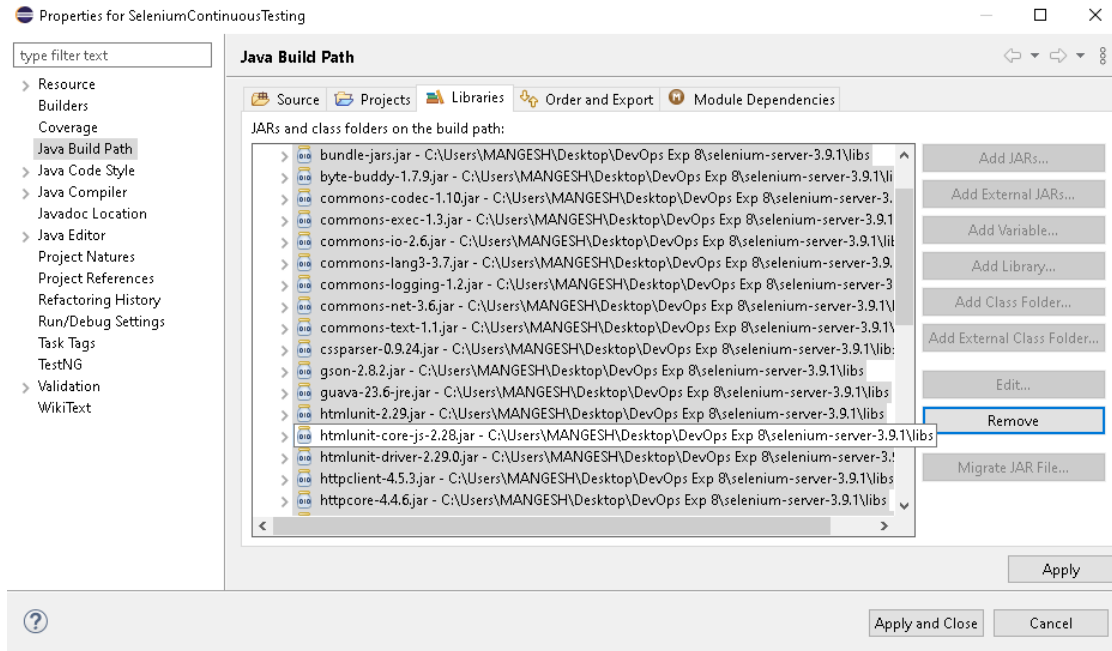
```

 driver.navigate().back();
 String title = driver.getTitle();
 System.out.println("Page Title is:-" + title);
 driver.navigate().to("https://www.facebook.com");
 driver.findElement(By.name("email")).sendKeys("abcd@gmail.com");
 driver.findElement(By.name("pass")).sendKeys("1234");
 driver.findElement(By.id("loginbutton")).click();
 driver.manage().timeouts().implicitlyWait(60, TimeUnit.SECONDS);
 driver.quit();
}
}

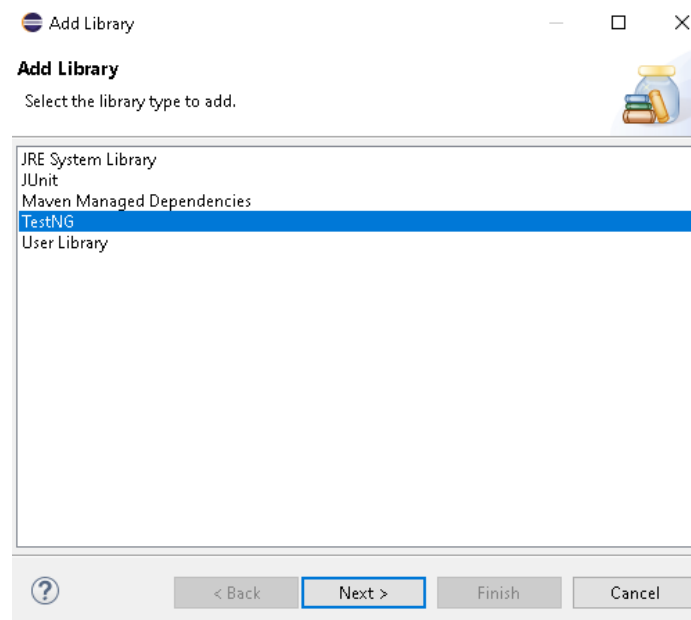
```



Step 1.7: Select Project ☐ Properties ☐ Java Build Path ☐ Select Add External JARs and Select all the Jar files in the unzipped Selenium Jar files also in Lib folder and Selenium standalone jar file. Apply changes



Step 1.8: Select and right click Project Properties Java Build Path Select Add Library and Select TestNG and click on Finish

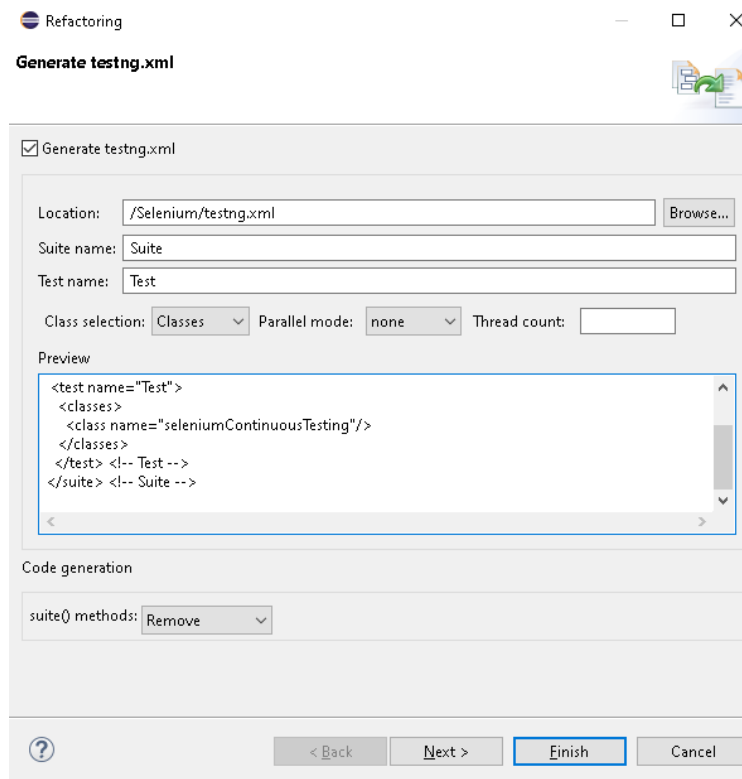


## Activity 2:

### Converting into XML files for executing TestNG test suite.

Step 2.1: Select and right click on Class file ☐ TestNG ☐ Convert to TestNG and type the below code in Preview

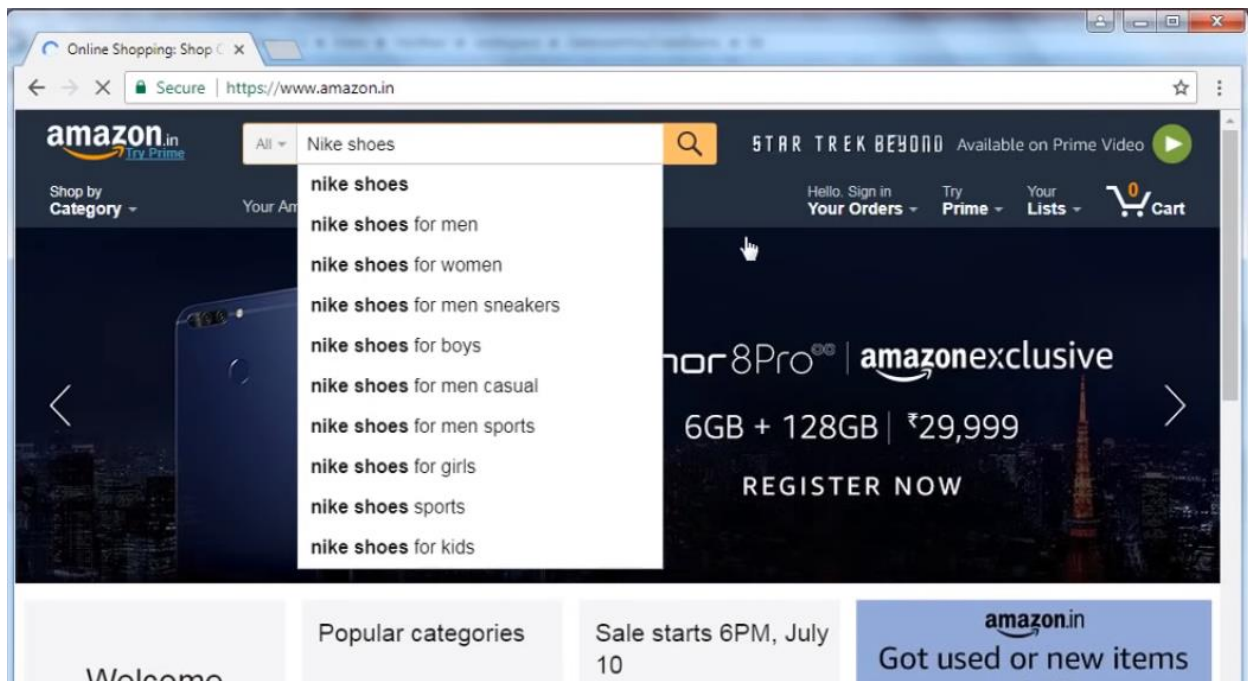
```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE suite SYSTEM "https://testng.org/testng-1.0.dtd">
<suite name="Suite">
 <test name="Test">
 <classes>
 <class name="seleniumContinuousTesting"/>
 </classes>
 </test> <!-- Test -->
</suite> <!-- Suite -->
```



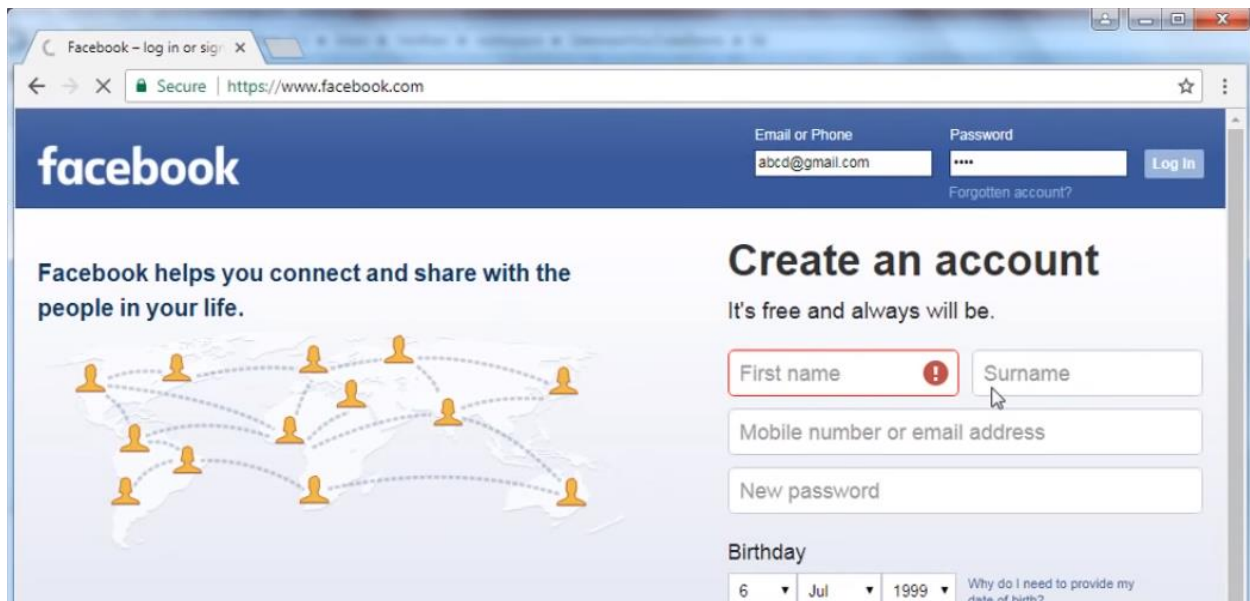
Step 2.2: Open command prompt and change the working directory to the eclipse workspace and enter the command

```
java -cp C:\Users\SHWETA\eclipse workspace\Selenium\bin;C:\Users\SHWETA\eclipse-
workspace\Selenium\lib* org.testng.TestNG testng.xml
```

Test Result: (Explanation: A new chrome browser opens up and automatically open amazon.in and enter nike shoes in the search bar and then navigates back again and then open facebook.com and login with email: abcd@gmail.com and password: 1234 and hit Login. This has been coded in the class file)







```
Command Prompt

C:\Users\MANGESH\eclipse-workspace\Selenium>python Script.py

C:\Users\RISHABH\eclipse-workspace\Selenium> java -cp C:\Users\RISHABH\eclipse-workspace\Selenium\bin;C:\Users\RISHABH\
eclipse-workspace\Selenium\lib* org.testng.TestNG testing.xml

[TestNG] Running:
 C:\Users\RISHABH\eclipse-workspace\Selenium\testing.xml

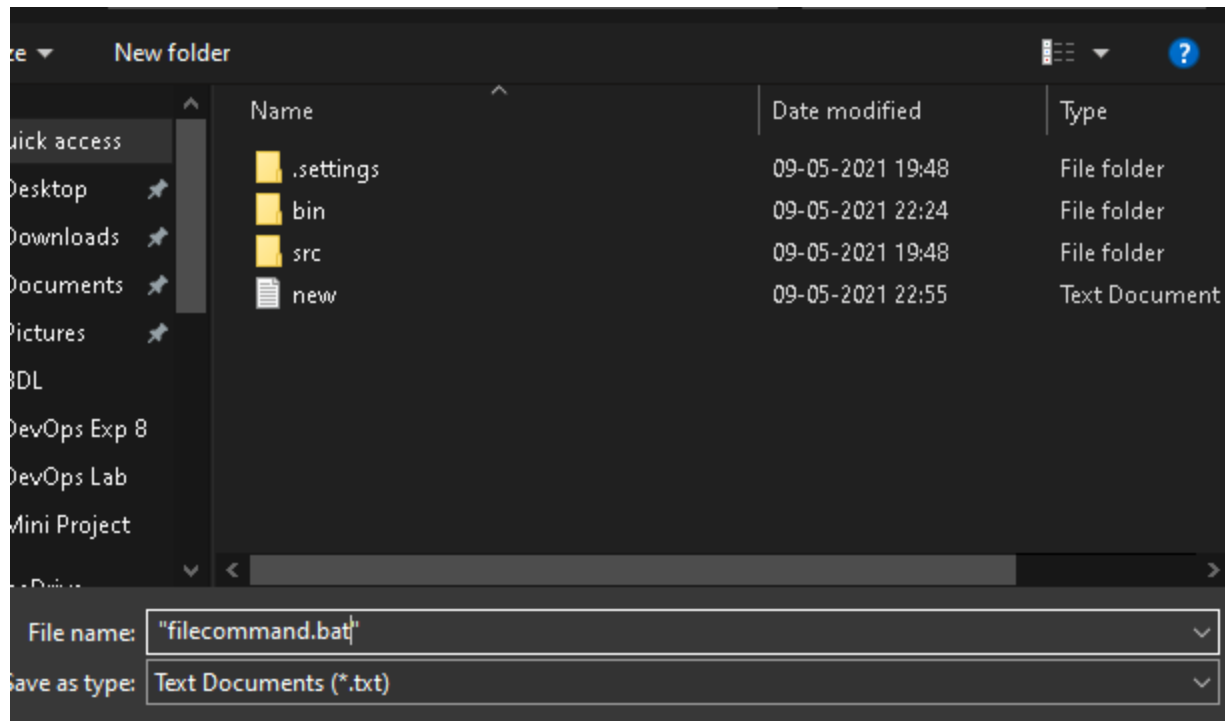
Starting WebDriver 90.0.4430.24 (2021-03-15) on port 38504
Only local connections are allowed.
Page Title is:-Online Shopping: Shop Online for Mobiles, Books, Watches, Shoes and More - Amazon.in
=====
Suite
Total tests run: 1, Failures: 0, Skips:0
=====

C:\Users\RISHABH\eclipse-workspace\Selenium>
```

### Activity 3: Create windows batch file

Create a Text Document and Type the command below and create a bat file with Save as file name as shown below to create a Windows batch file

```
java -cp bin;lib/* org.testng.TestNG testing.xml
```

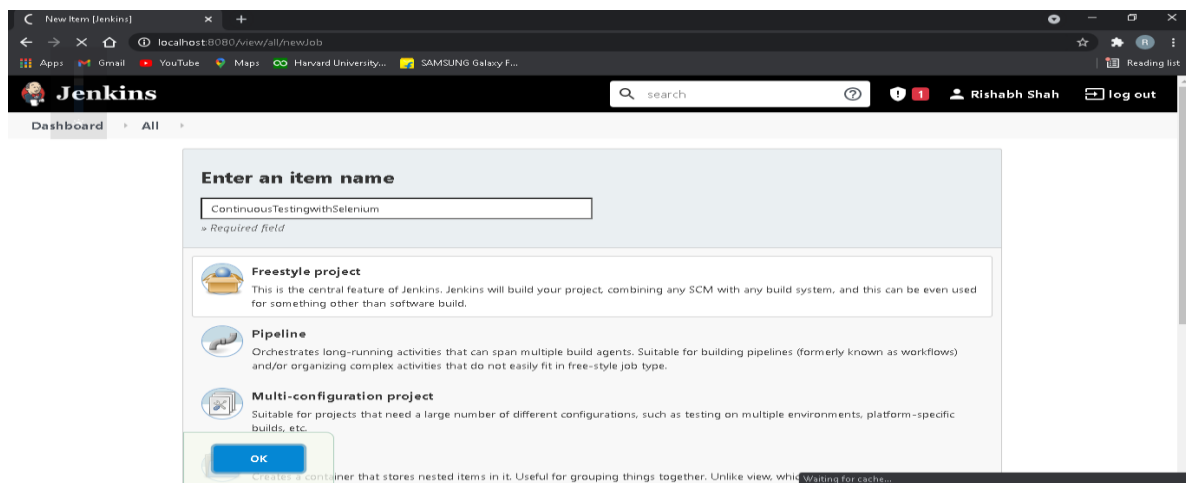


## Activity 4: Build Jenkins jobs by executing the batch file

Step 4.1: Install Jenkins and setup Jenkins by referring the below guide

<https://www.jenkins.io/doc/book/installing/windows/>

Step 4.2: Create a new Freestyle project on Jenkins by browsing to localhost:8080 or whichever port has been assigned to Jenkins during the installation.



Step 4.3: Under General Section ☐ Tab on Advanced and Select the checkbox “Use custom workspace” and in the Directory Form Fill enter the workspace location of Eclipse project

General

Source Code Management

Build Triggers

Build Environment

Build

Post-build Actions

[Plain text] [Preview](#)

☐ Discard old builds

☐ GitHub project

☐ This build requires lockable resources

☐ This project is parameterized

☐ Throttle builds

☐ Disable this project

☐ Execute concurrent builds if necessary

☐ Quiet period

☐ Retry Count

☐ Block build when upstream project is building

☐ Block build when downstream project is building

☒ Use custom workspace

Directory

C:\Users\MANGESH\eclipse-workspace\Selenium\filecommand.bat

Step 4.4: Under Build Section ☐ Select “Add build step” button ☐ Select Execute Windows batch command and in Post-build Actions Section ☐ Select Add “post-build action” ☐ Select “Email notification” and add your email address in the Recipients field. Finally, click on Apply and Save buttons.

General

Source Code Management

Build Triggers

Build Environment

Build

Post-build Actions

Build

Execute Windows batch command

Command

filecommand.bat

See the list of available environment variables

Advanced...

Add build step

Post-build Actions

E-mail Notification

Recipients

rushabshah120@gmail.com

Whitespace-separated list of recipient addresses. May reference build parameters like \$PARAM. E-mail will be sent when a build fails, becomes unstable or returns to stable.

table build

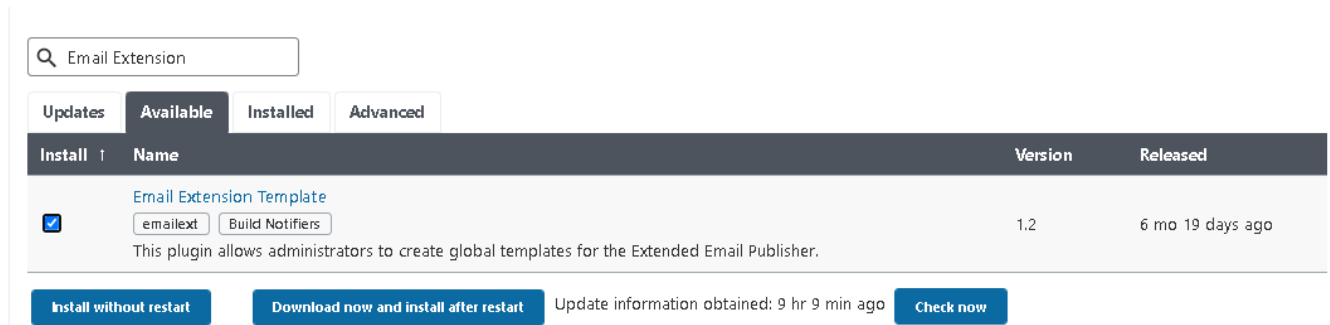
Save

Apply

151

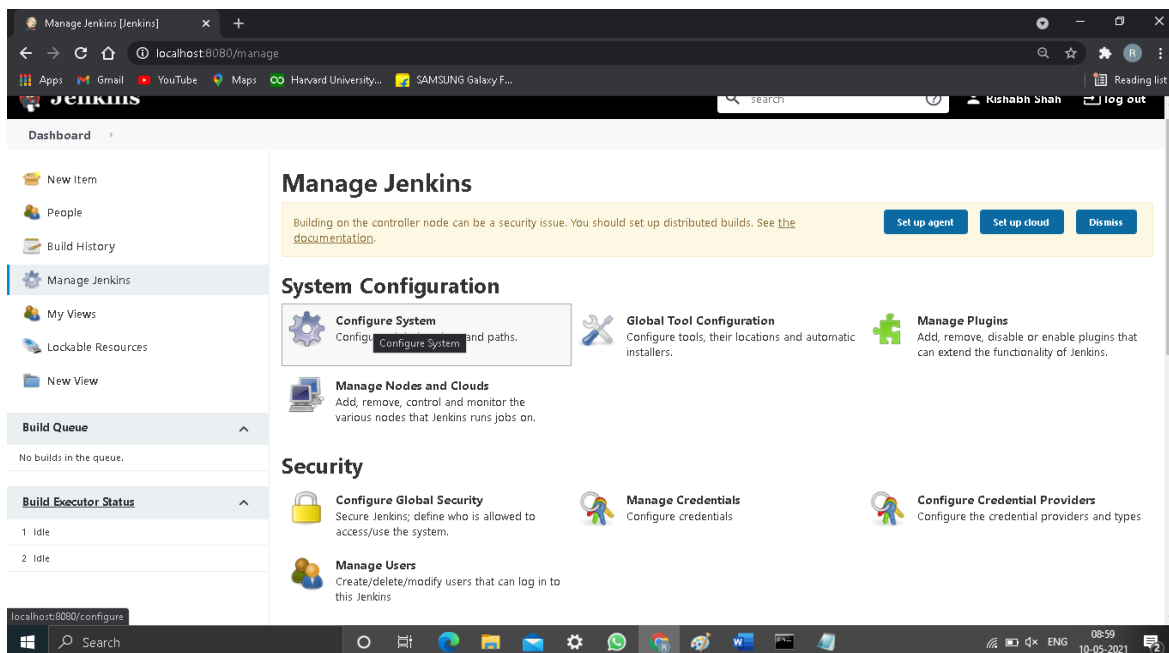
## Step 5: Schedule Jobs & configure email notification

Step 5.1: Go the Jenkins Dashboard ☐ Manage Jenkins ☐ Manage Plugins ☐ Search Email Extension Template ☐ click on the checkbox and click on Install without restart.



Step 5.2: Turn On “Less Secure app access” of your google account, in order to receive mails

Step 5.3: Go the Jenkins Dashboard ☐ Manage Jenkins ☐ Configure Systems



Step 5.4: Scroll down to E-mail Notification section at the bottom and add the details as shown below

### E-mail Notification

SMTP server

smtp.gmail.com

Default user e-mail suffix

☒ Use SMTP Authentication

User Name

rushabshah120@gmail.com

Password

\*\*\*\*\*

☒ Use SSL

☐ Use TLS

SMTP Port

465

Reply-To Address

noreply@gmail.com

Save

Apply

Step 5.5: Click on Test configuration by sending test e-mail, Enter mail address in Test e-mail recipient and hit Test Configuration Button.

☒ Test configuration by sending test e-mail

Test e-mail recipient

rushabshah120@gmail.com

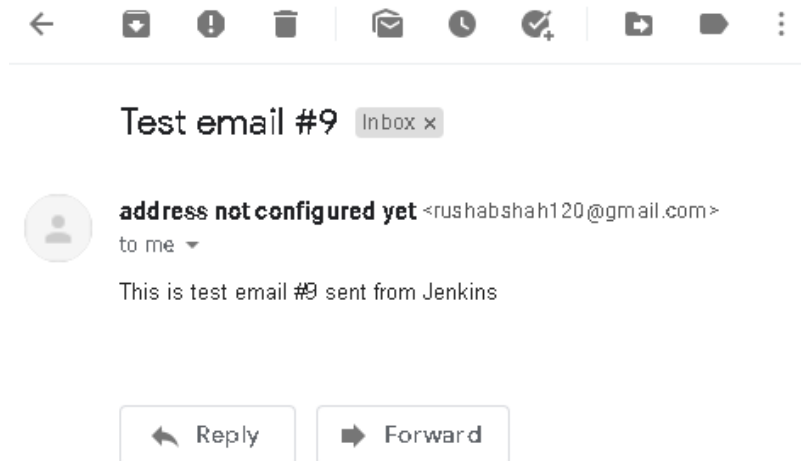
Email was successfully sent

Test configuration

Save

Apply

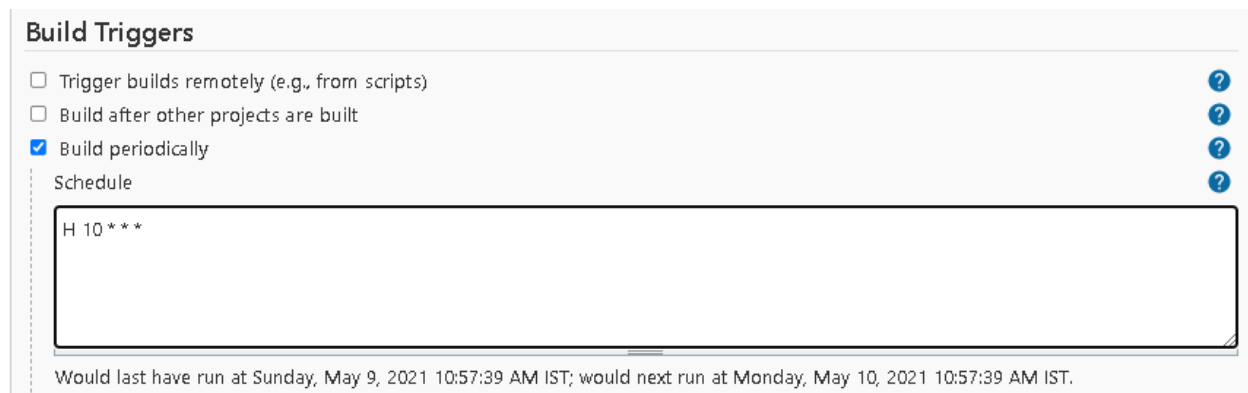
Step 5.6: Check the test mail in your Gmail inbox



Step 5.7: Go to the Dashboard ☐ Select the project ☐ Select Configure ☐ Build Trigger Section  
☐ Select Build periodically checkbox and Enter the Schedule details in cron format

H 10 \* \* \*

Meaning: In the morning 10 on every date of every month and every day of the week.



Step 5.8: Go to the Project ☐ Select Build Now



---

Step 5.11: Check your Gmail Inbox for failed Test Case mail

## Build failed in Jenkins: ContinuousTestingwithSelenium #3 Inbox x

**address not configured yet** <rushabshah120@gmail.com>

to me ▾

See <<http://localhost:8080/job/ContinuousTestingwithSelenium/3/display/redirect>>

Changes:

-----  
Started by user Rishabh Shah

Running as SYSTEM

Building in workspace <<http://localhost:8080/job/ContinuousTestingwithSelenium/ws/>>

[Selenium] \$ cmd /c call C:\WINDOWS\TEMP\jenkins565127751367542246.bat

<<http://localhost:8080/job/ContinuousTestingwithSelenium/ws/>>filecommand.bat>

<<http://localhost:8080/job/ContinuousTestingwithSelenium/ws/>>java> -cp bin;lib/\* org.testng.TestNG testing.xml

Error: Could not find or load main class org.testng.TestNG

Caused by: java.lang.ClassNotFoundException: org.testng.TestNG

Build step 'Execute Windows batch command' marked build as failure

## 3) Graded Lab Tasks

**Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.**

### Lab Task 1:

We have successfully completed the task of executing continuous testing with the help of Selenium to build test cases, Now use TestNG to create testing.xml file so that we can integrate Jenkins for continuous testing on specified time intervals and receive email notification if the test cases fail.



---

## Lab 12

### Creating Complete Jenkins Pipeline

#### Objective:

In this lab, We will go through Jenkins to implement using programming

#### Activity Outcomes:

The activities provide hands - on practice with the following topics

- Jenking Programming
- Continous Integration

#### Instructor Note:

---

## 1. Continuous Integration:

“Continuous Integration is a software development practice where members of a team integrate their work frequently, usually each person integrates at least daily - leading to multiple integrations per day. Each integration is verified by an automated build (including test) to detect integration errors as quickly as possible.” In simple way, Continuous integration (CI) is the practice of frequently building and testing each change done to your code automatically. Jenkins is a self-contained, open source automation server which can be used to automate all sorts of tasks related to building, testing, and delivering or deploying software. Our first job will execute the shell commands. The freestyle project provides enough options and features to build the complex jobs that you will need in your projects.

## 2. Solved Lab Activities

### Activity 1:

#### 1. Installation in a Docker container

```
docker pull jenkins/jenkins:lts

list images, jenkins should be available
docker images
```

docker ps will not show a running container, as we did not yet start it. Start an instance of Jenkins with the following command:

```
docker run -p 8080:8080 -p 50000:50000 -v [werjenkinshome]:/var/jenkins_home jenkins/jenkins:lts
```

Replace [werjenkinshome] with a path fitting for we, e.g. /home/vogella/jenkins\_home.

For example, on Windows:

```
docker run -u 0 -p 8082:8080 -p 50002:50000 -v //C:/Users/Lars/jenkins:/var/jenkins_home jenkins/jenkins:lts
```

If we need other tools installed, we can create our own Docker file with the necessary tooling installed. For example the following creates a new image based on Jenkins with Maven installed.

- 
- Create a new directory
  - Create a file called Dockerfile with the following content:

```
FROM jenkins/jenkins:lts
if we want to install via apt
USER root
RUN apt-get update && apt-get install -y maven
drop back to the regular jenkins user - good practice
USER jenkins
```

- Create a new image with wer Dockerfile

```
docker build -t jenkins-maven .
```

Now we can start wer new image jenkins-maven-latest.

```
docker run -u 0 -p 8082:8080 -p 50002:50000 -v //C/Users/Lars/jenkins:/var/jenkins_home jenkins-maven:latest
```

All the configuration and jobs will be stores in wer user defined directory.

## 2. Using .war file to start Jenkins

Download the *jenkins.war* file.. From this file we can start Jenkins directly via the command line with `java -jar jenkins*.war`.

If we start it locally, we find it running under the following URL: <http://localhost:8080/>

To run it in wer Tomcat server, put the .war file into the *webapps* directory. If we start Tomcat, wer Jenkins installation will be available under

```
http://localhost:8080/jenkins
```

If the jenkins.war is deployed in wer *webapps* directory, but can not be started and the tomcat manager says *FAIL - Application at context path /jenkins could not be started*, we may need to grant the permissions for `JENKINS_HOME`.

```
sudo mkdir .jenkins
```

```
sudo
```

```
chown tomcat7:nogroup .jenkins
```

## Activity 2:Configure Jenkins

**After installation, open a browser and connect to it. The default port of Jenkins is :8080, therefore on wer local machine we find it under the following URL:**

```
http://localhost:8080/
```

---

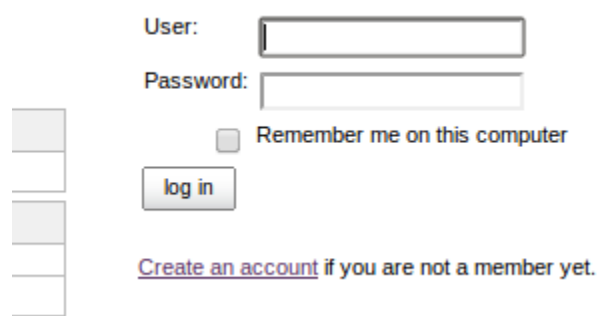
We will need to copy the initial password from the file system of the server.

Afterwards we can select to install Plugins. Select the *Install suggested Plugins* to get a typical configuration.

Create an admin user and press *Save and Finish*.

Select *Manage Jenkins* and then *Configure Global Security*. Select the *Enable security* flag. The easiest way is to use Jenkins own user database. Create at least the user "Anonymous" with read access. Also create entries for the users we want to add in the next step.

On the login page, select *Create an account* to create the users we just gave access.



The image shows a login form with the following elements: a 'User:' label followed by a text input field; a 'Password:' label followed by a text input field; a checkbox labeled 'Remember me on this computer'; a 'log in' button; and a link that says 'Create an account if you are not a member yet.' To the left of the form is a vertical sidebar with several grey rectangular buttons.

### Assign roles to users

If we want to create a role based authorization Strategy we first need to install the *Role-based Authorization Strategy* Plugin. Go to Manage Jenkins Manage Plugins Available enter *Role-based Authorization Strategy* in the *filter box* and select and install the Plugin. To see a list of commonly used Plugins go to Plugin management.

Now go to Manage Jenkins Manage and Assign Roles Assign Roles to grant users additional access rights.

Navigate to *Manage Roles* to define access restrictions in detail. *Pattern* is a regex value of the job name. The following grants unregistered users read-only access to wer build jobs that start with the L-, C-, I- or M- and only those.

#### **Project roles**

Role	Pattern	Job							
		Delete	Configure	Read	Discover	Build	Workspace	Cancel	ViewStatus
 Anonymous	[LCIM]?-.*	<input type="checkbox"/>	<input type="checkbox"/>	<input checked="" type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>

---

### Generate ssh key for Jenkins user

If we want to access a private Git repo, for example at GitHub, we need to generate an ssh key-pair. Create a SSH key with the following command.

```
sudo -u jenkins ssh-keygen
```

The public key must be uploaded to the service we are using, e.g., GitHub.

### Configure the default port of the Jenkins build server

The default port number can be changed in the config file at

```
sudo vim /etc/default/jenkins
```

Here we can set a different port number, e.g. HTTP\_PORT=8082

Now we need to restart Jenkins with

```
service jenkins restart
```

## 4. Setting up a Jenkins job

The build of a project is handled via *jobs* in Jenkins. Select **New Item**. Afterwards, enter a name for the job and select *Freestyle project* and press OK.

Enter a description for the job (project name) and configure how many builds should be retained and for how long.

The screenshot shows the Jenkins 'New Item' configuration page. The 'General' tab is selected. The 'Project name' field contains 'example-maven-project'. The 'Description' field contains 'Example project for the usage of Maven'. Below the description, there is a '[Plain text] Preview' link. The 'Discard old builds' checkbox is checked. Under 'Strategy', the 'Log Rotation' section is expanded, showing 'Days to keep builds' set to 10 and 'Max # of builds to keep' set to 10. Below this, there are several unchecked checkboxes: 'GitHub project', 'This project is parameterized', 'Throttle builds', and 'Disable this project'. The 'Execute concurrent builds if necessary' checkbox is checked.

Configure how the source code can be retrieved. If we are for example using Git, enter the URL to the Git repository. If the repository is not public, we may also need to configure the credentials.

**Source Code Management**

☐ None  
☒ Git

**Repositories**

Repository URL

Credentials

**Branches to build**

Branch Specifier (blank for 'any')

**Repository browser**

**Additional Behaviours**

☐ Subversion

Specify when and how the build should be triggered. The following example polls the Git repository every 15 min. It triggers a build, if something has changed in the repo.

**Build Triggers**

☐ Trigger builds remotely (e.g., from scripts)  
☐ Build after other projects are built  
☐ Build periodically  
☐ GitHub hook trigger for GITScm polling  
☒ Poll SCM

**Schedule**

Would last have run at Monday,

To trigger a build after a commit on the Git repository, select *GitHub hook trigger for GITScm polling* instead of *Poll SCM*.

---

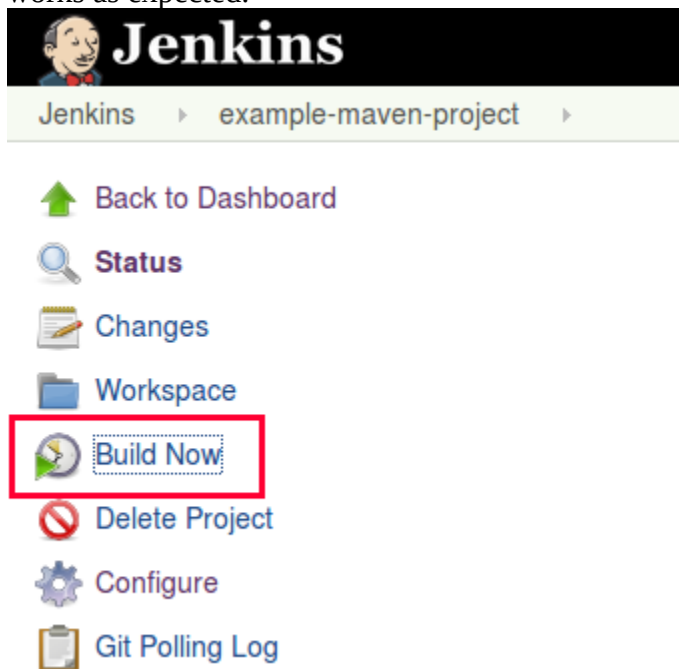
## Build

Invoke top-level Maven targets

Goals

Add build step ▾

Press Save to finish the job definition. Press Build Now on the job page to validate that the job works as expected.



After a while the job should go to green or blue (depending on wer configuration), if successful. Click on the job and afterwards on *Console Output* to see the log file. Here we can analyze the build errors.

### **Activity 3: Build Pipelines**

Jenkins pipelines help we align the build process of a project. This is done by specifying tasks and the order in which they are executed. There are all kinds of possible tasks that a Jenkins pipeline can do for we. For example, build assets, send an email on error, send the build artifacts via SSH to wer application server, etc.

Jenkins allows to specify pipelines using a Jenkinsfile. This is just a textfile that contains the necessary data for Jenkins to execute the pipeline.

Jenkins supports two different syntaxes.

1. Declarative (since Pipeline version 2.5)
2. Scripted

---

For this tutorial we will focus on the declarative approach.  
The following example shows a pipeline with 2 stages:

```
pipeline {
 agent any

 stages {
 stage('Build Assets') {
 agent any
 steps {
 echo 'Building Assets...'
 }
 }
 stage('Test') {
 agent any
 steps {
 echo 'Testing stuff...'
 }
 }
 }
}
```

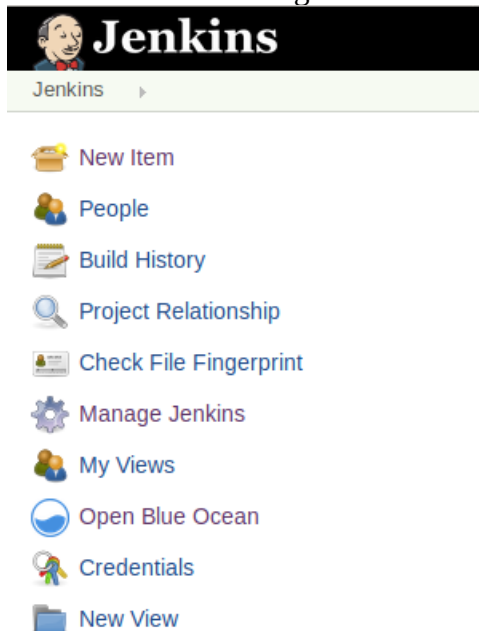
The agent directive tells Jenkins to allocate a workspace and an executor for the pipeline. Without it, the pipeline is not valid and therefore required.

### Setup using the Blue Ocean Plugin

The above process can also be done using the [Blue Ocean Jenkins](#) Plugin.

To install the Plugin go to **Manage Jenkins** **Manage Plugins** **Available** and select the *Blue Ocean* Plugin.

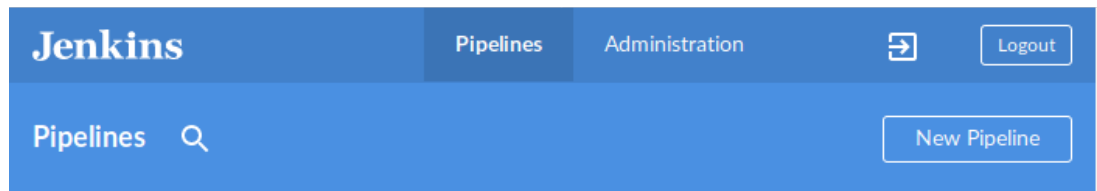
After the installation is finished we have an additional menu entry called Open Blue Ocean in our main *Jenkins* navigation.



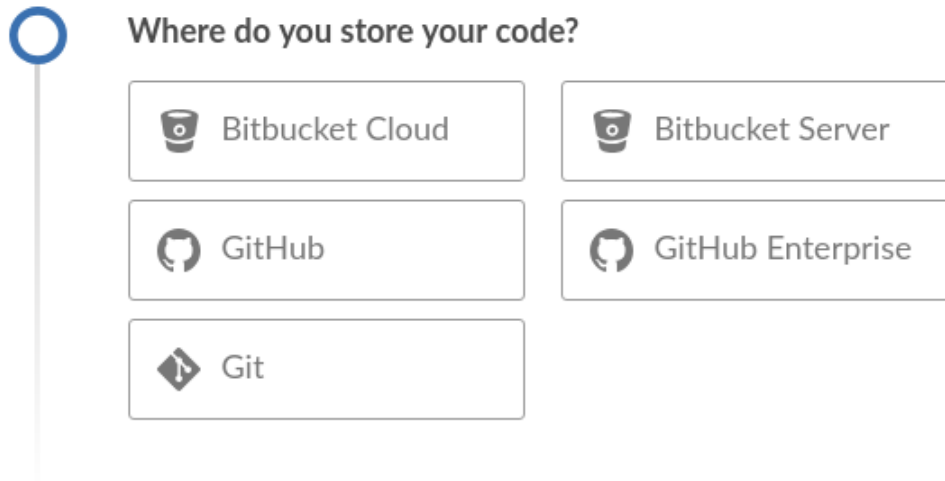
### Creating a new Pipeline

Click on New Pipeline to create a new Pipeline.





Select wer version control provider.



Depending on wer provider we will need to pass some kind of credentials.

The *Blue Ocean* application will provide a link to the GitHub page we need to visit. The necessary permissions that *Blue Ocean* needs to operate are already selected. Add a description and click on Generate Token at the bottom of the page.

Copy the generated token and paste it in the *Blue Ocean* mask.

Select the account the repository belongs to and select the repository.

#### Adding steps to wer Pipeline

In the next screen we will see a visual representation of wer Pipeline. Here we can add or remove steps.

✓

Where do you store your code?

 Bitbucket Cloud

 Bitbucket Server

 GitHub

 GitHub Enterprise

 Git

✓

Connect to GitHub

Jenkins needs an access token to authorize itself with GitHub.  
[Create an access token here.](#)

.....

✓

✓

Which organization does the repository belong to?

○

Choose a repository

Loaded 131 repositories

scripts

scripts

tutorials.scripts

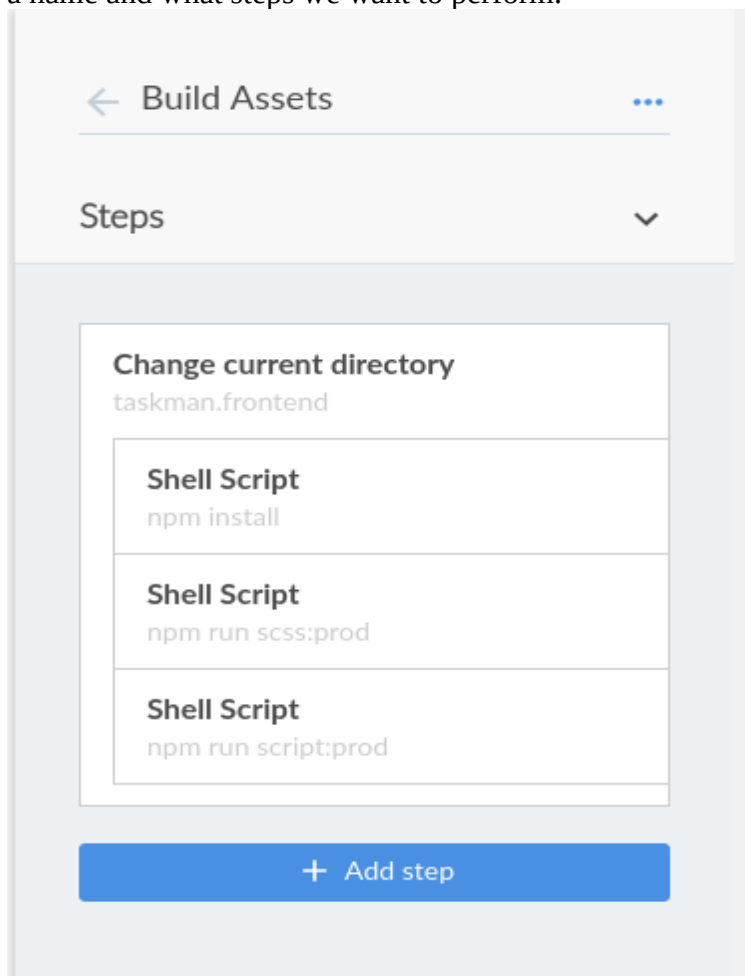
Create Pipeline

○

Complete



To create a new step click on + in the canvas. A menu will open on the right that lets we specify a name and what steps we want to perform.



After we have finished editing the Pipeline *Blue Ocean* will offer we to commit the newly created pipeline to wer repository.

Under the hood *Blue Ocean* only created a valid Jenkinsfile to be used by Jenkins.

---

After committing *Jenkins* will build the project using the newly modified Pipelines.

## 6. Restart Jenkins

After installing Plugins we will need to restart Jenkins. We can do so by adding restart as URL parameter,

or with the following command:

```
service jenkins restart
```

## 3) Graded Lab Tasks

*Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.*

Lab Task 1:

Use Jenkin pipeline created to automatic deployment of new code.

Lab Task 2:

Use Jenkin pipeline created to automatic running of test on that code.

---

## Lab 13

### Software Configuration Management using Puppet tool

#### Objective:

In this lab, We will go through puppet tool for software configuration management in DevOps

#### Activity Outcomes:

The activities provide hands - on practice with the following topics

- Puppet tool
- Software Configuration management in Cloud

#### Instructor Note:

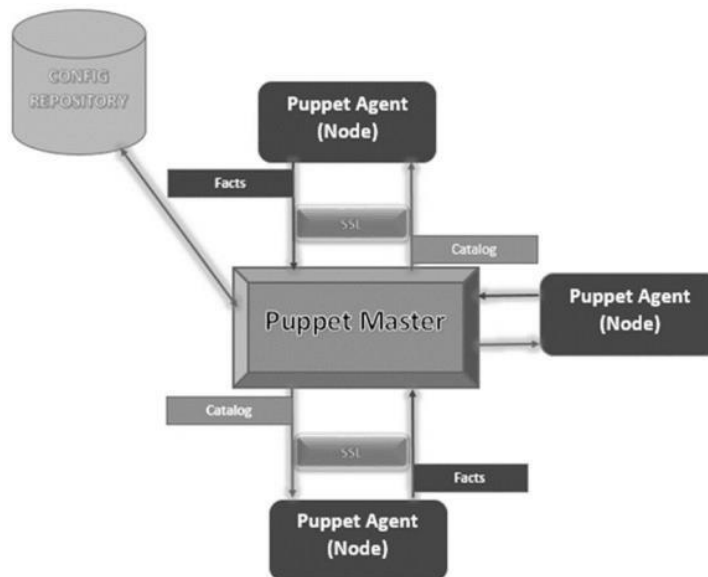
---

## 1. Puppet

Puppet is a configuration management tool that simplifies system administration. Puppet uses a client/server model in which your managed nodes, running a process called the Puppet agent, talk to and pull down configuration profiles from a Puppet master.

Puppet deployments can range from small groups of servers up to enterprise-level operations. This guide will demonstrate how to install Puppet 6.1 on three servers:

- A Puppet master running Ubuntu 18.04
- A managed Puppet Agent node running Ubuntu 18.04



## 2. Solved Lab Activities:

### Activity 1:

1. First install Puppet master and puppet slave in ubuntu. in the master virtual machine, theres no master certificate , we will confirm that by adding following code. In Puppet agent virtual machine, a puppet agent certificate and we will fetch puppet master certificate.

```
File Edit View Search Terminal Help
[root@PuppetAgent ~]# puppet agent -t
Info: Creating a new SSL key for puppetagent
Info: Caching certificate for ca
Info: csr_attributes file loading from /etc/puppet/csr_attributes.yaml
Info: Creating a new SSL certificate request for puppetagent
Info: Certificate Request fingerprint (SHA256): C5:F9:DE:75:94:2C:A2:4A:9A:F8:6C:3C:68:14:10:CF:BB:E6:DC:A4:C8:1C:D7:46:3B:B9:51:D2:1A:CF:B7:AB
Info: Caching certificate for ca
Exiting; no certificate found and waitforcert is disabled
[root@PuppetAgent ~]#
```

We can see the certificate in puppet master send by puppet agent and will sign that certificate.

```
File Edit View Search Terminal Help
[root@PuppetMaster ~]# puppet cert list
"puppetagent" (SHA256) C5:F9:DE:75:94:2C:A2:4A:9A:F8:6C:3C:68:14:10:CF:BB:E6:DC:A4:C8:1C:D7:46:3B:B9:51:D2:1A:CF:B7:AB
[root@PuppetMaster ~]# puppet cert sign puppetagent
Notice: Signed certificate request for puppetagent
Notice: Removing file Puppet::SSL::CertificateRequest puppetagent at '/var/lib/puppet/ssl/ca/requests/puppetagent.pem'
[root@PuppetMaster ~]# clea
```

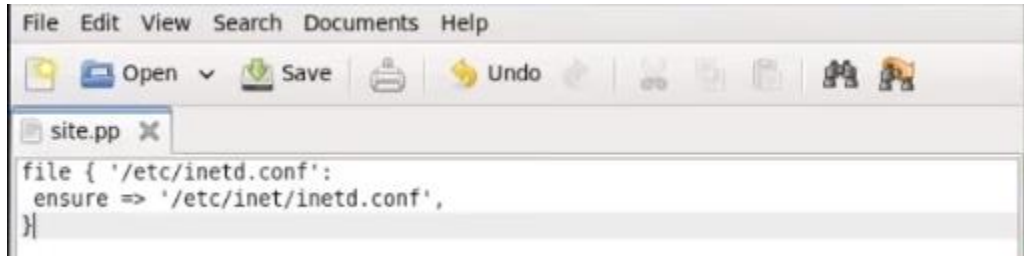
Now we will securely establish the connection between puppet master and puppet agent.

```
File Edit View Search Terminal Help
[root@PuppetAgent ~]# puppet agent -t
Info: Caching certificate for puppetagent
Info: Caching certificate_revocation_list for ca
Info: Caching certificate for puppetagent
Warning: Unable to fetch my node definition, but the agent run will continue:
Warning: undefined method 'include?' for nil:NilClass
Info: Retrieving pluginfacts
Info: Retrieving plugin
Info: Caching catalog for puppetagent
Info: Applying configuration version '1485495511'
Info: Creating state file /var/lib/puppet/state/state.yaml
Notice: Finished catalog run in 0.03 seconds
[root@PuppetAgent ~]#
```

Now, here we'll deploy php and mysql using puppet. We will first download the predefined modules of php and mysql which are present in puppet forge. We cannot directly deploy the modules, we need to declare the classes.

Now, we'll define the file in the manifest directory.

```
File Edit View Search Terminal Help
[root@PuppetMaster ~]# gedit /etc/puppet/manifests/site.pp
```



```
file { ['/etc/inetd.conf':
 ensure => '/etc/inet/inetd.conf',
}]
```

Now we must see the file updated in the puppet agent.

```
File Edit View Search Terminal Help
[root@PuppetAgent ~]# puppet agent -t
Info: Retrieving pluginfacts
Info: Retrieving plugin
Info: Caching catalog for puppetagent
Info: Applying configuration version '1485496165'
Notice: /Stage[main]/Main/File[/etc/inetd.conf]/ensure: created
Notice: Finished catalog run in 0.03 seconds
[root@PuppetAgent ~]#

File Edit View Search Terminal Help
[root@PuppetAgent ~]# ls /etc/inetd.conf
/etc/inetd.conf
[root@PuppetAgent ~]#
```

Now we'll write the code in order to deploy the apache in the same pp file.

```
File Edit View Search Documents Help
site.pp
package { 'httpd':
 ensure => installed,
}
service { 'httpd':
 ensure => running,
}
```

Puppet agent automatically pulls from the puppet master.

```
File Edit View Search Terminal Help
[root@PuppetAgent ~]# puppet agent -t
Info: Retrieving pluginfacts
Info: Retrieving plugin
Info: Caching catalog for puppetagent
Info: Applying configuration version '1485528946'
Notice: Finished catalog run in 0.24 seconds
[root@PuppetAgent ~]#
```

So we have successfully installed the apache on your localhost.

Now we'll download the mysql module from the puppet forge for that we'll add the following command and it will install and download the mysql module.

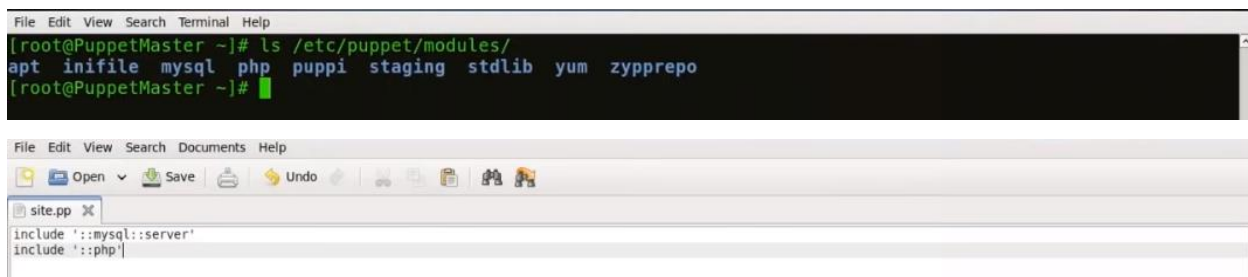
```
File Edit View Search Terminal Help
[root@PuppetMaster ~]# puppet module install puppetlabs-mysql --version 3.10.0
Notice: Preparing to install into /etc/puppet/modules ...
Notice: Downloading from https://forgeapi.puppetlabs.com ...
Notice: Installing -- do not interrupt ...
/etc/puppet/modules
├─ puppetlabs-mysql (v3.10.0)
├─ puppet-staging (v2.1.0)
└─ puppetlabs-stdlib (v4.15.0)
[root@PuppetMaster ~]# cle
```

After that, we will install and download the php module from the forge.

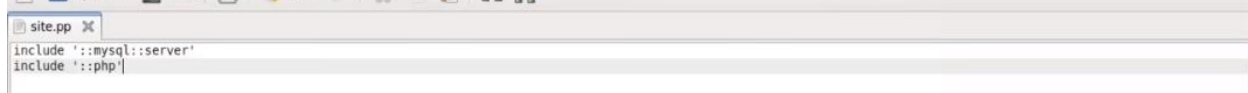
```
File Edit View Search Terminal Help
[root@PuppetMaster ~]# puppet module install mayflower-php --version 4.0.0-beta1
Notice: Preparing to install into /etc/puppet/modules ...
Notice: Downloading from https://forgeapi.puppetlabs.com ...
Notice: Installing -- do not interrupt ...
/etc/puppet/modules
├─ mayflower-php (v4.0.0-beta1)
├─ darin-zypprepo (v1.0.2)
├─ example42-yum (v2.1.28)
├─ example42-puppi (v2.2.1)
├─ puppetlabs-apt (v2.3.0)
├─ puppetlabs-stdlib (v4.15.0)
└─ puppetlabs-inifile (v1.6.0)
[root@PuppetMaster ~]#
```



We will check the downloaded modules from puppet forge by using following command.



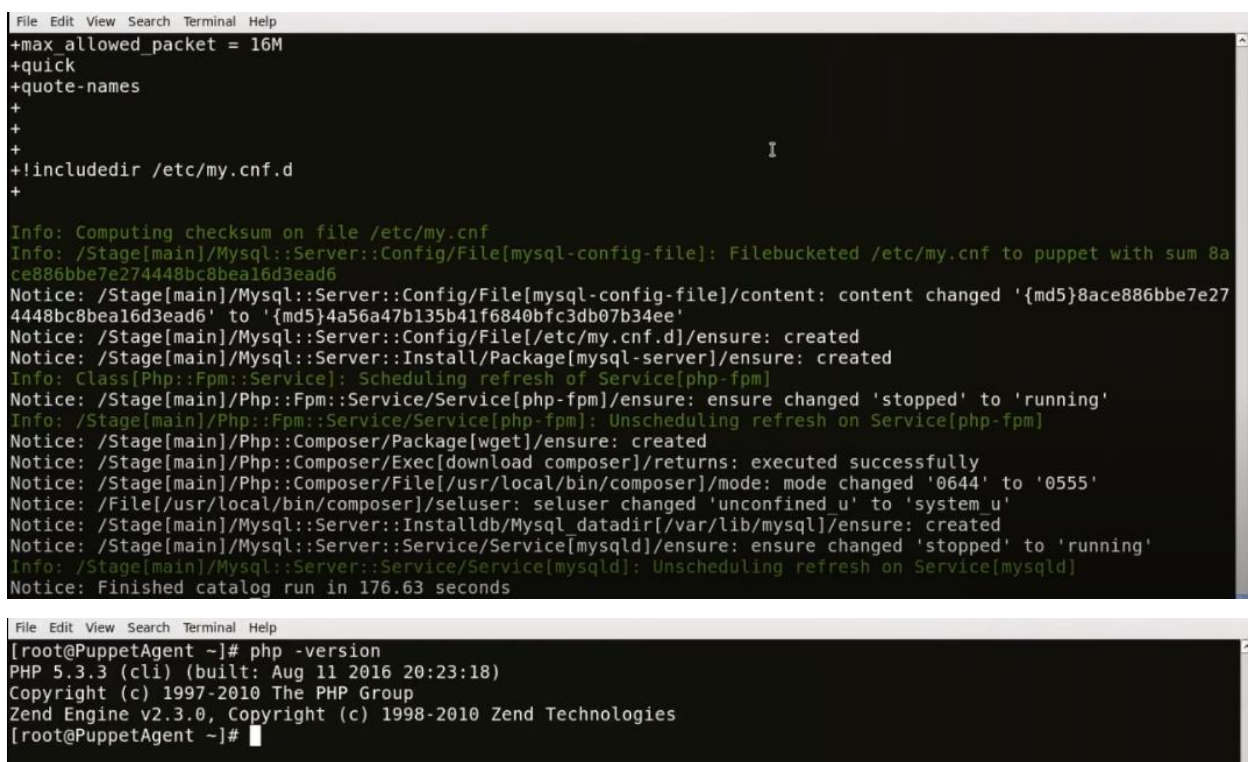
```
File Edit View Search Terminal Help
[root@PuppetMaster ~]# ls /etc/puppet/modules/
apt inifile mysql php puppi staging stdlib yum zypprepo
[root@PuppetMaster ~]#
```



```
File Edit View Search Documents Help
site.pp
include '::mysql::server'
include '::php'
```

Now we can clearly deploy the php and mysql modules in agent by using the following command.

*Puppet agent -t*



```
File Edit View Search Terminal Help
+max_allowed_packet = 16M
+quick
+quote-names
+
+
+!includedir /etc/my.cnf.d
+
Info: Computing checksum on file /etc/my.cnf
Info: /Stage[main]/Mysql::Server::Config/File[mysql-config-file]: Filebucketed /etc/my.cnf to puppet with sum 8a
ce886bbe7e274448bc8bea16d3ead6
Notice: /Stage[main]/Mysql::Server::Config/File[mysql-config-file]/content: content changed '{md5}8ace886bbe7e27
4448bc8bea16d3ead6' to '{md5}4a56a47b135b41f6840bfc3db07b34ee'
Notice: /Stage[main]/Mysql::Server::Config/File[/etc/my.cnf.d]/ensure: created
Notice: /Stage[main]/Mysql::Server::Install/Package[mysql-server]/ensure: created
Info: Class[Php::Fpm::Service]: Scheduling refresh of Service[php-fpm]
Notice: /Stage[main]/Php::Fpm::Service/Service[php-fpm]/ensure: ensure changed 'stopped' to 'running'
Info: /Stage[main]/Php::Fpm::Service/Service[php-fpm]: Unscheduling refresh on Service[php-fpm]
Notice: /Stage[main]/Php::Composer/Package[wget]/ensure: created
Notice: /Stage[main]/Php::Composer/Exec[download composer]/returns: executed successfully
Notice: /Stage[main]/Php::Composer/File[/usr/local/bin/composer]/mode: mode changed '0644' to '0555'
Notice: /File[/usr/local/bin/composer]/seluser: seluser changed 'unconfined u' to 'system_u'
Notice: /Stage[main]/Mysql::Server::Installdb/Mysql_datadir[/var/lib/mysql]/ensure: created
Notice: /Stage[main]/Mysql::Server::Service/Service[mysqld]/ensure: ensure changed 'stopped' to 'running'
Info: /Stage[main]/Mysql::Server::Service/Service[mysqld]: Unscheduling refresh on Service[mysqld]
Notice: Finished catalog run in 176.63 seconds

File Edit View Search Terminal Help
[root@PuppetAgent ~]# php -version
PHP 5.3.3 (cli) (built: Aug 11 2016 20:23:18)
Copyright (c) 1997-2010 The PHP Group
Zend Engine v2.3.0, Copyright (c) 1998-2010 Zend Technologies
[root@PuppetAgent ~]#
```

So, We have successfully php and mysql using puppet.

### 3) Graded Lab Tasks

*Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.*

Lab Task 1:

---

Configure a mongodb database and node application using puppet

## **Lab 14**

### **Software Configuration Management and provisioning using Ansible tool**

#### **Objective:**

In this lab, We will go through Ansible tool for provision and configuration management in cloud

#### **Activity Outcomes:**

The activities provide hands - on practice with the following topics

- Ansible
- Provision and Configuration management in DevOps

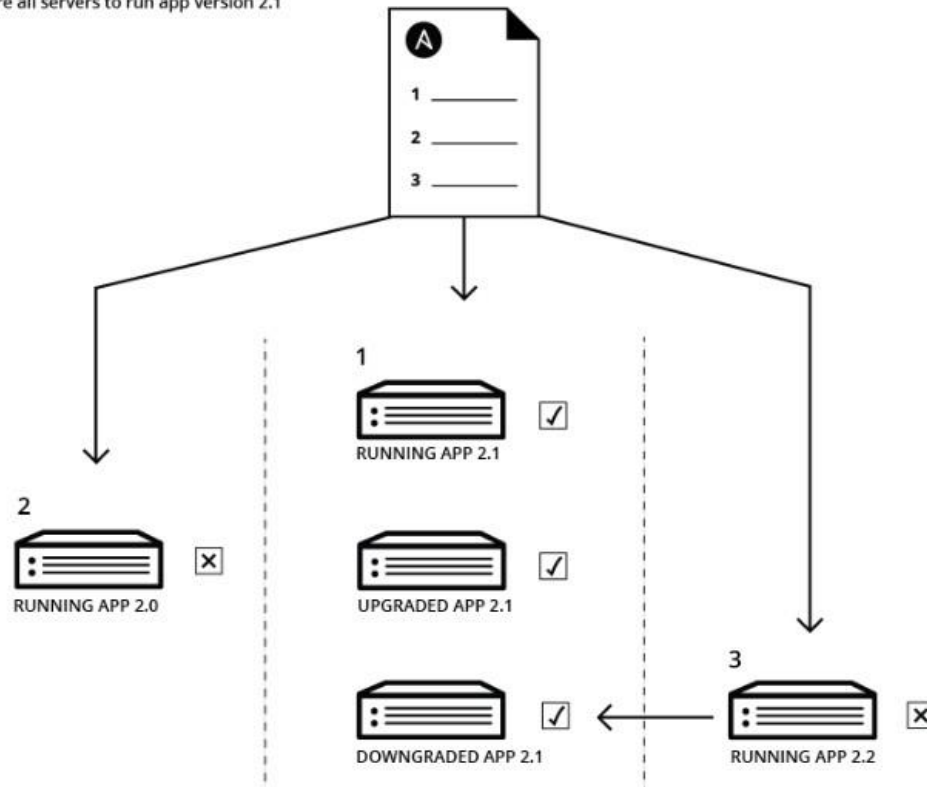
#### **Instructor Note:**

---

## 1. Software Configuration Management and provisioning using Ansible tool.

Ansible is a free and open source automation engine that removes the workload associated with repetitive tasks. It allows you to comfortably simplify your workload by automating different tasks in your IT environment. It makes use of SSH protocol to retrieve information from remote machines and manage them.

Configure all servers to run app version 2.1



## 2. Solved Lab Activities

### Activity 1:

How to install Ansible:

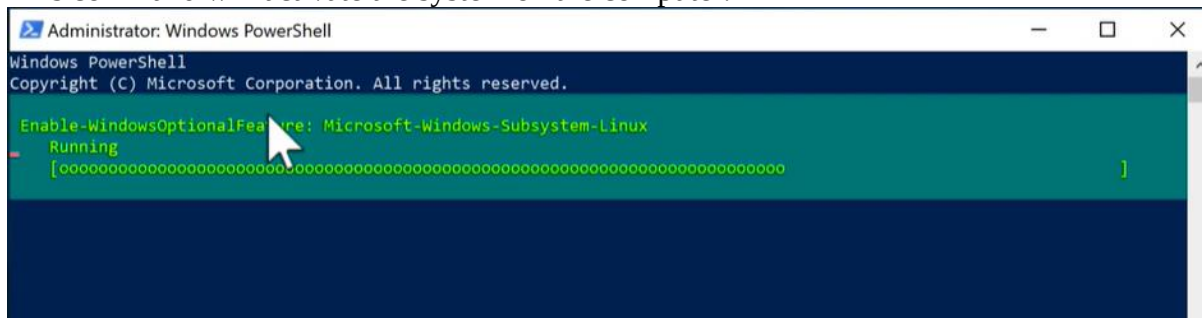
[https://docs.ansible.com/ansible/latest/installation\\_guide/intro\\_installation.html](https://docs.ansible.com/ansible/latest/installation_guide/intro_installation.html)

Step1: Go to Windows Powershell and run as administrator

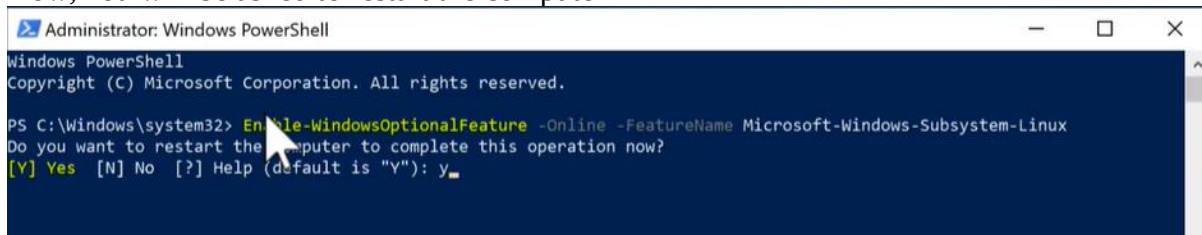


Step2: Run the command: **Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Windows-Subsystem-Linux**

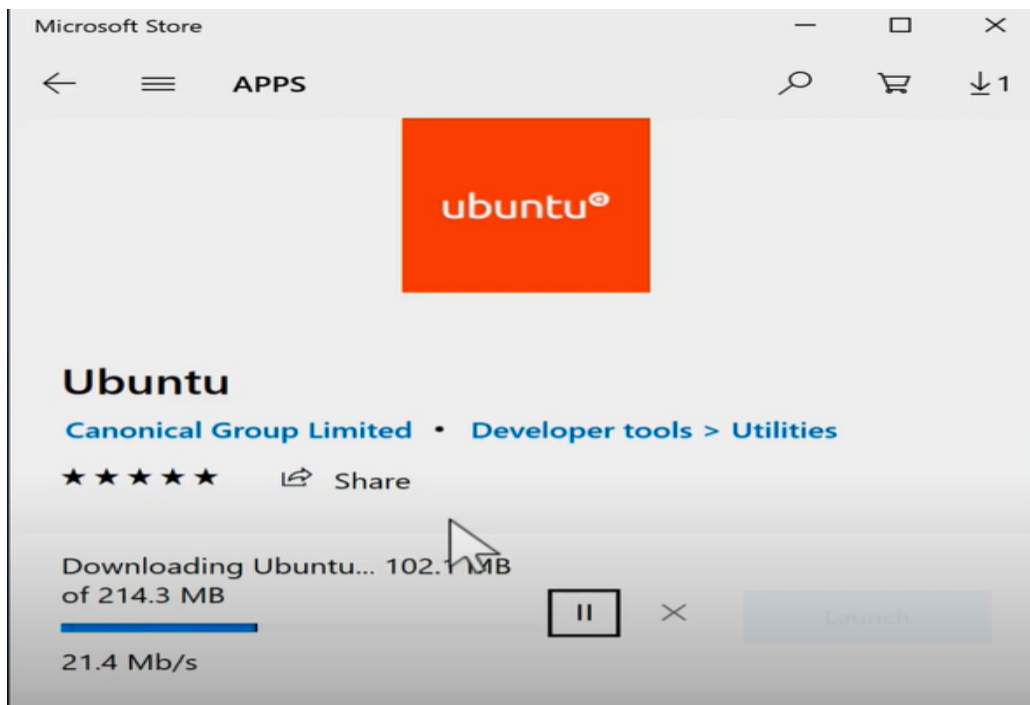
This command will activate the system on the computer.



Now, You will be asked to restart the computer



Download and install Ubuntu on your computer



After installation you will require to set the username and password for accessing Ubuntu

```
Installing, this may take a few minutes...
Please create a default UNIX user account. The username does not need to match your Windows username.
For more information visit: https://aka.ms/wslusers
Enter new UNIX username: maxim
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
Installation successful!
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
```

Now add the ansible repository to the packer manager

```
maxim@DESKTOP-GM07NR3:~$ sudo apt-add-repository ppa:ansible/ansible
[sudo] password for maxim:
Ansible is a radically simple IT automation platform that makes your applications and systems easier to deploy. Avoid
writing scripts or custom code to deploy and update your applications- automate in a language that approaches plain Engl
sh, using SSH, with no agents to install on remote systems.

http://ansible.com/
More info: https://launchpad.net/~ansible/+archive/ubuntu/ansible
Press [ENTER] to continue or Ctrl-c to cancel adding it.
```

Confirm and Press Enter



```

Get:1 http://ppa.launchpad.net/ansible/ansible/ubuntu bionic InRelease [15.9 kB]
Hit:2 http://archive.ubuntu.com/ubuntu bionic InRelease
Get:3 http://archive.ubuntu.com/ubuntu bionic-updates InRelease [88.7 kB]
Get:4 http://security.ubuntu.com/ubuntu bionic-security InRelease [88.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu bionic-backports InRelease [74.6 kB]
Get:6 http://archive.ubuntu.com/ubuntu bionic/universe amd64 Packages [8570 kB]
Get:7 http://archive.ubuntu.com/ubuntu bionic/universe Translation-en [4941 kB]
Get:8 http://archive.ubuntu.com/ubuntu bionic/multiverse amd64 Packages [151 kB]
Get:9 http://archive.ubuntu.com/ubuntu bionic/multiverse Translation-en [108 kB]
Get:10 http://ppa.launchpad.net/ansible/ansible/ubuntu bionic/main amd64 Packages [540 B]
Get:11 http://ppa.launchpad.net/ansible/ansible/ubuntu bionic/main Translation-en [344 B]
Get:12 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64 Packages [558 kB]
Get:13 http://archive.ubuntu.com/ubuntu bionic-updates/main Translation-en [207 kB]
Get:14 http://archive.ubuntu.com/ubuntu bionic-updates/restricted amd64 Packages [6996 B]
Get:15 http://archive.ubuntu.com/ubuntu bionic-updates/restricted Translation-en [3076 B]
Get:16 http://archive.ubuntu.com/ubuntu bionic-updates/universe amd64 Packages [746 kB]
Get:17 http://archive.ubuntu.com/ubuntu bionic-updates/universe Translation-en [195 kB]
Get:18 http://archive.ubuntu.com/ubuntu bionic-updates/multiverse amd64 Packages [6388 B]
Get:19 http://archive.ubuntu.com/ubuntu bionic-updates/multiverse Translation-en [3452 B]
Get:20 http://archive.ubuntu.com/ubuntu bionic-backports/main amd64 Packages [1024 B]
Get:21 http://archive.ubuntu.com/ubuntu bionic-backports/main Translation-en [448 B]
Get:22 http://archive.ubuntu.com/ubuntu bionic-backports/universe amd64 Packages [3468 B]
Get:23 http://archive.ubuntu.com/ubuntu bionic-backports/universe Translation-en [1604 B]
Get:24 http://security.ubuntu.com/ubuntu bionic-security/main amd64 Packages [283 kB]
Get:25 http://security.ubuntu.com/ubuntu bionic-security/main Translation-en [103 kB]
Get:26 http://security.ubuntu.com/ubuntu bionic-security/restricted amd64 Packages [4296 B]
Get:27 http://security.ubuntu.com/ubuntu bionic-security/restricted Translation-en [2192 B]
Get:28 http://security.ubuntu.com/ubuntu bionic-security/universe amd64 Packages [128 kB]
Get:29 http://security.ubuntu.com/ubuntu bionic-security/universe Translation-en [72.6 kB]
Get:30 http://security.ubuntu.com/ubuntu bionic-security/multiverse amd64 Packages [3748 B]
Get:31 http://security.ubuntu.com/ubuntu bionic-security/multiverse Translation-en [1952 B]
Fetched 16.4 MB in 20s (821 kB/s)
Reading package lists... Done

```

Lastly to install ansible run the command **sudo apt-get install ansible**

```

maxim@DESKTOP-GMO7NR3:~$ sudo apt-get install ansible
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following package was automatically installed and is no longer required:
 libfreetype6
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
 libpython-stdlib libpython2.7-minimal libpython2.7-stdlib python python-asn1crypto python-cffi-backend python-crypto
 python-cryptography python-enum34 python-httplib2 python-idna python-ipaddress python-jinja2 python-markupsafe
 python-minimal python-paramiko python-pkg-resources python-pyasn1 python-setuptools python-six python-yaml python2.7
 python2.7-minimal python2.7-stdlib python2.7-tk
Suggested packages:
 python-doc python-tk python-crypto-doc python-cryptography-doc python-cryptography-vectors python-enum34-doc
 python-jinja2-doc python-gssapi python-setuptools-doc python2.7-doc binutils binfmt-support
The following NEW packages will be installed:
 ansible libpython-stdlib libpython2.7-minimal libpython2.7-stdlib python python-asn1crypto python-cffi-backend
 python-crypto python-cryptography python-enum34 python-httplib2 python-idna python-ipaddress python-jinja2
 python-markupsafe python-minimal python-paramiko python-pkg-resources python-pyasn1 python-setuptools python-six
 python-yaml python2.7 python2.7-minimal python2.7-stdlib python2.7-tk
0 upgraded, 27 newly installed, 0 to remove and 0 not upgraded.
Need to get 16.4 MB of archives.
After this operation, 100 MB of additional disk space will be used.
Do you want to continue? [Y/n]

```

Ansible is successfully installed in the Ubuntu operating system. You can check this by running this command.

```

maxim@DESKTOP-GMO7NR3:~$ ansible --version
ansible 2.7.9
 config file = /etc/ansible/ansible.cfg
 configured module search path = [u'/home/maxim/.ansible/plugins/modules', u'/usr/share/ans
 ansible python module location = /usr/lib/python2.7/dist-packages/ansible
 executable location = /usr/bin/ansible
 python version = 2.7.15rc1 (default, Nov 12 2018, 14:31:15) [GCC 7.3.0]

```

---

### 3) Graded Lab Tasks

*Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.*

#### Lab Task 1:

You are required to run and configure Unix-like systems as well as Windows systems to provide infrastructure as code.